BT404 – MIDTERM MERGED PPT HANDOUTS (LECTURE 01 TO 92)

Food Biotechnology

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All About Biotech

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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

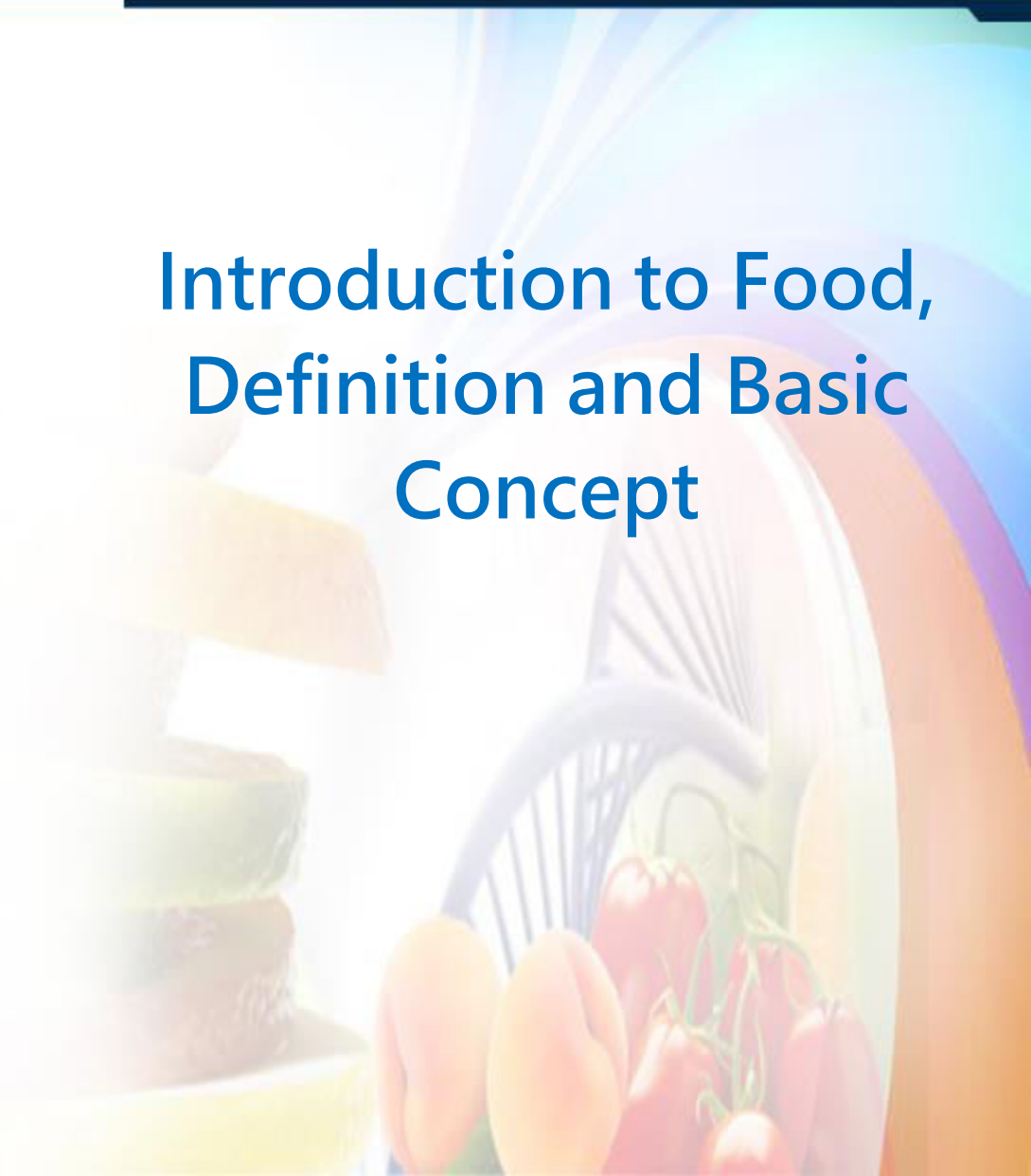
Food Biotechnology

Module #1



Food Biotechnology

Introduction to Food, Definition and Basic Concept



Introduction to Food, Definition and Basic Concept

Foods substances that are ingested by living organisms and can be metabolized to perform following functions:

1. source of energy;
2. source of raw materials for growth and development;
3. to supply minute amount of nutrients to regulate vital metabolic processes;

Introduction to Food, Definition and Basic Concept

4. to supply food components (phytochemicals) that

retard the development and progression of degenerative

diseases.



Introduction to Food, Definition and Basic Concept

The chemical components of food, which perform these functions are called *nutrients.*

Nutrition: The study of various nutrients, their functions, food sources and their utilisation by the human body and their effect on human wellbeing.

Introduction to Food, Definition and Basic Concept

Food Science The study of food science involves understanding the nature, composition and behaviour of food materials under varying conditions of growth, harvest, transportation, processing, storage, handling and use.



Introduction to Food, Definition and Basic Concept

Food Technology: is the application of food science & engineering principles to the selection, preservation, processing, packaging, distribution, and use of safe, nutritious, and wholesome food.



Introduction to Food, Definition and Basic Concept

Food Security:

Food security is achieved when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life. (800 million people suffer from hunger).

Introduction to Food, Definition and Basic Concept

Food Industry:

Food is not only our most important need, but the food industry is the largest of all industries, employing tens of millions of people in all areas such as growing, fishing, processing, transportation, storage, preparation, inspection, distribution, sales, and marketing.

Introduction to Food, Definition and Basic Concept

Food Biotechnology:

Application of technology to modify genes of animals, plants, and microorganisms to create new species which have desired production, marketing, or nutrition related properties. Called genetically engineered (GE) or genetically modified (GM) foods.

Food Biotechnology

Module # 2



Food Biotechnology

Food Composition

Food composition

Food composition

The nutrients

found in foods include carbohydrates, proteins, fats, minerals, vitamins, water and fibre

Food composition

Food composition

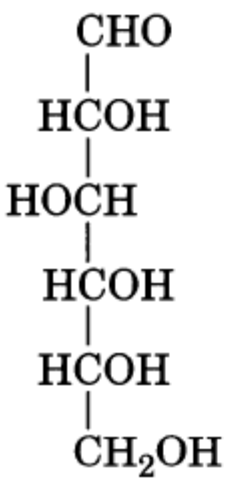
Carbohydrates: In our diet, about 65 to 80 per cent of the energy is supplied by carbohydrates.

Carbohydrates contain carbon, hydrogen and oxygen.

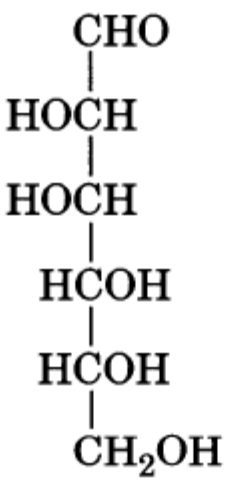
Monosacharides : are the the simplest units of the carbohydrates because these cannot be broken down any further by hydrolysis.

Food composition

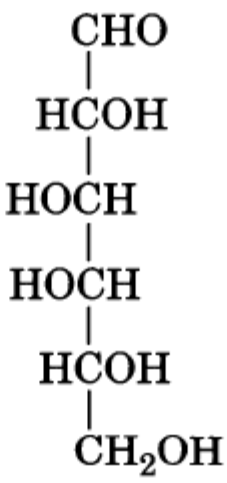
Food composition



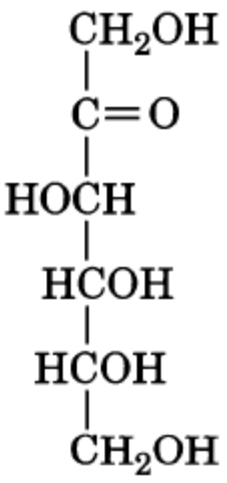
D–glucose



D–mannose



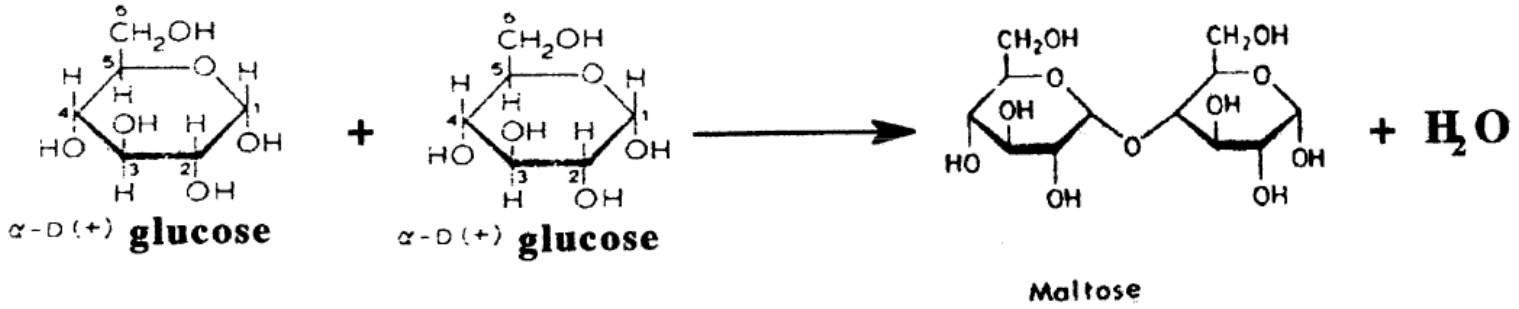
D–galactose



D–fructose

More complex sugars may be formed as two or three molecules of monosaccharides join together. When this happens, water molecules are formed:

glucose + glucose → maltose + water



Food composition

Food composition

Polysaccharides

Polysaccharides are polymers (compounds of many smaller molecules) of the ten or more simple monosaccharides joined together.

However most polysaccharides in nature consist of hundreds to several thousand of monosaccharide units..

Food Composition

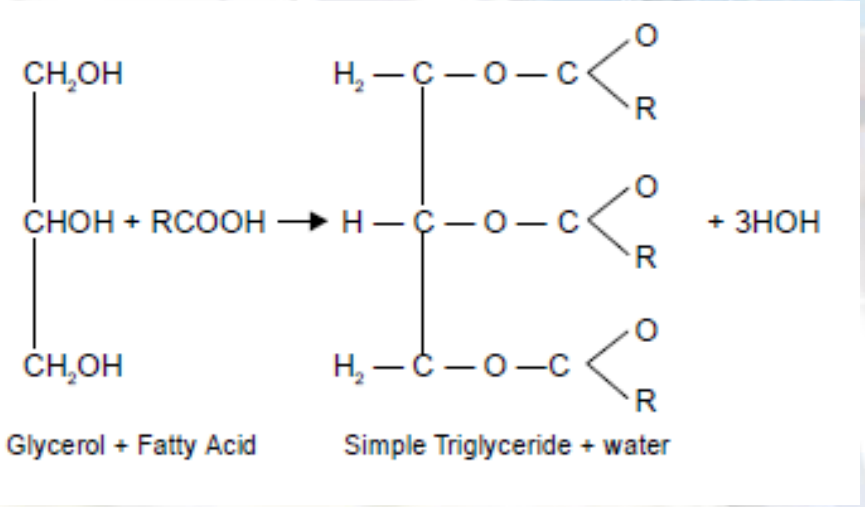
Fibers: Dietary fiber includes the nondigestible carbohydrates. These may be either water soluble or water-insoluble. The water-insoluble group, which includes wheat products and wheat bran. The water-soluble fibers, includes such as some cereal brans and pectin

Food Composition

Oils and fats belong to a naturally occurring substances called **lipids**. The common characteristics of lipids are:

1. They are soluble in organic solvents (ether, acetone etc.)
2. They are insoluble in water
3. Most of them are derivatives of fatty acids

Food Composition



Food Composition

Vegetable oils Groundnut, sesame, mustard, coconut, safflower, coconut, corn, cotton seed, soyabean and palm oil.

Animal fats Butter, ghee and cream from milk, lard from hogs and tallow from cattle.

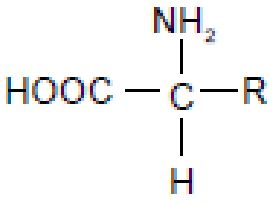
Manufactured fats *Vanaspati* and margarine.

Food composition

Proteins: Proteins are present in all living plant and animal tissues.

Composition Proteins, like fats and carbohydrates, contain carbon, hydrogen and oxygen. In addition they contain about sixteen per cent nitrogen, which distinguishes the proteins from carbohydrates and fats.

Some proteins contain sulphur, phosphorus, iron or other minerals also. Proteins are built from simpler compounds called *amino acids*. The amino acids contain a basic (amino — NH2) and an acidic (carboxyl — COOH) group in their molecules.



Food Composition

Water: Cellular material contains an abundance of water as it essential for survival of organisms due to its critical role in maintenance of biological functions.

Food Composition

Minerals:

Human beings require 14 different elements for health and growth. These include calcium, phosphorus, sodium, chlorine, potassium, magnesium and sulphur, which are present in appreciable amounts. the other iron, iodine, manganese, copper, zinc, cobalt and fluorine are found in traces.

Food Composition

Vitamins: are essential for the maintenance of good health as they catalyse various body processes, which help to utilize all the nutrients supplied and regulate a number of functions in the body. Vitamins are either fat-soluble and water-soluble. Fat-soluble vitamins include A, D, E and K, and water-soluble vitamins include the B-group and vitamin C.

Food Biotechnology

Module # 3



Food Biotechnology

Water Contents in Food

Water Contents in Food

Food composition

Moisture content is quantity of water present in various food

products and ingredients

Water Contents in Food

It exists in three forms in foods

1. Free Water
2. Chemically adsorbed water
3. Chemically bound water

Water Contents in Food

Composition

Importance of Water in Foods

1. Water as Quality Factor in Preservation :
such as in products like dried milk, dehydrated vegetables, spices and herbs.

2. Quality factor of products
Such as jams, jellies, fruit syrups, prepared cereals etc.

Food composition

3. Convenience in Transportation :
Reduced moisture food is helpful in packaging and shipping such as concentrated milk, liquid cane sugar, concentrated fruits juices.

4. Moisture as compositional standards
Such as in foods cheese, flour, pineapple, meat etc.

Food composition

5. Moisture as nutritional Standards

6. Moisture data as expressed in analytical results of many foods such as food contents on wet basis and dry basis

Water Contents in Food

Water Activity is a concept different than water contents and it has important implications such as non-enzymic browning, lipid oxidization, degradation of vitamins, enzymatic hydrolysis, protein denaturation, and starch gelatinization and retrogradation. Shelf life, Microbial growth

Food Biotechnology

Module # 4



Food Biotechnology

Carbohydrates

Food Composition

Carbohydrates

Monosaccharides:

are simple sugars ofter with suffix “ose” such as ketoses or aldoses.

Food Composition

Carbohydrates

Oligosaccharides:

Consistes of 2-10 units of sugars, mostly disaccharidese are very important such as

1. Sucrose,
2. Fructoses
3. lactoses.
4. Maltoses

Food Composition

Carbohydrates

Polisaccharides:

Polysaccharides are built of repeat units of monosaccharides

systematically named with the suffix *-an*. The generic name for polysaccharides is *glycan* and these can be *homoglycans* consisting of the one type of monosaccharide or *heteroglycans* consisting of

two or more types of monosaccharide.

Food Composition

Carbohydrates

Starch:

Starch occurs in the form of semi-crystalline granules

ranging in size from 2 to 100 µm, and consists

of two types of glucan: *amylose* and *amylopectin*.

Food Composition

Carbohydrates

Glycogen:

The polysaccharide that animals use for the shortterm storage of food energy in the liver and muscles is known as *glycogen*.Branching aids the rapid release of glucose since the enzymes that release glucose attack on the non-reducing ends, cleaving one glucose molecule at a time.

Food Composition

Carbohydrates

Cellulose:

The most abundant molecule in nature. Makes up dietary fibre.

Food Composition

Carbohydrates

Pectin:

Form gels in food products such as jams and preserves.

Carbohydrates

Gums:

High affinity with water and instead of making gels they form high viscosity liquids.

Food Biotechnology

Module # 4



Food Biotechnology

Carbohydrates

Food Composition

Carbohydrates

Monosaccharides:

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Food Composition

Carbohydrates

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Food Composition

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Food Composition

Carbohydrates

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Food Composition

Carbohydrates

Pectin:

Form gels in food products such as jams and preserves.

Carbohydrates

Gums:

High affinity with water and instead of making gels they form high viscosity liquids.

Food Biotechnology

Module # 6



Food Biotechnology

Fat contents in foods

Fat contents in foods

Lipids provide structure of cells and are source of energy, They are also carrier of fat soluble vitamins.

They are either fats or oils.

They are all fatty acid esters and may exist in form of triglycerides, phospholipids, sterols and tocopherols.

Food composition

Fat contents in foods

They are classified as

1. Simple lipids
2. Complex lipids
3. Derived lipids

Food composition

Fat contents in foods

Fatty acids:

Saturate and non saturated
Cis and trans

Food composition

Fat contents in foods

Systematic name	Trivial name	Lipid number
Butanoic acid	Butyric acid	4:0
Hexanoic acid	Caproic acid	6:0
Octanoic acid	Caprylic acid	8:0
Decanoic acid	Capric acid	10:0
Dodecanoic acid	Lauric acid	12:0
Tetradecanoic acid	Myristic acid	14:0
Hexadecanoic acid	Palmitic acid	16:0
cis-9-Hexadecenoic acid	Palmitoleic acid	16:1, *cis*-Δ^{9}
Octadecanoic acid	Stearic acid	18:0
cis-9-Octadecenoic acid	Oleic acid	18:1, *cis*-Δ^{9}
cis, cis-9,12-Octadecadienoic acid	Linoleic acid	18:2, *cis,cis*-$\Delta^{9,12}$
all-*cis*-9,12,15-Octadecatrienoic acid	Linolenic acid	18:3, *cis,cis,cis*-$\Delta^{9,12,15}$
Eicosanoic acid	Arachidic acid	20:0
all-*cis*-5,8,11,14-Eicosatetraenoic acid	Arachidonic acid	20:4, *cis,cis,cis,cis*-$\Delta^{5,8,11,14}$
cis-13-Docosenoic acid	Erucic acid	22:1, *cis*-Δ^{13}

Food composition

Fat contents in foods

Structure of Triglycerides:

MUFA

PUFA

Polymorphism

α, β, and β'.

Food composition

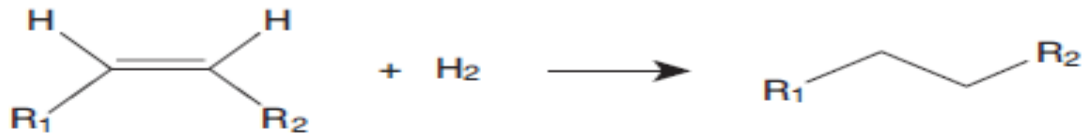
Processing of Fats and oils:

1. Hydrogenation
oils are exposed to hydrogen gas under conditions of high temperature (150–180◦C) and pressure (2–10 atm) in the presence of a nickel catalyst.

Food composition

Fat contents in foods

Hydrogenation (main reaction):

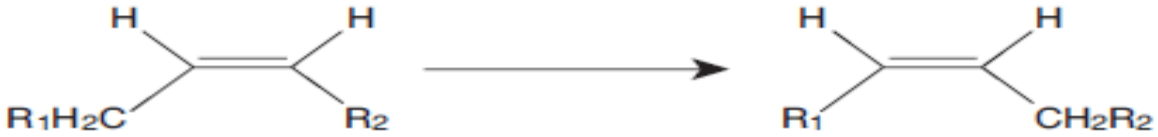


Isomerization (side reaction 1):



cis *trans*

Double bond migration (side reaction 2):



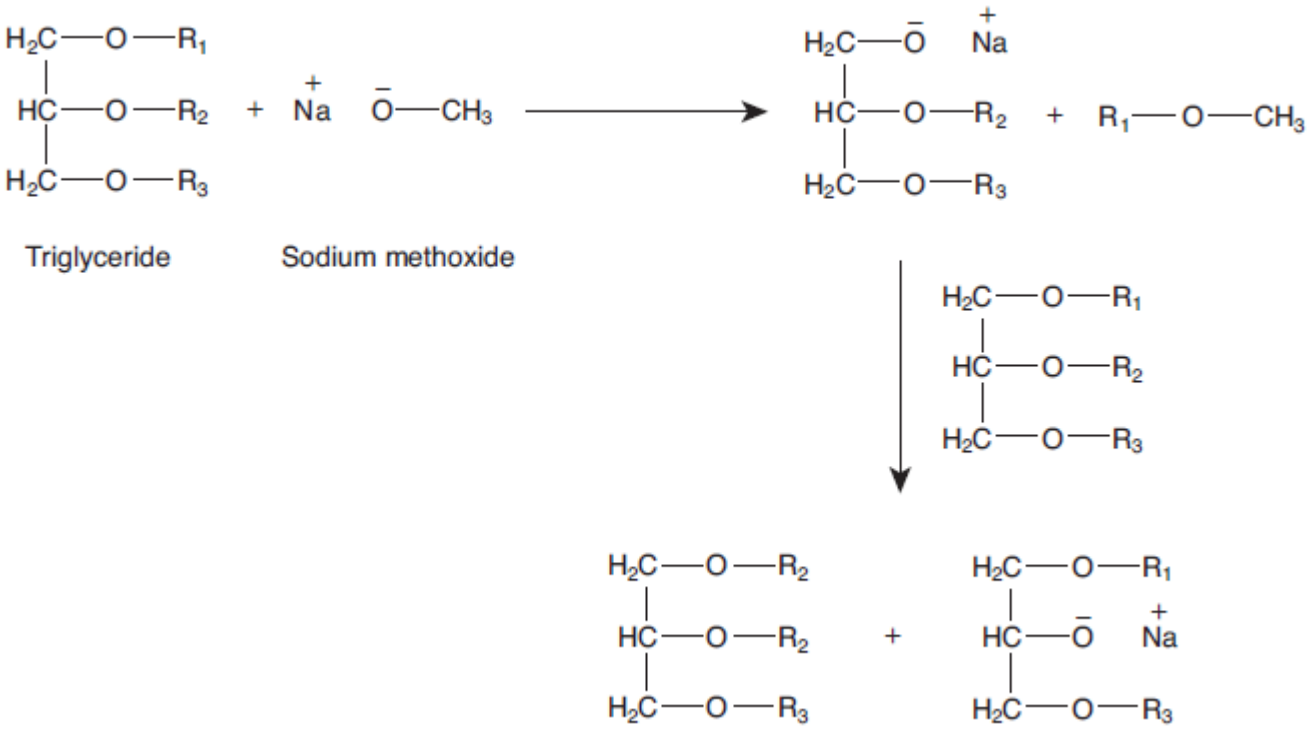
Fat contents in foods

Interesterification:

achieved through heating fats at high temperature (<200◦C) for long periods, but it is more efficient to employ catalysts that speed up the process (30 min) and lower the temperature required (50◦C). Sodium methoxide is the most popular catalyst

Food composition

Fat contents in foods



Food Biotechnology

Module # 7



Food Biotechnology

Minerals

d composition Minerals

After remove of all water organic matter from the food products what remains is the inorganic material or ash. The human body consists of about 3% minerals, most of which is in the skeletal system.

Minerals are generally categorized as "major“ and "trace" based on the amounts in the body.

Food Composition

Minerals

1. **Iron:**

is a trace element, Iron is an essential part of both the blood pigment hemoglobin and the muscle pigment myoglobin. It is included in some enzymes. Deficiencies

of iron cause anemia.

iron phosphates and iron phytates found in

Plants are poorly absorbed.

Toxicity is also possible.

Food Composition

Minerals

Calcium:

It is required for bone and tooth structure and is necessary for the function of nerves and musc1es. Calcium is also needed in the blood clotting mechanism. Deficiencies can lead to osteoporosis. Vitamin D and Lactose help absorption.

Food Composition

Minerals

Sodium:

In humans, sodium is required in the extracellular fluids to maintain osmotic equilibrium and body-fluid volume. Sodium has been shown to aggravate hypertension (high blood pressure) and it is recommended that daily consumption levels be kept between 1100 and 3300 mg.

Food Composition

Minerals

Chlorine

Chlorine is also involved in extracellular fluids as the major negatively charged ion.

It has a role in controlling blood pH and is necessary in the production of hydrochloric acid of gastric juice. If great losses of body fluids occur, as in vigorous exercise, the amount of chlorine may have to be replenished.

F coMinpositionMineralist

Potassium

is associated with the function of musc1es and nerves and with the metabolism of carbohydrates.

It is important in maintaining the fluid volume inside cells, and the acid-base balance. Good sources of potassium are meats, eggs, oranges, bananas, and fresh milk. Cell membranes are quite permeable to potassium, but as it leaks out, a highly active membrane pump returns it to the cell in exchange for sodium. If 6% of the potassium contained in the cells were to escape into the blood,
the heart would stop.

Fod composition Minerals

Phosphorus

About 85% is found combined with calcium. It is also part of the body's major buffers (phosphoric acid and its salts) and it is part of both DNA and RNA. It is part of some lipids .

It plays a key role in energy transfer as it is part ofthe energy carrier of the cells, ATP. Some sources of phosphorus are meats, fish, eggs, and nuts..

Food Composition

Minerals

Magnesium:

Magnesium is a minor component of bones and is present in soft tissue cells, where it is involved in protein synthesis.

Sulfur:

Sulfur is present in virtually all proteins Deficiencies of sulfur are associated with protein deficiency and if foods containing sufficient protein are eaten then sulfur amounts will be adequate.

Food Composition

Minerals

Trace elements:
Iodine , Flourine
Copper, Cobalt
Zinc, Mangnese
Selenium, Vanadium
Silicon, Tin,
Chromium
Aluminium boron cadmium.

Food Biotechnology

Module # 8



Food Biotechnology

Vitamins

Food Composition

Vitamins

Fat Soluble vitamins:

Vitamin A:

Carotene conents in plant foods converts to Vit A. Important for vision and resistance to infection. Bone and teeth formation.

1 IU= 0.3 RE

1 RE= 6 microgram b-carotene

Food Composition

Vitamins

Vitamin D:

Vitamin D (calciferol or activated ergosterol) is necessary for normal tooth and bone formation. It is now expressed in micrograms (Ilg) as cholecalciferol and 10 Ilg = 400 IU. Vitamin D, as vitamin A is also routinely added to milk or vegetable oil.

Food Composition

Vitamins

VitaminE

Vitamin E has four different forms (the tocopherols) that have the same name except with the prefixes alpha-, beta-, gamma-, and delta-. It serves as an antioxidant that serves to prevent the oxidation of body components.

Good sources of Vitamin E are corn oil, cottonseed oil, and peanut oil. Deficiency effects reproduction, injury to the central nervous system, growth retardation, muscular dystrophy, and interference with normal heart action. The RDA for vitamin E is now given in milligrams of alpha-tocopherol equivalents (a-TE).

Vitamins

VitaminK

Vitamin K is essential for clotting of blood. Cabbage, spinach, cauliflower, and liver are especially good sources and it is also synthesized by bacteria in the human intestine. Antibiotic therapy that destroys intestinal organisms can produce deficiencies of vitamin K.

Food Composition

Vitamins

Vitamin BI (Thiamin)

It is also necessary for nerve function, appetite, and normal digestion as well as growth, fertility, and lactation. The symptoms ofthiamin deficiency are retardation
of growth, palpitation and enlargement of the heart, hypertension, and beri-beri.

Food Composition

Vitamins

Vitamin B2 (Riboflavin)

Vitamin B2 makes up apart of enzyme systems involved in the oxidation and reduction of different materials in the body. Deficiency of riboflavin generally results in growth retardation and may result in vision impairment, scaling of the skin, and

lesions on mucous tissues. Neuritis is another effect of deficiency.

Food Composition

Vitamins

Niacin (Nicotinic Acid)

Niacin is part of enzyme systems regulating reduction reactions in the body. It is also a compound that dilates blood vessels. It is part of the coenzyme nicotinamide adenine dinucleotide (NAD) which is involved in glucose breakdown. Deficiency of niacin causes pellagra (a disease that causes diarrhea, dermatitis, nervous disorders, and sometimes death). Good sources are yeast, meat, fish, poultry, peanuts, legumes, and whole grain.

Vitamins

Vitamin B6 *(Pyridoxine)*

Other substances closely related to this vitamin are pyridoxal and pyridoxamine.

It removes CO2 from the acid group (COOH) of certain amino acids and transfers amine groups (NH2) from one compound to another in the body.

Food Composition

Vitamins

Biotin:

Biotin is reported to be a coenzyme in the synthesis of aspartic acid, involves fixation of carbon dioxide. Deficiency of this compound is unusual, although it can be tied up by a substance in raw egg whites called aviden and the deficiency has been demonstrated in feeding studies with mice. Because ofthe production ofbiotin by microbial flora of the intestines, the dietary requirement for this compound is unknown. Liver is an excellent source of biotin and peanuts, peas, beans, and whole cooked eggs are good sources.

Food Composition

Vitamins

Pantothenic Acid

Pantothenic acid, a vitamin required for normal growth, nerve development, and normal skin. Organs of animals (liver, heart, kidney), eggs, whole wheat products, and peanuts are excellent sources.

Food Composition

Vitamins

Choline

It is a component of ceIl membranes and brain tissue and it functions as part ofthe substance acetylcholine, which is one ofthe brain's principal neurotransmitters.

***Vitamin** B 12*
(Cyanocobalamin)
It is required for the normal development of red blood ceIls, and a deficiency in vitamin B12 causes acute pernicious anemia and a variety of other disorders.

Food Composition

Vitamins

Vitamin C:

Ascorbic acid is required for the formation of intercellular substances in the body including dentine, cartilage, and the protein network of bone. Hence, it is important in tooth formation, the healing ofbroken bones, and the healing ofwounds.

Food Biotechnology

Module # 9



Food Biotechnology

Fibers

Food composition

Fiber

Dietary fiber includes the nondigestible carbohydrates. It is either

1. Water soluble
2. Water insoluble

Food composition

Fiber

Cellulose:
main component of plants and industrially important material such as wood , paper and cotton.

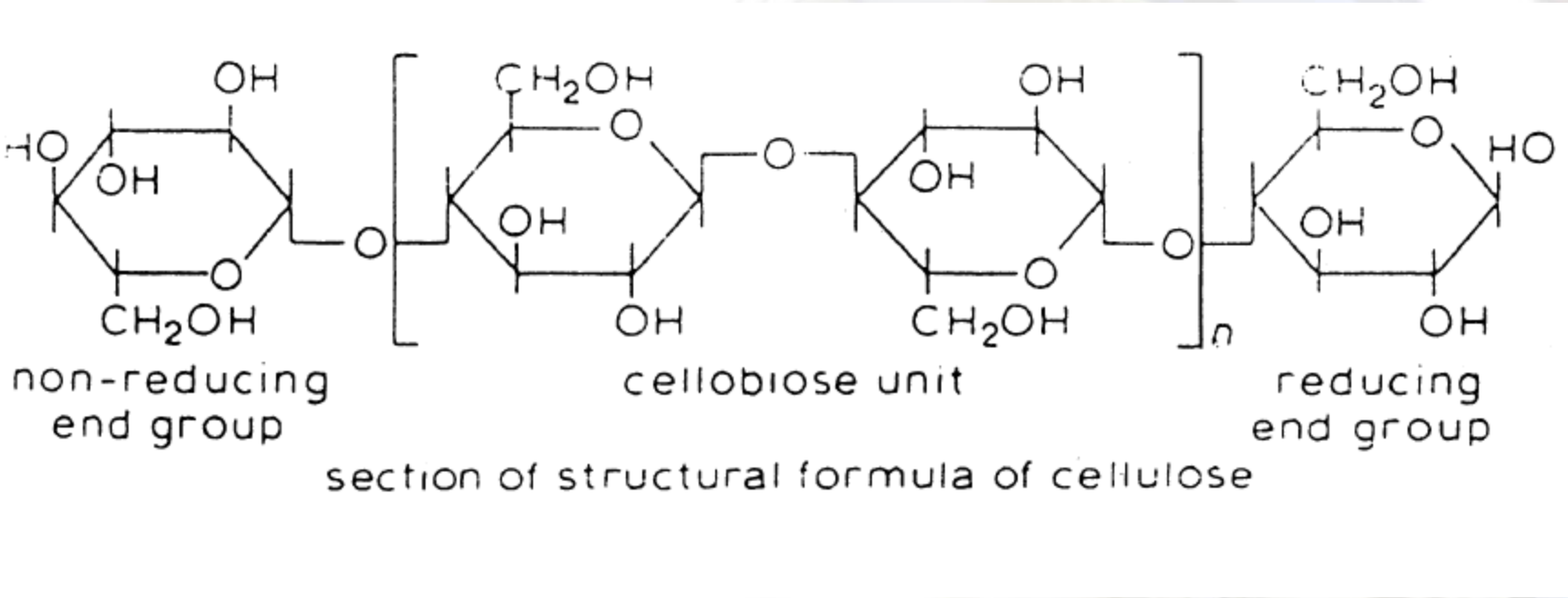
It is water insoluble and gives mechanical strength. The molecule is elongated and rigid. It needs enzyme cellulases to digest cellulose.

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Food composition

Fiber

Chemically it is linear polymer of D-glucose with beta 1,4 bonds



Food composition

Fiber

Treating cellulose with dilute acids yields microcrystalline celluloses for food applications.

Food composition

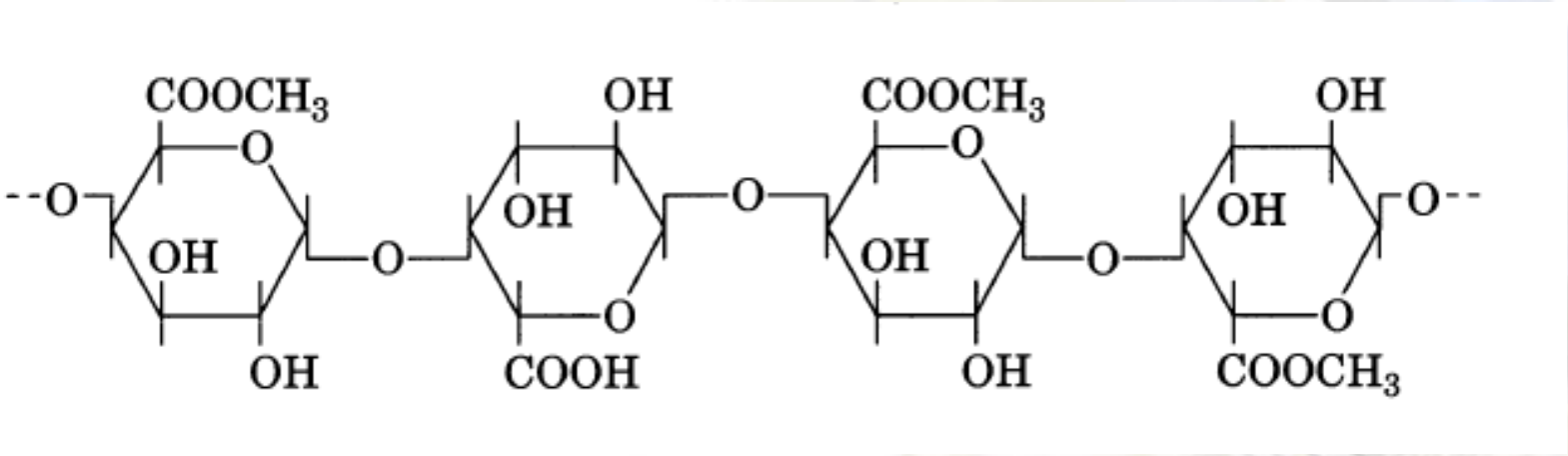
Fiber

Pectin:
Is water soluble fiber existing in intracellular spaces of plant tissues.
Structurally they are heteropolysaccharides

Food composition

Fiber

The unbranched chains consist of polygalacturonic acid molecules joined with a 1,4 linkage. The galacturonic acid units may be esterified (an acid is combined with an alcohol with methyl alcohol:



Food composition

Fiber

Slow set pectin have 50 to 70 % esterification and it sets slowly with addition of more sugar and acids.

Rapid set pectin has above 70% esterification and is used in products such as jams and preserves.

Low methoxy pectin is used as low calorie products because it needs less sugar and divalent ions such as calcium.

Food composition

Fiber

Gums are water soluble fibers. They are hydrophilic compounds that act as gelling , stabilizing and suspending agents with less concentration.
They can be
Exudate gums
Bean gums
Seed gums
Seaweed gums

They are also heteropolysaccharides with multiple applicatoins in variety of food products

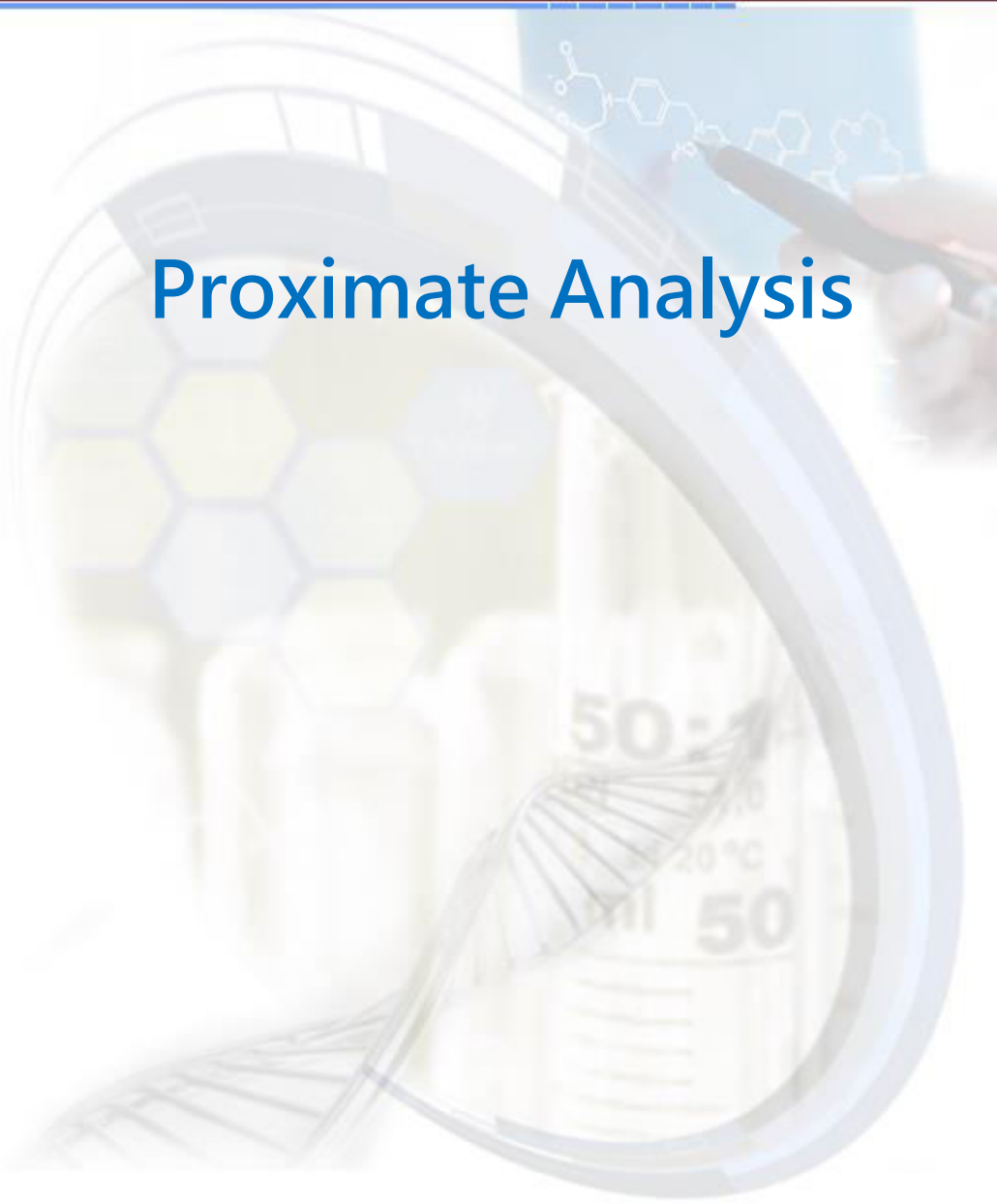
Food Biotechnology

Module # 10



Food Biotechnology

Proximate Analysis



Food Composition

Proximate Analysis

Food Matrix is complex and consists of carbohydrates, proteins, fats, minerals, vitamins, water and fibre.

Broader classes of these closely related groups are termed as proximate constituents. Proximate analysis puts together all these closely related groups together.

Proximate Analysis
osition

Crude Fiber:

Refers to organic residues of vegetable origin consisting of cellulose, lignins and pentosans.

Carbohydrates and proteins are removed by boilling acid and alkali treatment.

Quality factor in many foods and it is inversely proportional to food value of products.

Food composition

Proximate Analysis

Moisture contents:
60-95 % water contents in foods.
Fruits and vegetalbes and pulses all have varying degree of water.
Its measure of stability and quality.

It exists in various forms such as free, bound and hence removal of water needs varying treatment to measure moisture contents.
there are many variables to consider for accurate measurement during drying process.

Food composition

Proximate Analysis

Crude Fat:

Also called ether extract, it includes phospholipids, lecithing, sterol, fatty acids, waxes, carotenoids, chlorophyl and other pigments.

Soxlet method of extraction for fats.

Food composition

Proximate Analysis

Ash contents:

Inorganic material after burning organic matter in food. Original form of minerals may change to carbonates or oxide, or phosphates, chloride, sulphate form. Change in chemical nature may lead to certain volatiles such as sulphides and halides.

Procedures must be followed thoroughly. It is important for identification, quality, adulteration etc.

Food composition

Proximate Analysis

Crude Proteins:

Nitrogen is a central element of proteins but it is also found in other compounds such as nucleic acids, alkaloids, nitrogen lipids and vitamins etc.

Kjeldahl Method to determine proteins involves these steps

1. Sample digestion or wet oxidation
2. Distillation of ammonia
3. Titration of liberated ammonia

Food Biotechnology

Module # 11



Food Biotechnology

Estimation of moisture

Estimation of Moisture

There are different methods for moisture analysis depending upon type of food and requirement of analysis.

1. Drying methods
2. Distillation methods

Food composition

Estimation of Moisture

Oven methods:

Many different types of oven can be used.

Basic principle involves drying of food and calculating the weight loss referred to as moisture contents.

Need to consider type of foods, such as oily foods, spices or sugars they all need different treatments.

Estimation of Moisture

Air Oven Method:

1. Dry the flat bottom dish with tightly fitted lid at 100ºCwith heat first and weigh and tare the balance.
2. Add 2-10 g of food, loose the cover and heat for 2-3 hrs at 100ºC.
3. Remove and cool in dessicator, repeat the process till 2 mg different persists.
4. Calculate moisture loss

Food composition

Estimation of Moisture

$$\% \text{ Moisture (wt/wt)} = \frac{\text{wt } H_2O \text{ in sample}}{\text{wt of wet sample}} \times 100$$

$$\% \text{ Moisture (wt/wt)} = \frac{\text{wt of wet sample} - \text{wt of dry sample}}{\text{wt of wet sample}} \times 100$$

Food composition

Estimation of Moisture

Vacuum Oven:
Suitable for spices, meats or sugar samples.
Pressure of 100 mg or less
4-6 hrs heating and 60-70ºC.
Remove dish and cool in dessicator.
Again heat for 1 hour and cool in dessicator till 2-3mg difference exists.
Calculate moisture loss.

Food composition

Estimation of Moisture

Infra red drying:

Infra red lamps with a filament temperature of 2000-2500 K can penetrate heat to food and shorten drying time to 20-25 minutes only.

Food composition

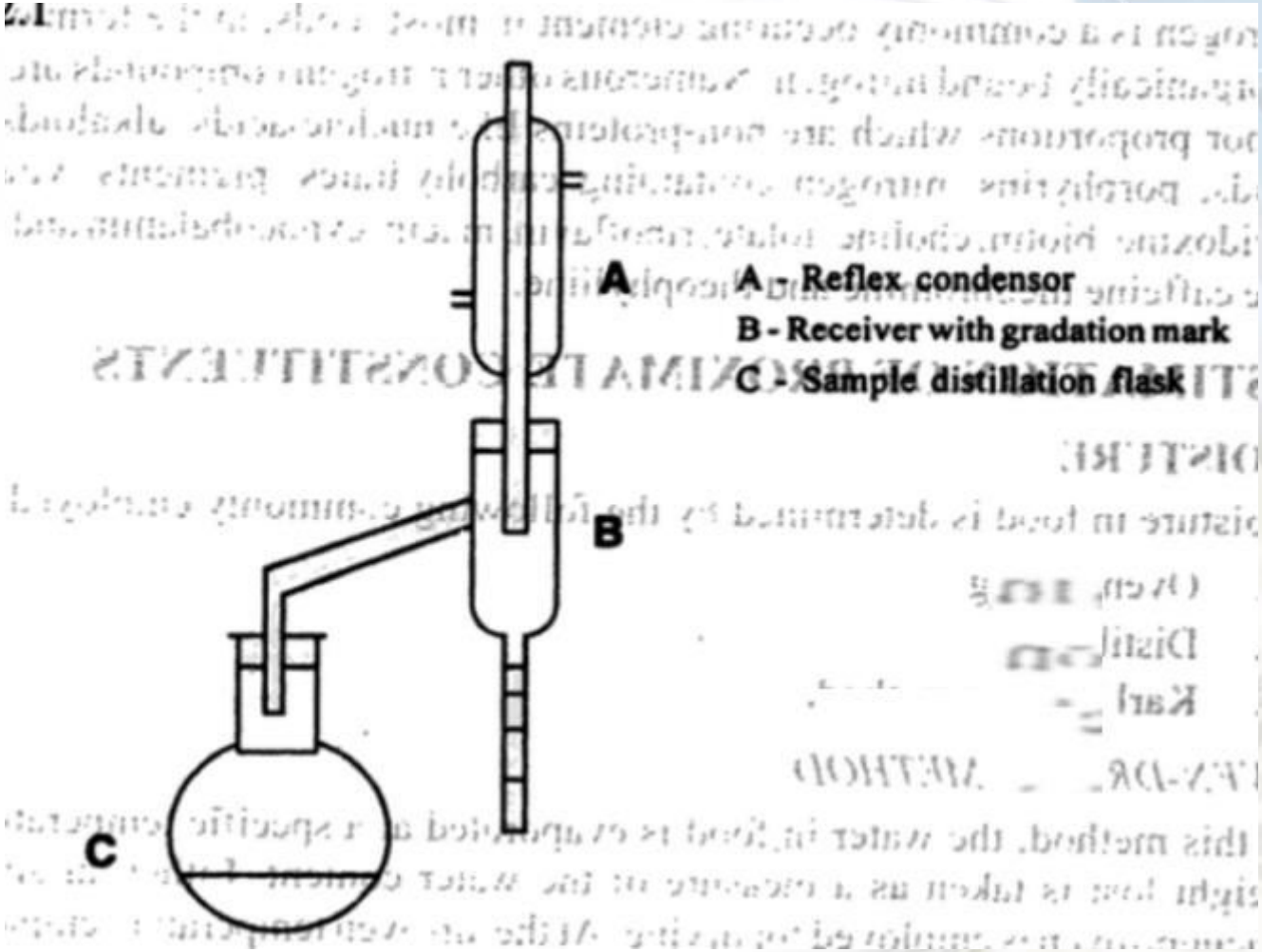
Estimation of Moisture

Distillation Method:

Food sample is heated with a solvent with higher boiling point and lower specific gravity than water. Solvent include xylene, toluene and carbon tetrachloride etc.
Water is received in a graduate cylinder and is measure directly from the graduated cylinder.

Food composition

Estimation of Moisture



Estimation of Moisture Fraction

Method:

1. Add 10-40 g sample into flask and 75-100 ml solvent to cover food product.
2. Distill @ 1-2 drop per minute unless all clear water is achieved.
3. Wash the condenser and continue condensing for 5 minutes.
4. Cool and read the volume of water.

Food Biotechnology

Module # 12



Food Biotechnology

Estimation of Ash in Foods

Food composition

Ash content

Total Ash:

Food sample is ignited in specific conditions and residue is weighed as total ash.

1. In silica dish weigh 2-3 g of food, pre dry if necessary.
2. Ignite over burner for 4-6 hrs /or use muffle furnace at 500ºC overnight ill white greyish ash achieved.

Food Composition

Ash content

3. Cool the dish in a dessicator and weigh the dish

4. Calculate the residue as total ash.

Food Composition

Ash content

$$\% \text{ ash (dry basis)} = \frac{\text{wt after ashing} - \text{tare wt of crucible}}{\text{original sample wt} \times \text{dry matter coefficient}} \times 100$$

Food composition

Ash content

Wet Ashing:

We use oxidizing agents or acids or their combinations to solubilize minerals without volatolization.

1. Weigh 1 g previously dried sample in Erlenmeyer flask (previously dried and acid washed).
2. Add 3 ml H_2SO_4 and 5 ml HNO_3. (with Blank)

Food composition

Ash content

3. Heat to 200ºC till fume cease.

4. Add HNO3 and then boil and cool at room temperature

5 . Add ultrapure water in flastk and make up volume for further mineral analysis.

Food composition

Ash content

Water Soluble Ash:

Ash soluble in water gives information about fruit contents in jams and preserves.

Solution of water is heated and filtered to know soluble and insoluble fractions.

F Composition

Acid Soluble Ash:

Acid insoluble ash is measure of added minerals to foods such as dirt, sand or talc.

Food Biotechnology

Module #13



Estimation of Carbohydrates

Sample Preparation

It depends on the nature of the food being analyzed.

Aqueous solutions require very little preparation prior to analysis.

nuts, cereals, fruit, breads and vegetables.

Estimation of Carbohydrates

Sample Preparation

In these foods it is usually necessary to isolate the carbohydrate from the rest of the food before it can be analyzed. by solvent extraction.

Estimation of Carbohydrates

Isolation method depends on type of carbohydrate food matrix the purpose of analysis, however, there are some procedures that are common to many isolation techniques. For example, foods are usually dried under vacuum (to prevent thermal degradation), ground to a fine powder (to enhance solvent extraction) and then defatted

Estimation of Carbohydrates

low molecular weight carbohydrates are extracted from foods by boiling a defatted sample with an 80% alcohol solution.
Monosaccharides and oligosaccharides are soluble in alcoholic solutions.

Estimation of Carbohydrates

The soluble components can be separated from the insoluble components by filtering. These two fractions can then be dried and weighed to determine their concentrations.

Estimation of Carbohydrates

Clarification:

Done to remove interfering substances for spectroscopic analysis or endpoint determinations. Heavy metal salts (such as lead acetate) which form insoluble complexes with interfering substances that can be removed by filtration or centrifugation.

Estimation of Carbohydrates

Ion-exchange.

Many monosaccharides and oligosaccharides are polar non-charged molecules and can therefore be separated from charged molecules by passing samples through ion-exchange columns.

Estimation of Carbohydrates

Ion-exchange.

By using a combination of a positively and a negatively charged column it is possible to remove most charged contaminants.

Estimation of Carbohydrates

Chromatographic and Electrophoretic methods

TLC, GC and HPLC are commonly used to separate and identify carbohydrates. Carbohydrates are separated on the basis of their differential adsorption on the basis of their partition coefficients, polarities or sizes, depending on the type of column used.

Estimation of Carbohydrates

HPLC is capable of rapid, specific, sensitive and precise measurements. In addition, GC requires that the samples be volatile, which usually requires that they be derivitized, whereas in HPLC samples can often be analyzed directly.

Estimation of Carbohydrates

HPLC and GC are commonly used in conjunction with NMR or mass spectrometry so that the chemical structure of the molecules that make up the peaks can also be identified.

Estimation of Carbohydrates

Carbohydrates can also be separated by electrophoresis after they have been derivitized to make them electrically charged, *e.g.,* by reaction with borates. A solution of the derivitized carbohydrates is applied to a gel and then a voltage is applied across it.

Estimation of Carbohydrates

The carbohydrates are then separated on the basis of their size: the smaller the size of a carbohydrate molecule, the faster it moves in an electrical field.

Estimation of Carbohydrates

Chemical methods

reducing nature of mono and oligosaccharides reacts with other components to yield precipitates or colored complexes which can be quantified. The concentration of carbohydrate can be determined gravimetrically, spectrophotometrically or by titration.

Methods For Estimation of Carbohydrates

Food composition

Non-reducing carbohydrates are first hydrolyzed to make them reducing. It is possible to determine the concentration of both non-reducing and reducing sugars by carrying out an analysis for reducing sugars before and after hydrolyzation.

Methods For Estimation of Carbohydrates

Food composition

Many different chemical methods are available for quantifying carbohydrates. Most of these can be divided into three catagories: titration, gravimetric and colorimetric.

Estimation of Carbohydrates

Titration Methods

The Lane-Eynon method is an example of a tritration method of determining the concentration of reducing sugars in a sample. A known amount of boiling copper sulfate solution and a methylene blue indicator. The reducing sugars in the carbohydrate solution react with the copper sulfate present in the flask.

Estimation of Carbohydrates

Once all the copper sulfate in solution has reacted, any further addition of reducing sugars causes the indicator to change from blue to white. The volume of sugar solution required to reach the end point is recorded.

A calibration curve by carrying out the experiment with a series of standard solutions of known carbohydrate concentration.

Estimation of Carbohydrates

Colorimetric Methods

Sugars react with the anthrone reagent under acidic conditions to yield a blue-green color.

Enzymatic Methods

Analytical methods based on enzymes rely on their ability to catalyze specific reactions.

Estimation of Carbohydrates

These methods are rapid, highly specific and sensitive to low concentrations and are therefore ideal for determination of carbohydrates in foods

Food Biotechnology

Module #14



Estimation of Proteins

Protein analysis is important for:

1.Blological activity

Some proteins, including enzymes or enzyme inhibitors, are relevant to food science and nutrition: for instance. the proteolytic enzymes in the tenderization of meats, pectinases in the ripening of fruits, and trypsin inhibitors in legume seeds are proteins.

Estimation of Proteins

2. Functional property investigation.

Proteins in various types of food have unique food functional properties: for example, gliadin and glutenins in wheat flour for breadrnaking, casein in milk for coagulation into cheese products, and egg albumen for foaming.

3. Nutrition labelling.

Estimation of Proteins

Protein analysis is required when you want to know:

1. Total protein content
2. Amino add composition
3. Content of a particular protein in a mixture
4. Protein content during isolation and purification of a protein

Estimation of Proteins

Protein analysis is required when you want to know:

5.Nonprotein nitrogen

6.Nutritive value (digestibility, protein efficiency ratio, or nitrogen balance) of a protein

Estimation of Proteins

Kjeldahl method :

About 1000 years old method , Developed by Kjeldahl in 1883 .

Applied on variety of substances e.g. meat ,grains ,waste water soil etc. With the passage of time technique and apparatus modified

The basic principle is same still now

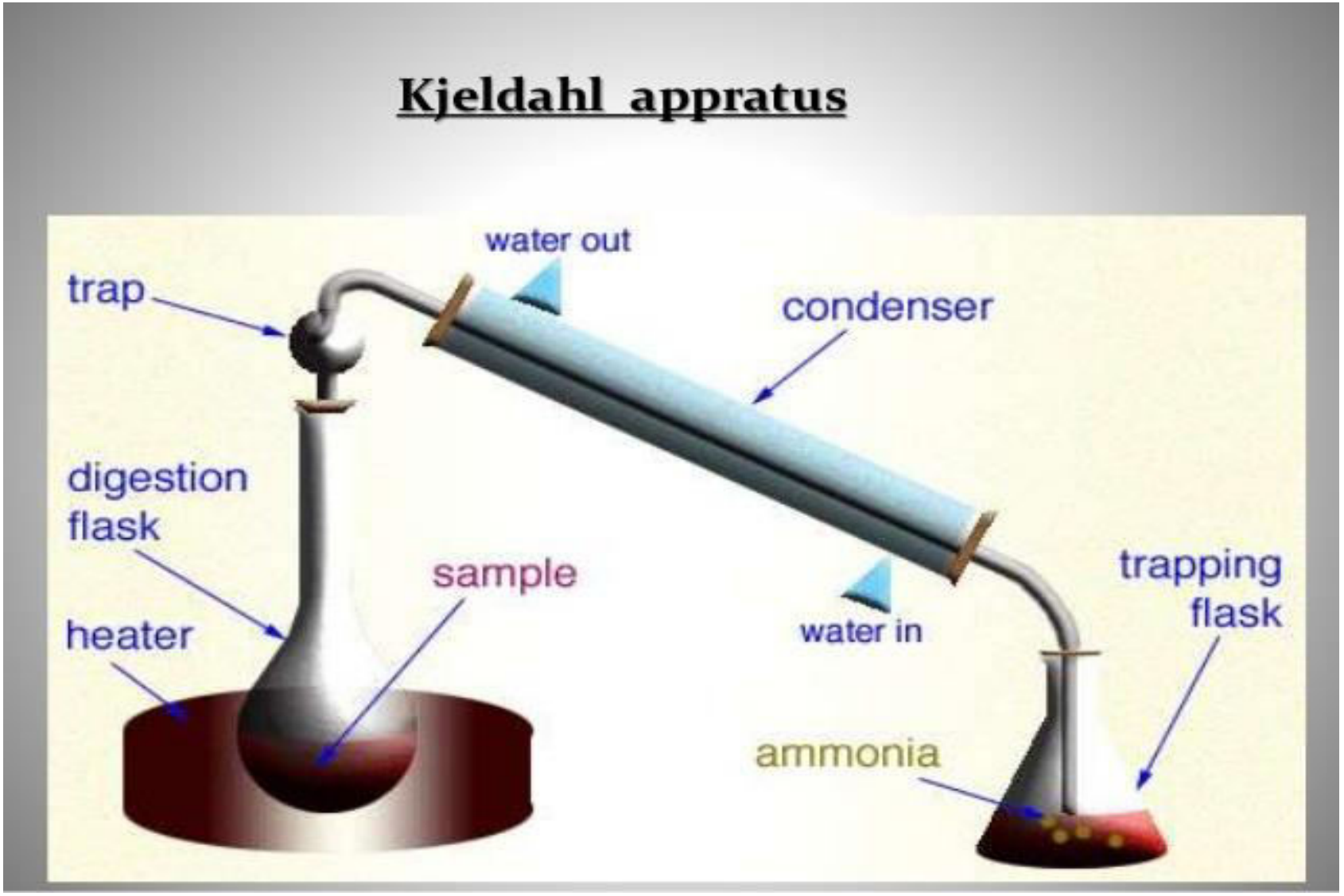
Estimation of Proteins

Estimation of nitrogen by digesting food in strong acid

Steps: The process is divided into 3 main steps.

1. Digestion
2. Distillation
3. Titration

Estimation of Proteins



Estimation of Proteins

Digestion

- Proteins are first digested
- Digestion is carried out with concentrated H_2SO_4
- In presence of an inorganic catalyst
- Digestion converts any nitrogen in food to ammonia and other organic matter to CO_2 and H_2O

Estimation of Proteins

Distillation

- Digestion flask is connected with receiving flask
- Ammonium is formed using distillation
- capture of ammonium occurs with a weak acid (boric acid)

Estimation of Proteins

Titration

- The third step is quantification of the ammonium done by titration
- Titration with a strong acid (sulfuric acid) & suitable indicator to determine end point of the reaction

Estimation of Proteins

Dumas method:

This is an automated instrumental technique

use to measure protein concentration of food

described by a scientist Dumas in 1831

Is older than the Kjeldahl

Estimation of Proteins

Sample of known mass is combusted at high temperature (900^0C) chamber in the presence of oxygen

This lead to the release of Carbon dioxide and water.

Estimation of Proteins

The method has three steps:

- Combustion
- Reduction
- Separation
- Detection

END

Food Biotechnology

Module #15



Estimation of Fat

Some of the most important properties of concern to the food analyst are:

Total lipid concentration

Type of lipids present

Physicochemical properties of lipids

Structural organization of lipids within a food

Estimation of Fat

Importance of lipid analysis

1. Nutritional importance (omega 3, saturated fat, cholesterol, fat sol vitamins).

2. Affect oxidative stability of foods (stability of many foods is affected by fatty acid composition or presence of enzymes that act on lipids such as lipoxygenase, lipase etc).

Estimation of Fat

3.Affect physical properties of foods (melting behavior in chocolate, margarine, etc.)

Estimation of Fat

Sample Selection and Preservations Depends on

Type of food – meat, milk, margarine.

The nature of the lipid component – volatility, physical state

Type of analytical procedure used – solvent extraction, instrumental technique

Estimation of Fat

Determination of Total Lipid Concentration

Reasons for analysis

- Economic
- Legal
- Health
- Quality
- Processing

Estimation of Fat

Solvent Extraction

Drying sample – solvent cannot easily penetrate foods containing water

Particle size reduction – finely ground

Acid hydrolysis – to release bound lipids into easily extractable forms

Solvent selection – choose the best solvent for the extraction

Estimation of Fat

Ideal Solvent For Fat Extraction

- High solvent power for lipids
- Low solvent power for nonlipids
- No residue
- Evaporate easily
- Low boiling point
- Non flammable / not explosive

Estimation of Fat

Ideal Solvent For Fat Extraction

- Nontoxic
- Low surface tension with food
- Single compound
- Non-hygroscopic

Estimation of Fat

Ethyl ether is used a lot but is

- Very flammable,
- Explosion hazard
- Forms peroxides
- Expensive.
- Petroleum ether is an excellent solvent for lipids
- More selective for more hydrophobic lipids
- Non hygroscopic
- Less flammable

Estimation of Fat

Batch Solvent Extraction

Mixing sample with organic solvent in separating funnel

Shake vigorously and allow the separation either by gravity or centrifugation.

Estimation of Fat

Batch Solvent Extraction

Aqueous phase is decanted off and the concentration of lipid in the solvent is determined by evaporating off the solvent and measure the mass lipid remaining

Have to be repeated a few times

Estimation of Fat

Semi-Continuous Solvent Extraction

Used to increase efficiency of lipid extraction from foods

Common method: Soxhlet extraction

Solvent extracts the lipids and carries them into the flask

The lipids still remain in the solvent due to low volatility

Estimation of Fat

Continuous Solvent Extraction

Commonly used is Goldfisch method

Similar to soxhlet method except the extraction chamber is designed.

Solvent trickles through the sample rather than building up around it

Estimation of Fat

Continuous Solvent Extraction

Disadvantage: channeling of the solvent can occur

i.e. solvent may take certain routes through the sample

Estimation of Fat

Accelerated Solvent Extraction

By increase the temperature and pressure normally used

The effectiveness of the lipid extraction increases as its temperature increases, but the pressure must also be increased to keep the solvent in the liquid state.

Advantage: reduce the amount of solvent

Estimation of Fat

Supercritical Fluid Extraction

Pressurized CO_2 is heated above a certain critical temperature to become supercritical fluid

This fluid behaves like a gas to easily penetrate into a sample and extract lipid while it also behaves like a liquid to dissolve a large quantity of lipids.

Estimation of Fat

Supercritical Fluid Extraction

The CO_2 extracts the lipid, and forms a separate solvent layer, which is separated from the aqueous components

Estimation of Fat

Non-solvent Liquid Extraction Methods

Babcock Method

A specified amount of milk is accurately pipetted into Babcock bottle. Sulfuric acid: breaks down the fat globule membrane than surrounds the droplet and thereby release the fat. Lipid is removed from the aqueous phase by centrifuging while it is hot (55-60°C).

Estimation of Fat

- Non-solvent Liquid Extraction Method
- Babcock method
- Surfactant Method
- Gerber Method

Food Biotechnology

Module #16



Estimation of Crude Fiber In Food

Definitions of Dietary Fiber

The term "dietary fiber" was originally synonymous With "roughage"—the nondigestible plant matter, principally of cell wall origin, that is responsible for fecal bulking.

Estimation of Crude Fiber In Food

Dimethyl Sulfoxide (DMSO)–Enzymatic–Chemical for NSP

The enzymatic–chemical methods were devised for specifically measuring NSP. They use enzymatic digestion to remove starch and protein.

Estimation of Crude Fiber In Food

Dimethyl Sulfoxide (DMSO)–Enzymatic–Chemical for NSP

And then measure remaining nondigestible polysaccharide specifically, as monosaccharide released by acid hydrolysis.

Estimation of Crude Fiber In Food

Enzymatic–Chemical/Gravimetric

The enzymatic–chemical/gravimetric method removes digestible food components enzymatically.

Estimation of Crude Fiber In Food

Enzymatic–Chemical/Gravimetric

And measures nondigested polysaccharides including undigested starch, chemically, but includes a lignin residue determined gravimetrically, with allowance for protein and ash content.

Estimation of Crude Fiber In Food

Detergent–gravimetric/Enzymatic–Gravimetric

This approach attempts to overcome the loss of soluble dietary fiber in the detergent extract of detergent methods. Duplicate samples are extracted, one with hot neutral detergent to measure detergent fiber, and the other with buffer and amyloglucosidase to measure soluble fiber.

Estimation of Crude Fiber In Food

Detergent–gravimetric/Enzymatic–Gravimetric

The detergent fiber and soluble fiber are added to obtain a measure of total fiber.

Estimation of Crude Fiber In Food

Enzymatic–Gravimetric

Method for measuring polysaccharides, lignin, and associated substances that are resistant to digestion by the alimentary enzymes of humans. Enzymatic digestion is used to remove most starch and protein.

Estimation of Crude Fiber In Food

Enzymatic–Gravimetric

Residual protein and ash content are deducted from the residue weight with the aim of obtaining a measure of NSP plus lignin by difference.

Estimation of Crude Fiber In Food

Nonenzymatic Gravimetric

The nonenzymatic gravimetric procedure was developed to give a rapid measurement of dietary fiber in samples, such as ripe fruit, that contained little starch and protein, and enough fiber relative to starch to enable starch digestion to be dispensed with.

Estimation of Crude Fiber In Food

Introduction

Some of the most important properties of concern to the food analyst are:

- Total lipid concentration
- Type of lipids present
- Physicochemical properties of lipids
- Structural organization of lipids within a food

Food Biotechnology

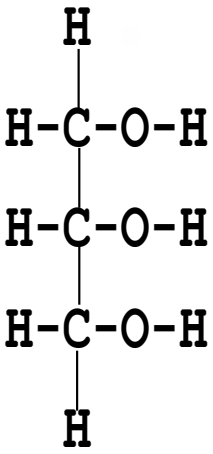
Module #17



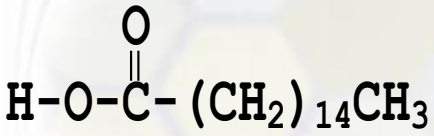
Fats and Oils

- Oils and fats – important for storing chemical energy in living things.
- Oils are liquids – fats are solids.
- Made from esters of propane-1,2,3,triol (glycerol) + long chain carboxylic acids RCOOH

Fats and Oils



Propane-1,2,3-triol

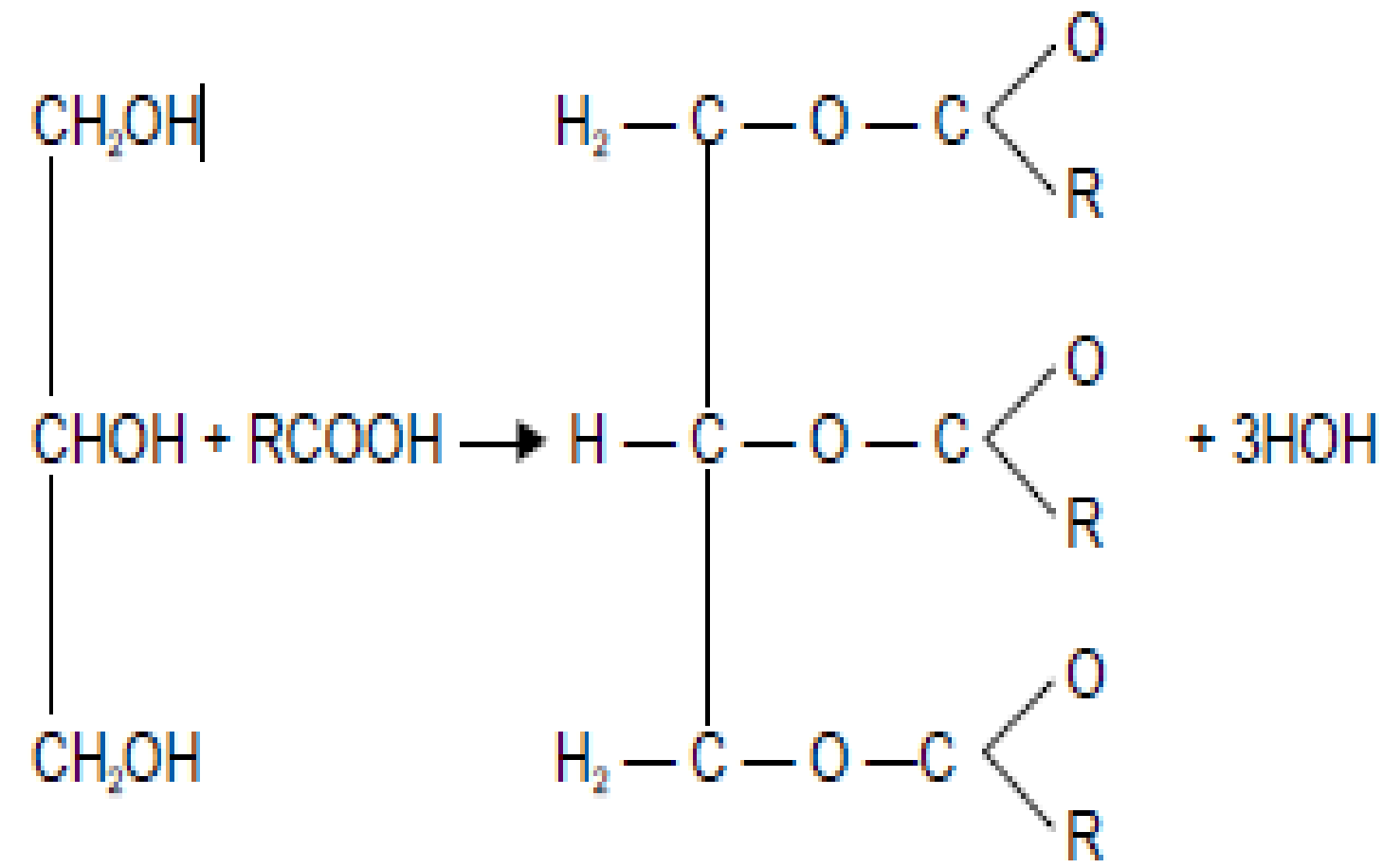


palmitic acid

Fats and Oils

S. no.	Food	Total fat (%)
1.	Ghee	99
2.	Butter	81
3.	Coconut	62
4.	Sunflower	52
5.	Gingelly	43
6.	Groundnut, Mustard	40
7.	Safflower	26
8.	Soyabean	20
9.	Fatty fish	5 to 21
10.	Egg	13
11.	Meat, Poultry	1 to 13
12.	Milk	1.5 to 7
13.	Cereals and Pulses	0.5 to 6
14.	Corn	4
15.	Vegetables	0.1 to 1

Fats and Oils



Fats and Oils

Composition and Classification

Oils and fats are composed of the elements carbon, hydrogen and oxygen. Fats are built up by linking together a number of individual fatty acids with glycerol.

Fats and Oils

Common Fatty Acids

Structure	Traditional name	Origin of name
$CH_3(CH_2)_{14}COOH$	palmitic acid	palm oil
$CH_3(CH_2)_{16}COOH$	stearic acid	suet (Greek: *stear*)
$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	oleic acid	olive oil
$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	linoleic acid	oil of flax (Latin: *linum*)

Fats and Oils

Selection and Buying of Oils and Fats

Criteria for selection of oil are

presence of the natural characteristic aroma, natural colour, clarity, freedom from admixture with other kinds of oils, freedom from solid particles and flat or rancid odour

Fats and Oils

Nutritive Value and Digestibility

•Fats and oils are concentrated sources of energy

- Each gram of pure fat or oil supplies nine kilocalories
- Fats impart special flavor and texture to our foods

Fats and Oils

Physical and Chemical Properties

- **Melting Point** All food fats are mixtures of triglycerides, and therefore, do not have a sharp melting point, but melt over a range of temperatures.

Fats and Oils

Physical and Chemical Properties

- **Creaming** of Fats Solid fats like butter and margarine can be creamed or made soft and fluffy by the incorporation of air

Fats and Oils

Plasticity of Fats:

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not flow at room temperature and are thus solid fats. The spreading quality of butter is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of one kind of a molecule.

Fats and Oils

Physical and Chemical Properties

- **Emulsification** The specific gravity of oils and fats is about 0.9, which indicates that they are lighter than water.

Fats and Oils

Physical and Chemical Properties

- **Emulsification**

They can form an emulsion with water when beaten up with it to form tiny globules in the presence of suitable emulsifying agent.

- E.g. butter

Fats and Oils

Fats are used in food preparation

(i) as a medium of cooking.

(ii) as a shortening biscuits, pastry and cakes.

(iii) to add richness and flavour

(iv) seasoning of vegetables and salads.

Fats and Oils

Effect of Prolonged Heating on Nutritive Value of Fats and Oils

- Thermal oxidation which is accelerated by the addition of soda and water. Consumption of thermally damaged fat is harmful.

Fats and Oils

Changes in Fats During Storage

•Fats and oils undergo certain undesirable changes during storage, which result in spoilage.

- Flavour Reversion
- Rancidity Spoilage

Food Biotechnology

Module #18



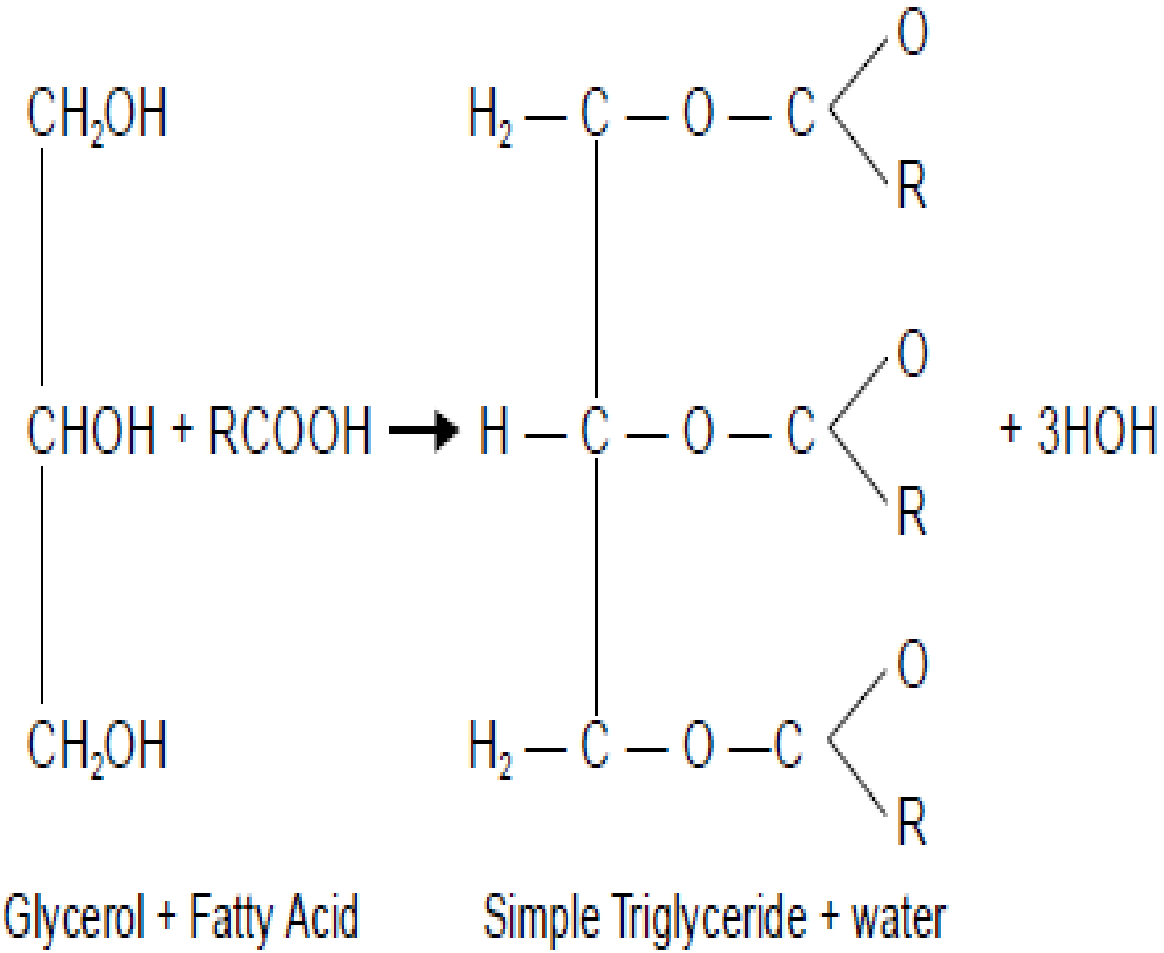
Food Biotechnology

Composition and Classification of fats & Oils

Composition and Classification of fats & Oils

Fats and oils are mixture of different triglycerides. Triglycerides are formed by mixing of glycerol and fatty acids with elimination of a water molecule.

Composition and Classification of fats & Oils



Composition and Classification of fats & Oils

Monoglycerides

Diglycerides

Simple triglycerides

Mixed triglycerides

Natural fats consists of mixed triglycerides and natural substances responsible for unique flavour or taste.

Composition and Classification of fats & Oils

Solid fats such as butter and ghee more saturated fatty acids.

Liquid fats such as soybean oil, canola oil etc. have less saturated fatty acids.

Composition and Classification of fats & Oils

No. of carbons	Fatty acid		Coconut oil (%)	Groundnut oil (%)	Corn (%)	Cotton seed (%)	Lard (%)	Mutton tallow (%)	Butter (%)
	Satura-ted	Unsatu-rated							
4	Butyric		–	–	–	–	–	–	3
6	Caproic		0.5	–	–	–	–	–	2
14	Myristic		18	–	–	1	1	2	10
16	Palmitic		9	6	6	21	28	34	30
18	Stearic		2	5	2	2	8	19	11
18		Oleic	7	61	37	25	56	43	30
18		Linoleic[2]	2	22	54	50	5	2	3
	Other acids		61.5	6	1	1	2	–	11
	Total		100	100	100	100	100	100	100

Composition and Classification of fats & Oils

Fatty acids are described as either short or long chain or saturated and unsaturated fatty acids.

small fatty acids such as acetic acid in vinegar, caproic and butyric acids in butter.

12-18 carbon chain length such as palmitic and stearic acids in lard and tallow.

Composition and Classification of fats & Oils

Fatty acids are mono and polyunsaturated fatty acids.

18 carbon oleic and linoleic found in olive and corn respectively.

Vegetable oils are more unsaturated and considered more healthy.

Composition and Classification of fats & Oils

END

Animal source of fats are more saturated and considered less healthy and the amount consumed should be restricted.

Trans fatty acid contents in hydrogenated fats are another factor making them more harmful for health.

Food Biotechnology

Module #19



Food Biotechnology

Nutritive value of Fats and Oils

Nutritive value of Fats and Oils

Fats and Oils are source of energy. One gram of fats and oils produces 9 kilocalories of energy so they are concentrated form of energy.

There are many other functions such as they act as carrier for fat soluble vitamins A, D, E and K.

Nutritive value of Fats and Oils

Fats and Oils have more palatable satiety value, means they take more time leaving the stomach.

Digestion of fats starts in small intestine call duodenum. Gall bladder produces bile and pancreas produces lipase through a common duct that goes to duodenum.

Nutritive value of Fats and Oils

Bile neutralizes acidic pH of food in duodenum so that the work of pancreas enzymes starts.

Lipids are emulsified and surface area is enlarged to facilitate work of pancreas enzyme to split triglycerides into glycerol and fatty acids and mono and diglycerides.

Nutritive value of Fats and Oils

Bile salts disperse glycerol and fatty acids into micelles.

Rest of mono and diglycerides are also broken down to glycerol and fatty acids by intestinal lipases, made into micelles and absorbed by intestine.

Essential and non essential fatty acids.

Nutritive value of Fats and Oils

Essential fatty acids are linoleic acids

(omega-6) and a-linoleic acids (omega-3).

Deficiency of EFAs results in impaired brain growth, mental retardation and learning difficulties, dermatitis (dryness of skin), hair loss, and poor wound healing.

Nutritive value of Fats and Oils

Saturated fatty acids can raise LDL (bad Cholestrol), that may lead to heart disease.

Similarly trans fats can build up plaque in arteries and lead to heart diseases.

Food Biotechnology

Module #20



Food Biotechnology

Physical properties of fats and oils

Physical properties of fats and oils

Melting Point

All food fats are mixtures of triglycerides, and therefore, do not have a sharp

melting point, but melt over a range of temperatures.

Physical properties of fats and oils

Creaming of Fats:

Solid fats like butter and margarine can be creamed or made soft and fluffy

by the incorporation of air. Fat and sugar are usually creamed together in the preparation of cakes

Physical properties of fats and oils

Plasticity of Fats

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not

flow at room temperature and are thus solid fats. The spreading quality of butter

is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of

one kind of a molecule. They, therefore, do not have a sharp melting

Physical properties of fats and oils

Emulsification

Lipids can form an emulsion with water when beaten up with it to form tiny globules in the presence of suitable emulsifying agent. Butter is an emulsion, so also is cream. The presence of minute amounts of milk protein helps to stabilise these emulsions. Lecithin, a phospholipid from egg yolk helps to stabilise mayonnaise.

Physical properties of fats and oils

Smoke Point:

The smoke point is the temperature at which a fat or oil gives off a thin bluish
smoke. Foods that are fried are added to the hot oil just before it reaches its smoke point. Fats and oils with low molecular weight fatty acids (those with a short chain length) have low smoke point.

Physical properties of fats and oils

Normally, oils that are selected for deep fat frying are those, which have a high smoke point. If oils with low smoke points are used for deep fat frying, then the foodstuff is fried at a lower temperature. Repeated use of the same sample of oil for frying results in a decrease in its smoke point and ultimately in its decomposition.

Food Biotechnology

Module #21



Food Biotechnology

Refining of Crude Oils

Refining of Crude Oils

Most of oil used now a days is refined.

In the refining process the impurities in oils are removed.

The impurities in the oils include moisture, free fatty acids, colouring pigments, resins, gums and sometimes vitamins. The colour, flavour, odour and clarity as also the shelf-life of the oil are affected by impurities.

Refining of Crude Oils

Refining is a 5 step process

1. Degumming :

Some impurities in the crude oil form gums in the presence of water. Such impurities are removed by adding water to the warm oil and centrifuging it to remove the denser gum particles. Thus, a clarified oil is obtained as an upper layer, which is siphoned off.

Refining of Crude Oils

2. Neutralising

All crude oils contain some free fatty acids formed due to spoilage. The oil is

neutralized with caustic soda solution and the insoluble soaps formed from the free fatty

acids, which settle at the bottom are removed..

Refining of Crude Oils

3. Washing and Drying:

The last traces of soap from the oil are removed by washing. The lower aqueous soap layer is run off and the oil layer is dried under vacuum. The oil thus obtained has a yellowish colour and a distinct odour. To remove these, the oil is bleached and deodourised..

Refining of Crude Oils

4. Bleaching

To absorb the colouring matter from the oil, adsorbing substances such as activated carbon and fuller's earth are added to the warmed oil. The mixture is stirred with maintenance of partial vacuum. After, all the coloured matter is adsorbed, the mixture is put through filter presses to get a clear colourless oil.

Refining of Crude Oils

5. Deodourising

Finally the oil is deodourised by injecting steam with agitation to remove all odourous material as vapour. The deodourised oil is packed as such or blended with other oils and packed. The refined oil has to be stored under an inert gas such as nitrogen or vacuum packed to prevent oxidation.

Food Biotechnology

Module #22



Food Biotechnology

Different Uses/Products of Fats and Oils

Different Uses/Products of Fats and Oils

Role in Cookery:

The role of different types of fats and oils in cookery is largely based on their composition and properties. Thus, liquid fats or oils with a high smoke point are used for deep-fat frying purposes and likewise, solid fats like butter and margarine are used as shortening and tenderizing agents in foods.

Different Uses/Products of Fats and Oils

Fats are used in food preparation

- (i) as a medium of cooking.
- (ii) as a shortening as in biscuits, pastry and cakes.
- (iii) to add richness and flavour
- (iv)seasoning of vegetables and salads.

Different Uses/Products of Fats and Oils

As a Medium of Cooking:

Foods get cooked in fat in shorter time than when cooked in water. Fried foods, *have a crisp* texture and a delectable flavour. The high temperature used in frying destroys harmful bacteria, thus making the food safe for consumption.

Different Uses/Products of Fats and Oils

As a Medium of Cooking:

Some fat is absorbed by the food and the calorific value of the food is increased when it is cooked in fat or oil

Different Uses/Products of Fats and Oils

As a Shortening:

In many preparations, such as cakes, biscuits *fats or oils are added to improve* the texture. The fat covers the surface of the flour particles and prevents the sticking of particles together.

Different Uses/Products of Fats and Oils

As a Shortening:

Many factors such as the nature of the fat or oil, the amount added, the temperature, presence of other ingredients, manipulation and the extent of mixing, affect the shortening power.

Different Uses/Products of Fats and Oils

As a Seasoning:

Fats and oils are used to season most food preparations. In sweet preparations, fats, such as butter, ghee, vanaspati are used, as they have mild flavour, which blends with the sweet preparation.

Food Biotechnology

Module #23



Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- Fats and oils undergo certain undesirable changes during storage, which result in spoilage.
- Flavour Reversion
- Rancidity Spoilage

Storage and Shelf life of Fats and Oils

- **Flavour Reversion**

The fats undergo a peculiar change before the onset of rancidity. The characteristic flavour is lost and the fat or oil has a flat taste and a greasy feel on the tongue. This is known as flavour reversion and precedes rancidity changes.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- *Rancidity*

Spoilage of fats results in off flavour and renders the fat inedible. These changes are known as rancidity of fats. Fats and oils can get rancid by the action of

(a) air (oxidation),

(b) water (hydrolysis) and

(c) Enzymes (enzymatic breakdown).

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- **Hydrolysis** is the decomposition of fats (triglyceride molecules) to glycerol and free fatty acids. Presence of moisture, microorganisms and the enzyme lipase hastens the hydrolytic breakdown. This kind of spoilage is known as hydrolytic rancidity.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- **Oxidation** of unsaturated fats leads to oxidative rancidity. Thus, oils or fats containing more double bonds (unsaturation) are more likely to develop oxidative rancidity than those with few double bonds.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

- **Oxidation**

This oxidation is a chain reaction initiated by the production of free radicals.

Storage and Shelf life of Fats and Oils

Changes in Fats During Storage

Enzymatic breakdown normally accompanies hydrolysis

Storage and Shelf life of Fats and Oils

Prevention of Fat Spoilage

- Keep fats and oils in dry, tightly covered containers to ensure exclusion of air and moisture.

Storage and Shelf life of Fats and Oils

- Keep the container sealed until needed.
- Keep fat in a container having a narrow opening to prevent undue exposure.
- Store in a cool, dry place away from cooking area, where the temperature and humidity fluctuations are not great.

Storage and Shelf life of Fats and Oils

Prevention of Fat Spoilage

Addition of antioxidants, such as tocopherols, and other phenolic compounds such as BHA1, BHT2, or propyl gallate, are used to retard rancidity in commercial fatty products

Food Biotechnology

Module #24



Introduction To Dairy Products

TYPES OF DAIRY PRODUCTS

- Milk
- Cream
- Cultured Dairy Products
- Frozen Dairy Products
- Concentrated Dairy Products
- Non-Dairy Products
- Butter
- Cheese

MAIN DAIRY FOODS







YOGHURT

CHEESE







MILK

FROMAGE FRAIS



CREAM



ICE CREAM

Introduction to Dairy Products

Milk

Can be plain or flavored (chocolate, strawberry, etc.)

Usually fortified with VITAMIN D

It is generally then processed in the following ways before it is sent to stores:

Pasteurized: Process of heating to destroy harmful bacteria

Introduction to Dairy Products

Milk

Homogenized: Process of agitating milk to help distribute the fat throughout so it's uniform in texture (not clumpy

Introduction to Dairy Products

Milk

Other types of milk

Flavoured Milk – usually UHT MILK, either semi-skimmed or skimmed.

Dried Skimmed Milk, Evaporated Milk, Condensed Milk full cream or skimmed

Introduction to Dairy Products

Milk

UHT Milk – this milk owes its long life to ultra high temperature

132*c for one second followed by packaging

Introduction to Dairy Products

Cream

Cream is a more concentrated form of milk

Once a cow is milked, the solids float to top (milkfat) and they are skimmed off and this becomes cream!

It comes in the following varieties:

Heavy (whipping) Cream

Higher percentage of fat (85% cream, 15% milk)

Introduction to Dairy Products

Cream

Light (whipping) Cream

Lower percentage of fat (70% cream, 30% milk)

Half & Half

Even less fat (half 50% cream, half 50% milk)

To Make Whipped Cream:

Use cold bowl and whip cream until frothy… to sweeten, gradually add sugar little by little

Introduction to Dairy Products

Cream

DO NOT OVERBEAT, or it will deflate and turn into butter

Introduction to Dairy Products

Cultured Products

Made from cultured, or specially grown bacteria

Usually thick in texture & tangy in flavor

Examples of cultured dairy products include:

Yogurt

Introduction to Dairy Products

Cultured Products

Yogurt

This is the dairy product with the lowest amt of fat

Can be substituted for sour cream to reduce the fat in a recipe

Sour Cream

Buttermilk

Introduction to Dairy Products

Frozan Products

Dairy products that have been prepared and stored at very low temperatures

ICE CREAM

Made from milk, cream, sugar and flavoring ,Generally has about 6-8 grams of fat

REDUCED FAT has about 4-5 grams of fat, LOWFAT has less than 3 grams of fat

NONFAT has less than 0.5 grams of fat

Introduction to Dairy Products

Frozan Products

SHERBERT

Made from milk, sugar and fruit juice

FROZEN YOGURT

Made from cultured dairy product, sugar & flavoring

Introduction to Dairy Products

Concentrated Products

Dairy products that have had the water or liquids removed to increase the density

Examples include:

EVAPORATED MILK

Has some water removed, Can be reconstituted and used as fresh milk.

Available in cans

SWEETENED CONDENSED MILK

Has water removed and sweetener added

Used most commonly in baking

Cannot replace fresh milk or evaporated milk

Available in cans

Introduction to Dairy Products

Concentrated Products

NONFAT DRY MILK POWER

Used by chefs because it does not spoil and it costs less than fresh milk

Can be reconstituted and used as fresh milk

Comes powdered in packets (boxes)… think hot cocoa

Introduction to Dairy Products

Butter & Products

Made by churning cream (either sweetened cream or sour cream) into butter

It is usually then colored artificially and either salted or left unsalted and packaged then sold

Whipped butter just incorporates more air into the churning process resulting in a less dense end product

Introduction to Dairy Products

Butter & Products

It can be frozen for a longer shelf-life but should ideally be refrigerated

Can spoil if left out (resulting in BITTER BUTTER… remember Betty?!)

Offers saturated fat (animal product)

Introduction to Dairy Products

Cheese

TYPES:

UNRIPENED sold immediately, not allowed to age

Ex. Cottage cheese, cream cheese, ricotta cheese

Better for cooking because they're more blendable

RIPENED curds are packaged and aged (sometimes for years)

Ex. Cheddar, Muenster, Provolone, Swiss…

The softer the cheese, the better it is for you… while all cheeses have saturated fats, harder cheeses have higher levels

Introduction to Dairy Products

Cheese

PROCESSED chemically made or altered

Ex. Velveeta, cheese sauces, imitation cheese

These tend to create really smooth, creamy cheeses & cheese sauces

Food Biotechnology

Module #25



Microbiology of Milk

- Milk is a highly nutritious media for bacteria
- Can spoil easily
- bacteria can enter milk during milking process and handling
- Refrigeration improved milk industry, but still other improvements needed to be done to reduce No's of non pathogenic bacteria

Microbiology of Milk

- **Non Pathogens in milk**
- Air borne contaminants : Pseudomonas, Flavobacterium, Alcaligenes, coliforms, micrococcus, fungi
- Psychrotrophs :grow at 5°c and less and make up a large number of the microbes in milk, they are natural soil inhabitants and wide spread.

Microbiology of Milk

- **Non Pathogens in milk**

Spoilage of refrigerated milk consists usually of bitter, rancid, fruity flavours

Microbiology of Milk

- **Sources of Pathogens in milk**
- Animal itself : Mycobacterium bovis, Brucella species, Coxiella burneti ,Campylobacter sp.
- Humans : Salmonella, Shigella
- Environment : Bacillus anthracis, Clostridium perfringens, Clostridium botulinum

Microbiology of Milk

Pasteurization of Milk

Pasteurization:

Kills and removes all pathogens(100%)

Kills most non pathogens (99%) Surviving organisms are spore formers, thermoduric and thermophilic organisms

Microbiology of Milk

- **Pasteurization of Milk**

Common thermoduric organisms:

Bacillus, Micrococci, Lactococci, Arthrobacter

Microbacteria, Corynebacteria

Pasteurization provide a product with average shelf life 7-10 days(LTLT-HTST)

Microbiology of Milk

- **Pasteurization of Milk**

Types and numbers of bacteria in milk depend on the microbial load before pasteurization

Microbiology of Milk

- **Pasteurization of Milk**

No of non pathogens in raw milk is 500,000 to 3,000,000 organisms/ml of milk

While No of non pathogens in pasteurized milk is 2,000 to 20,000 organisms /ml

Food Biotechnology

Module #26



Microbiology of Milk

- Milk is a highly nutritious media for bacteria
- Can spoil easily
- Initially milk may have few organisms , but bacteria can enter milk during milking process and handling
- Refrigeration improved milk industry, but still other improvements needed to be done to reduce No's of non pathogenic bacteria

Microbiology of Milk

- **Non Pathogens in milk**
- Air borne contaminants : Pseudomonas, Flavobacterium, Alcaligenes, coliforms, micrococcus, fungi
- Psychrotrophs :grow at 5°c and less and make up a large number of the microbes in milk, they are natural soil inhabitants and wide spread.

Microbiology of Milk

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.

Microbiology of Milk

- **Sources of Pathogens in milk**
- Animal itself : Mycobacterium bovis, Brucella species, Coxiella burneti ,Campylobacter sp.
- Humans : Salmonella, Shigella
- Environment : Bacillus anthracis, Clostridium perfringens, Clostridium botulinum

Microbiology of Milk

Pasteurization of Milk

LTLT : Low temp. Long time 63°C 30 minutes

HTST: High temp. Short time72°C 15 seconds

UHT : Ultra high temp.,130-135 °C 1 second

Microbiology of Milk

Pasteurization of Milk

Pasteurization:

Kills and removes all pathogens(100%)

Kills most non pathogens (99%) Surviving organisms are spore formers, thermoduric and thermophilic organisms

Microbiology of Milk

- **Pasteurization of Milk**

Common thermoduric organisms:

Bacillus, Micrococci, Lactococci, Arthrobacter

Microbacteria, Corynebacteria

Pasteurization provide a product with average shelf life 7-10 days(LTLT-HTST)

Microbiology of Milk

- **Pasteurization of Milk**

Types and numbers of bacteria in milk depend on the microbial load before pasteurization

Microbiology of Milk

- **Pasteurization of Milk**

No of non pathogens in raw milk is 500,000 to 3,000,000 organisms/ml of milk

While No of non pathogens in pasteurized milk is 2,000 to 20,000 organisms /ml

Microbiology of Milk

- **Pasteurization of Milk**

Different methods to count non pathogens in milk such as:

Viable count(pour plate method).

Dye reduction test Direct (clump) microscopic count

Microbiology of Milk

Viable count method

Pour plate method

- Deliver 1ml of milk to 9ml of sterile saline tube(10-1), mix and transfer 1ml diluted milk from tube 1 to tube 2 (10-2).
- Deliver 1ml of milk to an empty Petri dish ,1ml of diluted milk 10-1,10-2 to empty Petri dishes .
- Pour 9ml of molten nutrient agar to each Petri dish, mix well

Microbiology of Milk

- Leave to solidify, then incubate at 37 c 48 hrs.
- Count No of colonies and multiply by dilution factor to determine No of viable non pathogens on milk
- A plate containing more than 300 colonies should not counted, plate with more dilution is counted instead.

Microbiology of Milk

Dye reduction test

A quick test for determination of microbial load of milk

Use Methylene blue or resazurin dyes

Add 1ml of dye to 10ml of milk, incubate 1 hr at 36c

If there is change in color , it means high bact. Load .It is not good quality milk, shorten expiration date.

Microbiology of Milk

- If no change in color (still blue), low bact. Load , good quality milk , expand expiration period
- A long history of use in the dairy industry, especially to measure microbial quality from raw milk
- Outbreaks have been associated with cheeses

Microbiology of Milk

- Salmonella serovar Zanzibar in goat milk cheese
- L. monocytogenes in Mexican-style cheese

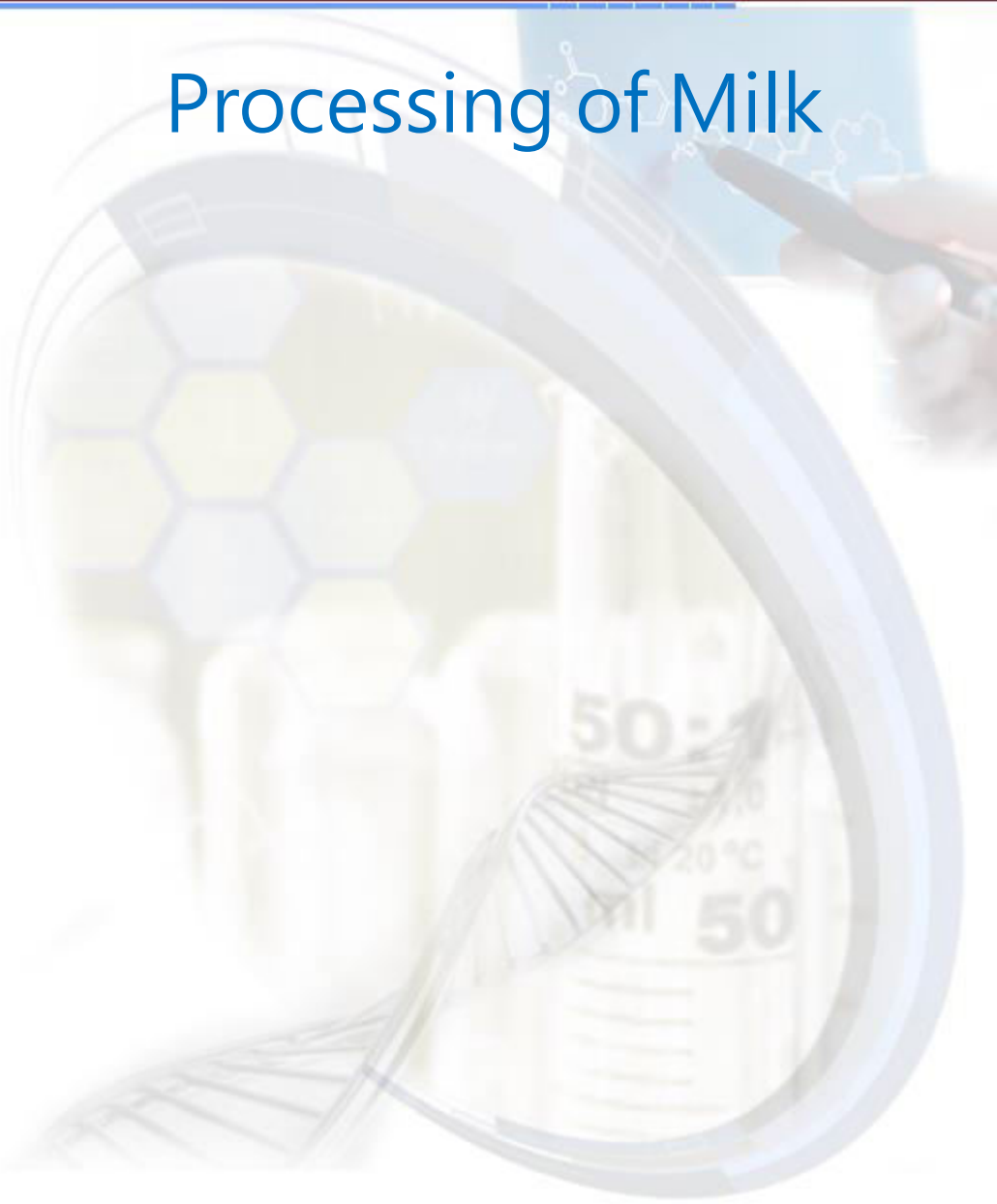
Food Biotechnology

Module #27



Food Biotechnology

Processing of Milk



Processing of Milk

Introduction:

Raw milk is associated with pathogenic diseases such as tuberculosis, brucellosis or typhoid fever, among others.

Hygienic milk production, proper handling and storage of milk, and appropriate heat treatment can reduce or eliminate pathogens in milk.

In many pasteurization of milk is mandatory.

•

Processing of Milk

- Milk may be given various treatments like heating, concentrating, drying or altering its pH to obtain a number of different products to preserve milk and add variety to our meals.

This is done by:

1. exposing milk to high temperatures at which microorganisms are killed,

Processing of Milk

2. binding or reducing the water present in milk and thus making moisture unavailable to micro-organisms.

3. by increasing the acidity of milk to a level that does not permit growth of spoilage microorganisms, and yet imparts a desirable sour flavour to the product.

Processing of Milk

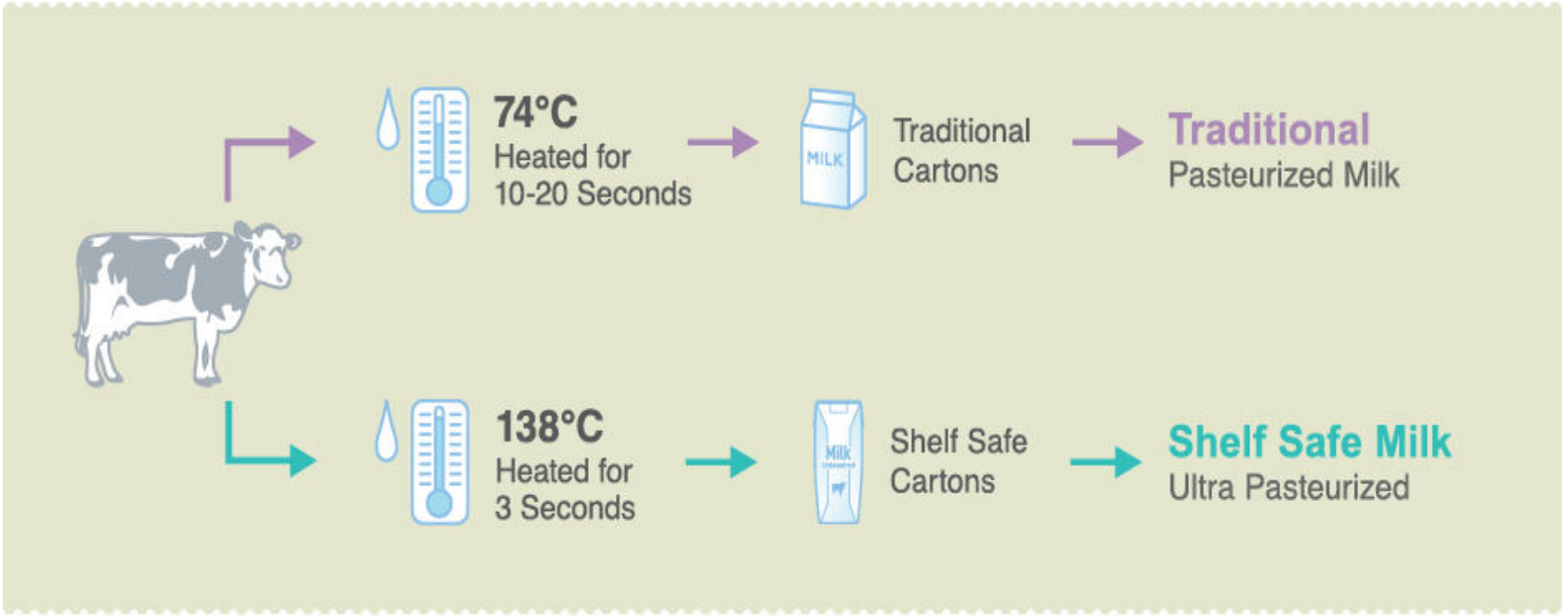
Types of Processing Methods:

Pasteurization

- ***Holding Method*** - 62.7°C for 30 minutes.
- ***Higher Temperature Short Time Method*** **(HTST)**- 71.6°C at least for 15 seconds.
- ***Ultra High Temperature Process***- Milk is heated to a temperature of 89–90°C or more for 1 second or less.

All followed by cooling at 10°C

Processing of Milk



Processing of Milk

Concentrated Milks:

- Such as evaporated milk, sweetened condensed milk and dried milk made from whole or skimmed milk.
- Varying amounts of water are removed to make the concentrated milk products.
- These products have a longer shelf-life and some can be reconstituted to their original form.

Processing of Milk

Evaporated Milk:

- more than half the water from milk under vacuum is removed.
- It is necessary to fore-warm milk 10 minutes to prevent coagulation of casein during sterilization after it is canned.

Processing of Milk

Condensed Milk:

- Is concentrated to about 1/3 of its original volume and has about 15 percent sugar added to the milk.
- Since the finished product contains at least 40 per cent of sugar, which acts as a preservative, it is not necessary to sterilize it before canning.

Processing of Milk

Dehydrated Milk Products:

- Includes whole milk powder, skimmed milk powder, infant milk foods and malted milk.

Processing of Milk

Skimmed Milk Products:

- In production of skimmed milk powder, the milk fat is removed or skimmed from the milk, before the milk is dehydrated to a moisture content less than 4 per cent.

Food Biotechnology

Module #28



Food Biotechnology

Other Dairy Products

Other Dairy Products

Curd (Yogurt):

- Curds is a very popular milk product in India.
- It ranks second to milk in the extent of consumption.
- Milk is boiled and cooled to about 50°C and a teaspoon of curd from an earlier batch of curd is added and mixed thoroughly.
- The lactic acid bacteria present in this sample of curd curdle the milk.

Other Dairy Products

- They utilize the lactose in milk and break it down to lactic acid.
- The formation of lactic acid increases the acidity of milk. When the pH reaches 4.6, the milk protein, casein, coagulates as curd.
- Since the process is gradual, the milk serum is bound in the mass of coagulated proteins.

Other Dairy Products

- The optimum temperature for the formation of curd is 35-40°C.
- The time needed for curd formation varies from 8-12 hours depending on atmospheric temperature.

Other Dairy Products

- Once made, curd keeps well at refrigerator temperature of 2–3 days. It is used as dressing on salads made from fresh vegetables.
- It combines well with plain cooked rice.
- Preparation of curd is one method of prolonging the shelf-life of milk by several days.

Other Dairy Products

- Soured milk does not readily undergo proteolysis and other undesirable changes.
- Curds can be held at refrigerator temperature for several days without loss of acceptability.
- It is reported that the riboflavin and thiamin content increases during curd formation.

Other Dairy Products

- It is also reported that fermented milk inhibits the growth of *Bacillus typhosus, Bacillus dysenteriae*, and *Vibrio cholerae* to a certain extent.

Other Dairy Products

Paneer:

- Is prepared by addition of lemon juice or citric acid to hot milk and precipitating the casein.
- The liquid released in this process is known as *whey*, which contains most of the soluble nutrients from milk.
- Paneer contains about 18 per cent protein and is a good source of it.

Other Dairy Products

Cheese:

- The milk is subjected to several process steps to produce cheese.
- The milk it held at about 27°C in vats and a lactic acid culture is added.
- When the milk gets acidic, rennet is added to
it and the milk is allowed to coagulate.

Other Dairy Products

- The curd formed is cut into strips and heated to about 37°C with stirring to separate the whey.
- The whey is drained.
- Salt is mixed with the curd and it is pressed to remove further amounts of whey.

Other Dairy Products

- The cheese formed is coated with paraffin to prevent loss of moisture and allowed to ripen at low temperature.
- Cheese contain about 24 per cent protein and is thus a concentrated source of protein.

Other Dairy Products

Buttermilk

- is made by adding water to curd and churning to remove fat in the form of butter.
- The energy value is thus reduced due to fat removal.
- The concentration of the other nutrients is reduced in proportion to the dilution.
- It is known to re-establish intestinal flora after an attack of diarrhoea.

Food Biotechnology

Module #29



Food Biotechnology

Detection of proteases in Milk

Detection of proteases in Milk

Introduction:

Proteases are enzymes that degrade proteins. Raw milk may have proteolytic activity from endogenous or indigenous origin. Wherever they come from, proteases can cause unpleasant flavours and odours in milk.

Detection of proteases in Milk

- Several bacteria present in raw milk can produce proteases. Among them, those from psychrotrophic microorganisms (*Bacillus subtilis*), which produce enzymes under refrigeration, cause the most serious problems.
- It secretes a thermo-resistant protease which may cause proteolysis in pasteurized or sterilized milk.

Detection of proteases in Milk

Sample preparation

Milk sample need to be chosen that is required for analysis.

The standard protease from Bacillus subtilis is needed and required concentrations are made for calibration curve.

Detection of proteases in Milk

Enzyme determination

Reagents

- Disodium hydrogenphosphate,
- Sodium dihydrogenphosphate,
- Trichloracetic acid.

•Solutions

•Prepare a pH 7.2 buffer solution by mixing 36 ml of sodium hydrogenphosphate (5 mM) and 14 ml of sodium dihydrogenphosphate (5 mM) and diluting with distilled water to 1000 ml.

•This buffer is used to prepare a 1% azocasein solution.

•A 5% aqueous solution of trichloracetic acid was also required.

Detection of proteases in Milk

- **Determination of protease activity**
- The assay for proteolytic activity involves combining 1 ml of the azocasein solution with 100 ml of the sample which contains the enzyme.
- The contents of the tubes are mixed and incubated at 35.5 degree C for 15 min. The reaction is stopped by the addition of 2 ml of 5% TCA.

Detection of proteases in Milk

The absorbance of the supernatant is recorded at 345 nm in a UV/Visible spectrophotometer.

Activity calculated against standard curve.

END

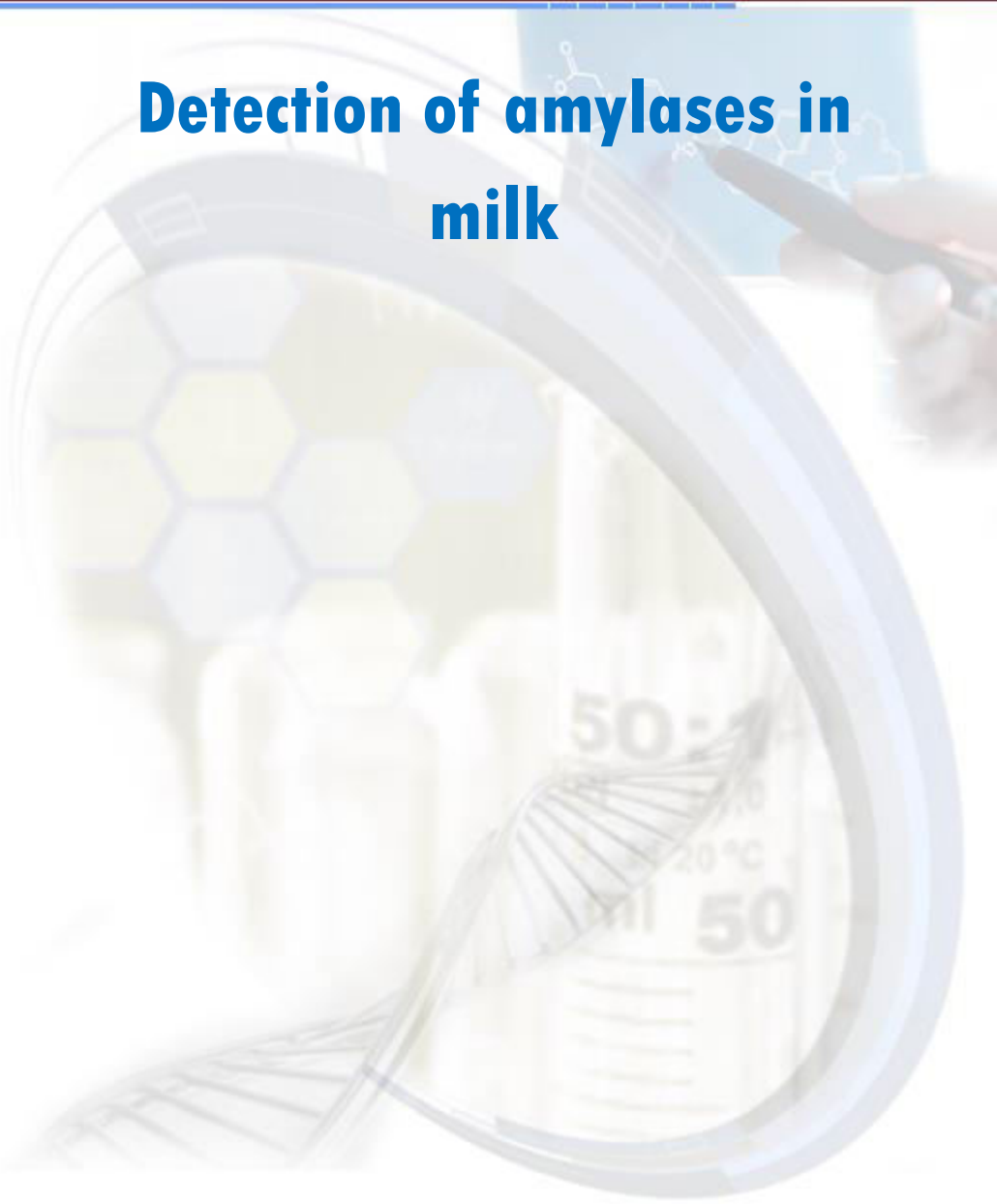
Food Biotechnology

Module #30



Food Biotechnology

Detection of amylases in milk



Detection of amylases in milk

Procedure:

- Amylase activity in milk is a critical quality parameter.
- The amylase activity in milk is often referred as diastatic power and refers to the production of reducing substances by the action of alpha & beta amylases.

Detection of amylases in milk

- The measurement involves digestion of the soluble matter with a malt infusion (extract) and following increase in reducing substances by measuring reduction of Fehling's Solution.
- Specifically measuring alpha amylase activity (dextrinizing activity) is more complicated and is based on using a limit dextrin as substrate.

Detection of amylases in milk

- Limit dextrin is prepared by action of beta amylase (free of alpha amylase activity).
- The beta amylase clips maltose units off the nonreducing end of the starch molecule until an alpha-1,6-brach point is encountered.
- The resulting product is a beta limit dextrin that serves as the substrate for the endo cleaving alpha amylase.

Detection of amylases in milk

- A malt infusion is added to the previously prepared limit dextrin substrate and aliquots removed periodically to a solution of dilute iodine.
- The alpha amylase activity is measured by changed color of the starch iodine complex in the presence of excess beta amylase used to prepare the limit dextrin.
- The color is compared to a colored disc on a comparator.

Detection of amylases in milk

- This is continued until the color is matched to a color on a comparator.
- The time to reach that color is dextrinizing time and is a measure of alpha amylase activity, a shorter time representing a more active preparation.

END

Food Biotechnology

Module #31



Food Biotechnology

Determination of iodine number of fats



Determination of iodine number of fats

Principle:

- The iodine number is a measure of degree of unsaturation, that is, number of carbon double bonds in relation to the amount of fat or oil.
- Iodine value is defined as the grams of iodine absorbed per 100 gram sample.

Determination of iodine number of fats

- The higher the amount of unsaturation the more iodine is absorbed.
- A quantity of fat or oil dissolved in solvent is reacted with a measured amount of iodine or some other halogen. Halogen addition to double bonds takes place.

Determination of iodine number of fats

Procedure:

- A solution of potassium iodide is added to reduce excess ICl to free iodine.
- The liberated iodine is then titrated with a standardized solution of sodium thiosulfate using a starch indicator, and the iodine value is calculated.

Determination of iodine number of fats

$$\underset{\text{(excess)}}{ICl} + R{-}CH = CH{-}R \rightarrow R{-}CHI{-}CHCl{-}R + \underset{\text{(remaining)}}{ICl}$$

$$ICl + 2KI \rightarrow KCl + KI + I_2$$

$$I_2 + \underset{\text{(blue)}}{\text{starch}} + 2Na_2S_2O_3 \longrightarrow 2NaI + \underset{\text{(colorless)}}{\text{starch}} + Na_2S_4O_6$$

$$\text{Iodine value} = \frac{(B - S) \times N \times 126.9}{W} \times 100$$

where:

Iodine value = g iodine absorbed per 100 g of sample
B = volume of titrant (ml) for blank
S = volume of titrant (ml) for sample
N = normality of $Na_2S_2O_3$ (mol/1000 ml)
126.9 = MW of iodine (g/mol)
W = sample mass (g)

Determination of iodine number of fats

- Calculated iodine value is obtained from fatty acid composition using below equation for triacylglycerol.
- A similar equation allows calculation of iodine value of FFAs.

Iodine value (triglycerides)

= (% hexadecenoic acid × 0.950)

+ (% octadecenoic acid × 0.860)

+ (% octadecadienoic acid × 1.732)

+ (% octadecatrienoic acid × 2.616)

+ (% eicosenoic acid × 0.785)

+ (% docosenoic acid × 0.723) [6]

Determination of iodine number of fats

Applications:

Is used to characterize oils for hydrogenation process in refining, and as an indication of lipid oxidation, since there is a deadline in unsaturation during oxidation.

The calculated value tends to be low for material with a low iodine value and for oils with greater than 0.5% unspecifiable material (e.g, fish oil).

Food Biotechnology

Module #32



Food Biotechnology

Introduction to meat

Meat: Introduction

Meat is defined as the flesh of animals used as food.

meat is a very good source of essential amino acids, and, to a lesser extent, of certain minerals. Although vitamins and essential fatty acids are also present, meat is not usually relied upon for these components in a wellbalanced diet

Meat: Introduction

Meat is one of the animal protein foods used in diet

Meat protein has high biological value.

The proportion of nutrients in meat depend upon the kind of animal, the species and the type of cut.

Meat: Introduction

The average protein content of meat ranges from 16 to 23 per cent and the average fat content ranges from 10 to 40 per cent. Unless the cooking water is discarded, minerals and water soluble vitamins are not lost to a great extent.

Meat provides us vitamin B-complex and some vitamin A, depending on the cut.

Meat: Introduction

Meat is a good source of phosphorus, iron and some trace elements.

The meat of sheep, under 12 months of age, is sold as *lamb*.

After 12 months, it is called *mutton*.

Meat: Introduction

Veal is the meat from cattle 3 to 14.

Between 14 and 52 weeks, the meat is called *calf*.

Meat after one year after birth is called *beef*.

Meat: Introduction

Variety of animal organs are served in food preparations like;

Liver

Kidney

Tongue

Brain etc.

These are less costly than muscle meat.

Meat: Introduction

Meat is also turned into sausages;

Ground beef

Chicken and turkey meat

Theses sausages are of two types;

Ready to eat

Ones which require cooking

Food Biotechnology

Module #33



Structure of Muscles

- ***Muscle Tissue;***
  - Smallest unit of a muscle is a muscle fiber.
  - Fibers join into bundles called *Fasciculi* by connective tissues.
  - The *Fasciculi* then combine with fat deposits are covered with membrane and attached to the bone.

Structure of Muscles

Connective Tissues;

- Bind the meat fibers together.
- Join meat muscle to the bone.
- Are of two types; 1)Collagen 2) Elastin
- Collagen is converted into gelatin during cooking.

Structure of Muscles

Connective Tissues;

- Amount of connective tissues depend on the part of the animal e.g in leg it is in greater amount.

Structure of Muscles

Fatty Tissues;

- These are made up of connective tissues with embedded fat cells.
- Fat is deposited under the skin, around the glands, organs and between and within the muscle fibers.

Structure of Muscles

Fatty Tissues;

- Fat distribution in lean part of meat is called *marbling* and well marbled is desired by consumers.

Structure of Muscles

Bones;

- Bone is an indicator of the part of the animal from which the cut of meat is taken.
- Bones are either long or short.
- Long bones are hollow and contain yellow marrow.
- Other bones, which contain red marrow, are spongy inside.

Structure of Muscles

Pigments;

- *Myoglobin* is the pigment in meat that gives the red colour to meat.
- The greater the amount of myoglobin, the darker the colour of meat.
- As the animal ages the amount of myoglobin increases.

Structure of Muscles

Pigments;

- Raw meat changes its colour from red to brownish red upon conditions.

Structure of Muscles

Pigments;

Myoglobin (red pigment)

- Upon refrigeration changes to brown red
- Upon heating changes grayish brown

END

Food Biotechnology

Module #34



Postmortem changes in Meat

When the animal is slaughtered certain type of changes start to take place:

- Stiffening of meat called *Rigor mortis*
- Formation of lactic acid lowering pH
- Swelling of muscle fibers with fluid causing the meat to become hard.

Postmortem changes in Meat

When the animal is slaughtered certain type of changes start to take place:

- The circulatory system stops functioning but the enzymes continue to remain active.
- Oxidation of glycogen brings about production of lactic acid.

Postmortem changes in Meat

- The amount of glycogen stored in a particular animal decides the amount of lactic acid produced after slaughter.
- A lower glycogen storage produces sticky and gummy meat which is undesirable.

Postmortem changes in Meat

- Stiff muscles of the animal start softening after holding for some time.
- This process is called *aging*.
- Tenderness in the meat increases with holding time. Holding conditions should be carefully controlled.

Postmortem changes in Meat

Uncontrolled conditions may allow putrefactive bacteria to grow. Controlled conditions of holding are:

I. Temperature of holding should have a range of (34–36°F)

II. Carbon dioxide in the atmosphere

III. 70 per cent humidity

Postmortem changes in Meat

This process keeps the meat:

- In a good condition for 3 to 6 weeks.
- After this *aging process*, the meat is ready for sale.
- This process of aging is also called *ripening of meat.*

end

Food Biotechnology

Module #35



Preservation of Meats

Meat is a highly perishable products and soon becomes unfit to eat and possibly dangerous to health through microbial growth, chemical change and breakdown by endogenous enzymes.

Preservation of Meats

Chilling/refrigeration slows or limit the spoilage rate at temperature below the optimal range can inhibit the microbial growth, enzymatic as well as chemical reactions. Storage of fresh meat is done at a refrigeration temperature of 2 to 5°C. Chilling is critical for hygiene, safety, shelf life, appearance and nutritional quality

Preservation of Meats

Curing

- Meat is treated with salt, sugar, sodium nitrate and spices to give different flavors.
- Smoke is used some times to give a desirable flavor and to improve the keeping quality of meat.

Preservation of Meats

Freezing:

- This is a common method of preserving meat.
- Nutrient loss is least during freezing.
- The one undesirable effect, freezing has on meat is that, the meat when cooked, is less juicy.
- This method is commonly used for storing cooked meat.

Preservation of Meats

Canning

- Meat can also be preserved by this method.
- Higher temperatures which are used during canning bring about some undesirable changes in flavor and texture.

Preservation of Meats

Radiation and Sterilization

- This is a new method of preservation of meats.
- The meat is sterilized and bacteria present killed.
- According to some people, this meat is not palatable.

end

Food Biotechnology

Module #36



Food Biotechnology

Cooking and Effect of Heat on Meats

Cooking and Effect of Heat on Meats

- Cooking of meat or application of heat brings majorchanges into meat.
- These include colour and texture changes
- Food safety is achieved by reducing bacterial load.

Cooking and Effect of Heat on Meats

- The method of cooking depends on the type of meat.
- Dry heat methods like roasting, broiling, pan frying are used for tender cuts or young animals.

Cooking and Effect of Heat on Meats

- Moist heat methods like pressure cooking, closed pan cooking, steaming are used for less tender cuts.

Cooking and Effect of Heat on Meats

- Most common cooking methods appear to be roasting, broiling and pan-frying.

Cooking and Effect of Heat on Meats

Effect of Heat on Meat

- Thiamin in meat is partially lost during cooking.
- If the food is heated for a long time, the loss is extensive.
- If water or juices, in which the meat is cooked, are discarded, then the loss of the vitamin increases.

Cooking and Effect of Heat on Meats

- Low temperatures are preferred for meat cookery.

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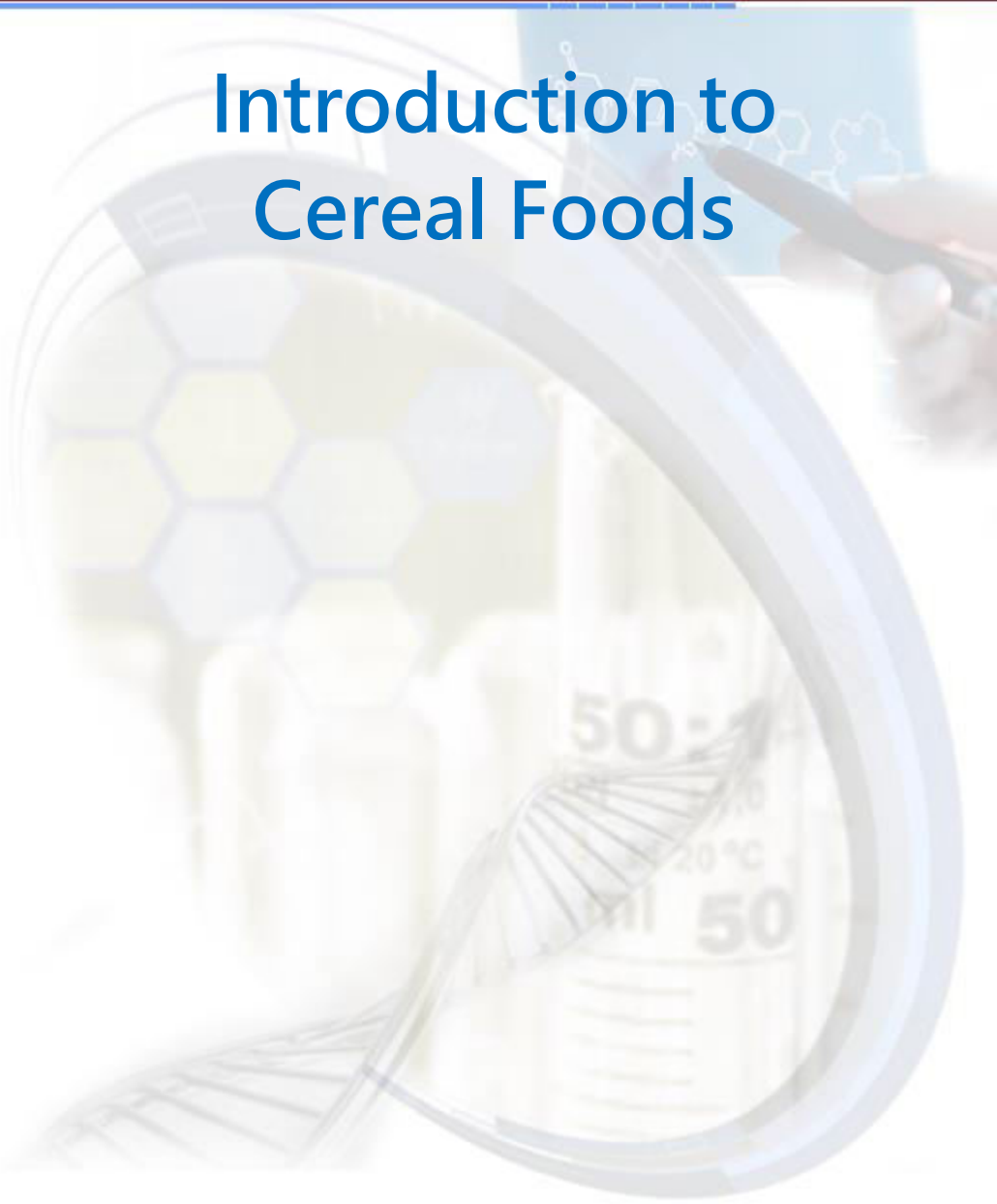
Food Biotechnology

Module # 37



Introduction to Cereal Foods

Introduction to Cereal Foods



Introduction to Cereal Foods

- Seeds of the grass family
- Major staple food in many countries
- Approx. 65-75 % carbohydrates
- Rice contains about 78% carbohydrates

Introduction to Cereal Foods

- Easy to package and transport
- Used to produce a variety of highly desirable foods
- For humans and animals
- Eaten in many ways

Introduction to Cereal Foods

- Most common cereal foods include:

- Rice
- Wheat
- Maize (corn)
- Millets such as:
  - Jowar
  - Bajra
  - Ragi

Introduction to Cereal Foods

- Can be processed further to make:

- Flour
- Starch
- Oil
- Bran
- Syrup
- Sugar

Introduction to Cereal Foods

- Advantages:
- Inexpensive
- Rich source of carbohydrates
- Gives high yields per acre
- Once harvested; excellent storage capabilities

Introduction to Cereal Foods

- Advantages:
- Can be grown in a variety of areas
- Some even in adverse soil and climatic conditions

End

Food Biotechnology

Module # 38



Cereal Foods

Composition and Structure

Cereal Foods

- Cereal grains are composed of:

- Bran
- Germ
- Endosperm

Cereal Foods

- Bran is outer chaffy coat that covers the kernel
- Endosperm is the large central portion
- Germ is a small structure at the lower end of the endosperm

Cereal Foods

- Bran

- Eliminated when grains are harvested
- Constitutes 5% of the kernel
- Discarded during milling
- High fiber and mineral content
- Good source of thiamin and riboflavin

Cereal Foods

- Aleurone layer

- Located just underneath the bran
- Rich in protein, phosphorus and thiamin
- Some fat
- Makes up 8% of the whole kernel
- Lost in the milling process

Cereal Foods

- Endosperm

- Constitutes about 84-85% of the kernel
- Contains most of the starch and protein
- Little fiber and mineral matter
- Traces of fat
- Low vitamin content

Cereal Foods

- Germ

- Makes up 2-3% of whole kernel
- Rich in fat, protein, minerals and vitamins
- Serves as a store of nutrients for the seed
- Some of it is lost during milling

Cereal Foods

- Bran is indigestible to humans
- Bran goes mainly to animal feeds
- Germ is removed for its high oil content

- Dietary guidelines recommend more fiber in foods: use of bran

Cereal Foods

- Composite Proximate Analysis:

- Protein: 11%
- Fat: 3%
- Carbs: 68%
- Moisture: 12%
- Fiber: 6%

Cereal Foods

- Nutritional Value:
- Varies with each type of cereal grain
- Starch: supplies energy
- Lacks amino acid : Lysine
- Whole cereal vs. Refined cereal

Food Biotechnology

Module # 39



Food Biotechnology

Nutritional Importance of Wheat Crop

Nutritional Importance of Wheat Crop

- Wheat Crop
- Widely cultivated
- Two crops in a year:
- Winter wheat
- Summer wheat
- Several thousand different varieties of wheat

Nutritional Importance of Wheat Crop

- Commonly milled into flour
- Can be grounded to obtain whole wheat flour
- Refined flour (*maida*); used in bread and other baked products
- Whole wheat flour (*atta*); used in preparation of *chapatis, puris, parathas*

Nutritional Importance of Wheat Crop

- Compared to refined wheat flour;
- Whole wheat flour contains more:

- Protein
- Iron
- Thiamin

Nutritional Importance of Wheat Crop

- There are two types of wheat:

- Hard wheat
- Soft wheat

- Hard wheat is higher in protein
- Produce more elastic doughs
- Better suited for making bread

Nutritional Importance of Wheat Crop

- Common used products of wheat:

- Vermicelli
- Broken wheat (*dalia*)
- Noodles
- *Suji*
- Cereals
- Canned soups
- Pasta

Nutritional Importance of Wheat Crop

- Wheat flour may be enriched with:
- Iron
- Calcium
- Can also be enriched by adding small amounts of vitamins

Nutritional Importance of Wheat Crop

- Flour may be used as thickener in canned stews, soups and gravies
- Breakfast cereal products are made from wheat

Food Biotechnology

Module # 40



Food Biotechnology

Nutritional importance of Maize

Nutritional importance of Maize

- Maize:
- Also called Corn
- Mostly used as:
- Animal feed
- Corn starch
- Corn syrup
- Bio-fuel

Nutritional importance of Maize

- Many types of corn are grown:

- Sweet corn
- Popcorn
- Dent corn
- Flour corn

Nutritional importance of Maize

- Number of varieties of corn are present
- Depending on the characteristics of starch present;
- Usually classified as starchy or waxy corn
- Field corn

Nutritional importance of Maize

- Field corn is:
- Lower in protein as compared to wheat
- Deficient in amino acids e.g Lysine
- Varieties have been developed to produce high lysine corn

Nutritional importance of Maize

- Most of corn crop is kept as animal feed

- Considerable amounts are used to produce cornstarch, corn syrup, high-fructose corn syrup an other varies sugar derivatives

Nutritional importance of Maize

- Popcorn
- Puffed cereal

- Specific variety of corn

- Internal moisture creates pressure owing to increased temperature

- Outer shell burst, expanding the grain

Food Biotechnology

Module # 41



Food Biotechnology

Nutritional Importance of Rice

Nutritional Importance of Rice

- Rice:
- Seed of the grass species:
- *Oryza sativa* (Asian rice)
- *Oryza glaberrima* (African rice).

Nutritional Importance of Rice

- Rice cultivation

- Flooding of fields
- Requires ample water
- Labor-intensive

Nutritional Importance of Rice

- Rice is available as

- Milled
- Unpolished
- Polished
- Parboiled
- Flaked
- Puffed

Nutritional Importance of Rice

- Degree of milling and polishing affects the nutrient content
- Highly polished milled rice has very little thiamin
- Parboiled and hand pounded rice contains significant amounts of thiamin

Nutritional Importance of Rice

- Grain types
- This long varieties preferred for table use
- Medium and short varieties used for preparation of rice made after grinding

Nutritional Importance of Rice

- Milling removes the bran, aluerone layer and some germ
- Polishing further removes more than half the mineral matter and most of the vitamins
- Parboiling before milling

Nutritional Importance of Rice

- Rice kernels may be enriched
- Mixed with powder containing vitamins and mineral
- Coated with waterproof edible film to prevent washing off

Nutritional Importance of Rice

- Rice is lower in total protein as compared to wheat
- Does not contain sufficient amounts of certain amino acids
- Usually cooked with twice its own volume of water

Food Biotechnology

Module # 42



Food Biotechnology

Nutritional Importance of Oats

Nutritional Importance of Oats

- Oats
- *Avena sativa*
- Used as livestock feed
- Oatmeal suitable for human consumption

Nutritional Importance of Oats

- Oats most commonly
- Rolled and crushed to make oatmeal
- Grounded into flour
- Used in variety of baked goods such as oatmeal cookies

Nutritional Importance of Oats

- Can grow in colder and wetter climates as compared to wheat
- Harvested in the same manner as wheat
- Moisture content at time of harvest should be no more than 13%

Nutritional Importance of Oats

- Oats can be milled
- May be produced for use as an ingredient of bread
- Or thickener of soups

Nutritional Importance of Oats

- Oat bran has water soluble fiber
- Linked with lower cholesterol levels

Food biotechnology

Module # 43



Food biotechnology

Nutritional Importance of Barley

Nutritional Importance of Barley

- Barley
- *Hordeum vulgare*
- Do not bake as well as wheat products
- Contains little or no gluten
- Not as popular as wheat

Nutritional Importance of Barley

- Hardy grain
- Grows in climate too cold for wheat
- Grows in soils too poor to grow wheat
- Requires a shorter growing time than wheat

Nutritional Importance of Barley

- Spring and winter varieties
- Used as feed for cattle and poultry
- Small amounts of barley flour are produced
- Used for production of malt for brewing

Nutritional Importance of Barley

- Malt
- Grain is soaked in water until moisture content reaches 50%
- Allowed to sprout
- Sprouted barley kiln dried for 24 hours

Nutritional Importance of Barley

- Enzymes that hydrolyze starch to maltose
- Maltose utilized by yeast to produce
- Ethyl alcohol
- Carbon dioxide

Nutritional Importance of Barley

- Malt
- used in the brewing industry
- convert the starches present in barley, rye, rice, corn, or other grains to maltose
- which can then be utilized by yeasts.

Nutritional Importance of Barley

- Malt may also be used in bread baking
- the yeast produce carbon dioxide for leavening
- alcohol produced is largely dissipated during the heating involved in baking.

Food Biotechnology

Module # 44



Food Biotechnology

Other Cereal Grains

Other Cereal Grains

- Other cereal grains are:

- Sorghum
- Buckwheat
- Millet
- Triticale

Other Cereal Grains

- Sorghum

- Four general classes

- 1. sweet sorghum
- 2. broom corn
- 3. grass sorghum
- 4. grain sorghum

Other Cereal Grains

- Some varieties yield glutinous starch
- Used as an alternate to tapioca in WW II
- Deterrent to use sorghum grain for production of starch is its pigmentation of the grains pericarp

Other Cereal Grains

- Buckwheat
- Not a true cereal grain
- Buckwheat belongs to the botanical family *Polygonaceae*
- All cereal grains belong to the botanical family *Gramineae*

Other Cereal Grains

- Dried to about 12% moisture
- Cleaned
- Graded by size
- Milled similarly to wheat

Other Cereal Grains

- Only a few varieties used
- Silverhull used for producing flour because of its higher yield of endosperm
- Most of the flour used for making pancakes

Other Cereal Grains

- Millet
- Used for food in Asia and some of Europe
- Used as hay in Europe and USA
- Some varieties used as food seed for birds and poultry

Other Cereal Grains

- Triticale
- Hybrid of wheat and rye
- Combines high total content of protein in wheat
- With high lysine content of rye
- More adaptable to unfavorable conditions

end

Cereal Foods

Module # 45



Cereal Foods

Processed Cereal Products

Cereal Foods

- A number of processed cereal products are made from grains
- Include products like:

- Cracked wheat
- Semolina
- *Atta*
- *Maida*
- Rice flakes
- Puffed rice

Cereal Foods

- Processes involved:

- Grinding
- Roasting
- Pounding

- Processes increase surface area of product exposed to atmosphere

Cereal Foods

- Processes:
- Decrease preparation time
- Reduce shelf life
- Whole grain shelf life is almost a year or more
- Processed cereal foods shelf life varies from two weeks to a few months

Cereal Foods

- Cracked Wheat

- Also called *dalia*
- Whole wheat coarsely ground
- Losses during milling are very little
- Very nutritious food

Cereal Foods

- Good quality indicated by sweet taste and absence of sour, moldy odor and flavor
- Spoils quickly
- Can be cooked to make and made into *upma, shira* or porridge

Cereal Foods

- Semolina
- *Suji* or *Rawa*
- Fine grain varieties used for preparation of *halwas*
- Large grain varieties used for preparation of *upma*

Cereal Foods

- Selected on basis of:

- Uniformity of size
- Freedom from insect infestation
- Freedom from oxidized or moldy odor
- Grit and bran

Cereal Foods

- *Maida*
- Finely extracted white wheat flour
- Free from bran
- Lower content of protein, iron and B-vitamins

Cereal Foods

- Lower shelf life than semolina
- Large surface area permits faster rate of food spoilage
- Good quality should be free from insect infestation, bad odors and lump formation

Cereal Foods

- Processed rice products
- Rice flakes
- Rice puffs
- Ready-to-eat rice products

Cereal Foods

- Rice flakes are free from bran, broken particles, fragments of the seed coat, insect, stones and bad odor
- Rice flakes retain large parts of iron and B-vitamins
- Rice Puffs are selected for crispiness, freedom from stone, seed coats, sand and dirt

Food Biotechnology

Module #46



Food Biotechnology

Spoilage is defined as ‘’undesirable changes in the properties of a product i.e. Food (Food Spoilage) , Wood, Marble etc.’’

Food spoilage may be

→Physical

→Chemical

→Biological

EFFECT OF SPOILAGE ON FOOD

Food spoilage can effect food by changing its

→ Nutrition Value

→ Organoleptic properties

→Color

but main drawback or hazard of food spoilage is that safety is compromised

FOOD SPOILAGE AND BIODEGRADATION

Main difference between Food Spoilage and Biodegradation is that biodegradation occurs when

→ Complex molecules and broken down to simpler one and it is crucial for maintenance of ecosystem whereas,

→Food spoilage make food unsafe for consumption

TYPES OF FOOD SPOILAGE

Food spoilage types are

→ Chemical Food Spoilage

1) Biochemical assimilatory Food Spoilage

2) Biochemical Dissimilatory Food Spoilage

→Physical Food Spoilage

1) Mechanical Food Spoilage

2) Fouling /Smelling

TYPES OF FOOD SPOILAGE

Food spoilage types are

→ Chemical Food Spoilage

1)Biochemical Assimilatory Food Spoilage :-

The organism uses the material as food i.e. an energy source.

2)Biochemical Dissimilatory Food Spoilage :-

The chemical change in the food is as a result of waste products from the organisms in question.

TYPES OF FOOD SPOILAGE

END

→ Physical Food Spoilage

1)Mechanical Food Spoilage–

Material is physically disrupted/damaged by the growth or activities of the organisms. ●

2)Soiling/fouling –

Material or product is not necessarily unsafe, but as its appearance has been compromised, it is rendered unacceptable.

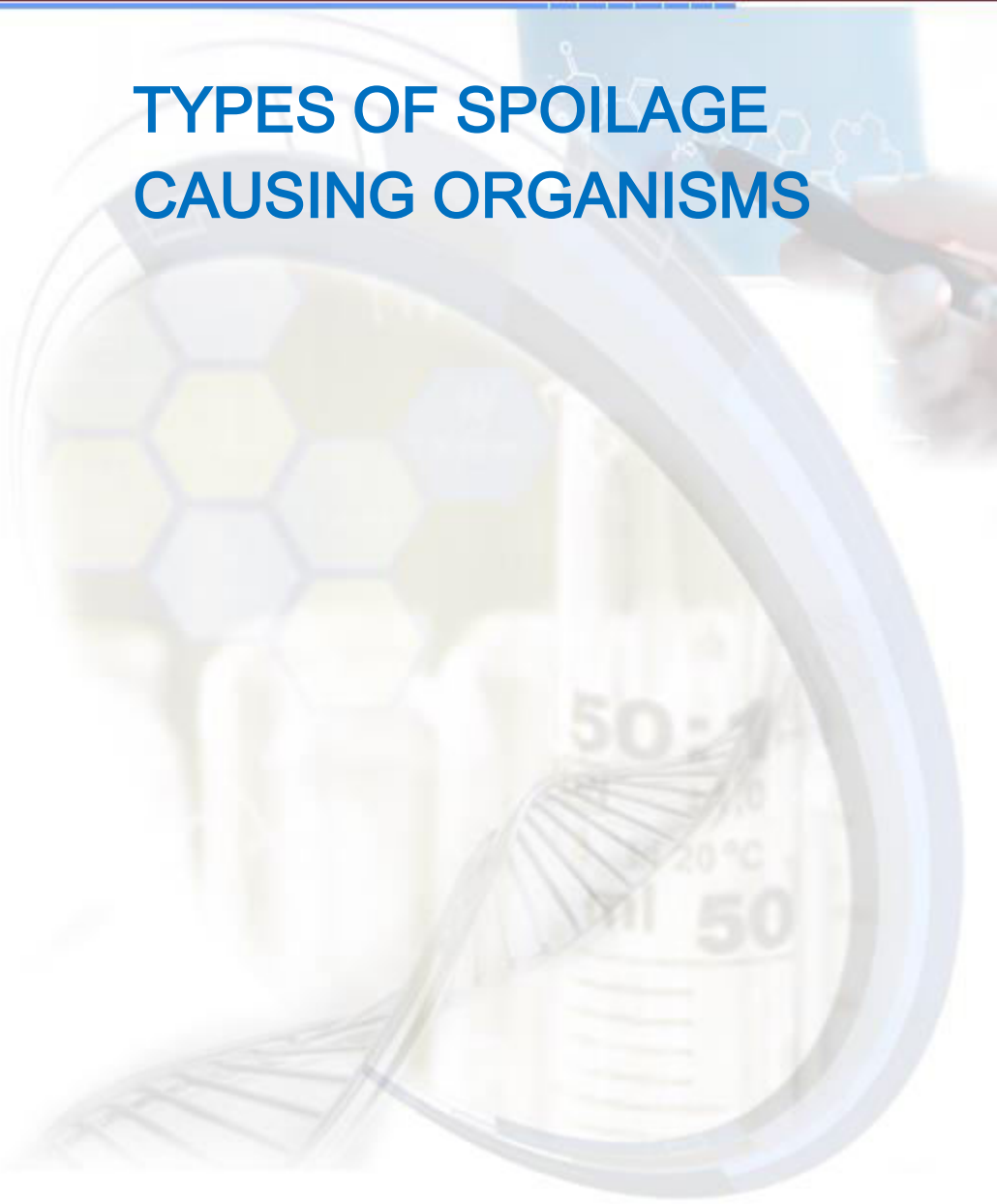
Food Biotechnology

Module #47



Food Biotechnology

TYPES OF SPOILAGE CAUSING ORGANISMS



TYPES OF SPOILAGE CAUSING ORGANISMS

Introduction :-

Many living organisms (Micro, Macro) cause food spoilage and make it unfit for consumption e.g

→ Bacteria

→Fungi

→Insects

→Rodents

→Birds

1) Bacteria

→ Bacteria greatly spoils the food and cause huge food losses

Example :-

→ Clostridium botulinum (Canned foods i.e. Beverages)

→ Micrococcus (Rancidity)

→ Lactobacillus (Gas formation)

→ Flavobacterium (Decoloration)

FUNGI / MOULD

Fungi and mold are so dangerous as they attack on Agriculture Food crops i.e. Wheat , Rice , Pulses etc.

Examples:-

→Black rot of wheat

→ Black spots on rice kernels

Mould also spoils bakery products i.e. Bread

INSECTS

→Insects can use food (Cereals,pulses,fruits,vegetables etc.) for their nourishment and in return spoil our food and cause huge food losses

→ They deform the food

→ Soften the tissues

RODENTS AND BIRDS

Rodents i.e. Rat/Mouse , Rabbit And Birds greatly spoil and destroy our food (Cereals,
Fruits ,Vegetables,Meat ,

cheese etc.)and cause huge food losses

→Deform food

→ Cause Food Borne illness

→ Make Food Rotten

Food Biotechnology

MODULE 48



Bacterial Food Spoilage

BACTERIAL FOOD SPOILAGE

Bacteria :-

Bacteria are a large diverse group of microscopic, prokaryotic, unicellular organisms. They can be of various shapes (spherical, rod-like or spiral) and may be motile or non-motile. They include both autotrophic and heterotrophic species, and can be aerobic or anaerobic, and many species can thrive under either condition

TYPES OF FOOD SPOILER BACTERIA

Food Spoiler bacteria classified on the basis of their activity or the end product / effect they produce after spoilage .

→Lipolysis bacteria

→ Pigment former bacteria

→ Gas former Bacteria

LIPOLYSIS BACTERIA

→ Lipolysis is the breakdown of fats and other lipids by hydrolysis to release Fatty Acids

AND

→ Bacteria which cause lipolysis is called lipolysis bacteria e.g. Lactobacillus,leuconostic

→ This bacteria make food containing fats rancid and food give off odour and smell and become spoil

PIGMENT FORMER BACTERIA

→ Sometimes discoloration of food occurs which lowers it aesthetic look and make it unpleasant to consume this is done by Pigment former bacteria

Examples :-

1) Flavobacterium
2) Micrococcus
3) Serratia

GAS FORMING BACTERIA

→Sometimes a canned product swells due to production of gas in it by gas former bacteria

e.g. Beverage can swells due to the activity of Clostridium Botulinum

Examples :-

1)Lactobacillus

2) Leuconostoc

3) Proteus

BENEFICIAL BACTERIAS

→ Despite of spoiling activities some bacteria have beneficial effects for use

Example :-

LAB (Lactic Acid Bacteria) produce lactic acid which is used as starter culture for the production of yoghurt from Milk

→ Proper care should be taken to prevent food spoiling bacteria to spoil food

Food Biotechnology

Module 50



Food Biotechnology

Factors Affecting Microbial Growth



FACTORS AFFECTING MICROBIAL GROWTH

There are many complex reactions and conditions that either inhibit or encourage microbial growth. The availability of oxygen, the temperature (hot or cold), light and other radiation, moisture and dryness, the activity of natural enzymes and the amount of spoilage micro-organisms that are present will all affect the growth of spoilage organisms

FACTORS AFFECTING MICROBIAL GROWTH

Factor affecting Microbial growth are

→ pH

→ Moisture

→Humidity of environment

→Temperature

→Availability of Oxygen

FACTORS AFFECTING MICROBIAL GROWTH

1) pH

→ Micro-organisms are sensitive of pH

→ Best growth at neutral pH 7

→Below pH 4 only few micro-organism can grow

→ pH < 4.6 (no spore forming bacteria can grow)

→ Bacteria are more sensitive to pH as compared to Mould and yeast

FACTORS AFFECTING MICROBIAL GROWTH

2) Water Activity :-

Water activity (aw) is a measure of the water that is available to micro-organisms

→ Pure water aw (1.0)

→Fresh fruits aw (0.97)

→ Bacteria cannot grow at aw (<0.91)

→ Yeast and moulds need less aw to grow

→ Lowest mould growth reported at aw at (0.61)

FACTORS AFFECTING MICROBIAL GROWTH

3) Humidity of the environment :-

The humidity of the environment is important as it affects the a_w of the food as well as the moisture on its surface. Food can pick up moisture from the atmosphere. Under conditions of high relative humidity storage (e.g. in a refrigerator), surface spoilage can take place, unless food is adequately protected by packaging.

FACTORS AFFECTING MICROBIAL GROWTH

4) Temperature :-

Most crucial factor for microbial growth

→ Lowest microbial growth temperature reported is -34 °C and highest is 90 ° C

→ Psychrotrophs optimum temperature(20-30 ° C)

→ Thermophiles optimum temperature (30-40 ° C)

→ Mesophiles optimum temperature is (20-30 ° C)

So if we lowers of increases temperature below or above the range of microbial growth microbial growth can be controlled

FACTORS AFFECTING MICROBIAL GROWTH

5) Oxygen availability :-

Bacteria are classified on the basis of their oxygen demand for growth e.g.

→ Aerobic bacteria (Need oxygen)

→ Anaerobic bacteria (No need of oxygen)

→ Facultative Anaerobic bacteria (survive with or without Oxgyen)

So if O_2 level controlled then microbial growth can be stopped

Food Biotechnology

Module 51



EFFECT OF pH ON MICROBIAL GROWTH

pH :-

pH refers to the hydrogen ion concentration of a solution, a measure of the solution's acidity. It is defined as the negative logarithm of the concentration of H+ ions.
$pH = -\log_{10}[H^+]$

where [H+] is the concentration of H+ ions in moles per litre.

EFFECT OF pH ON MICROBIAL GROWTH

→Different micro-organisms grows at different pH range

→so if we change the (Lower or higher) the pH of a food product below or above the growth range of micro-organisms then we can control their growth and hence preserve food

→so firstly we have to know the pH range at which micro-organisms grows easily at pH range at which their growth stops

EFFECT OF pH ON MICROBIAL GROWTH

pH and microbial growth :-

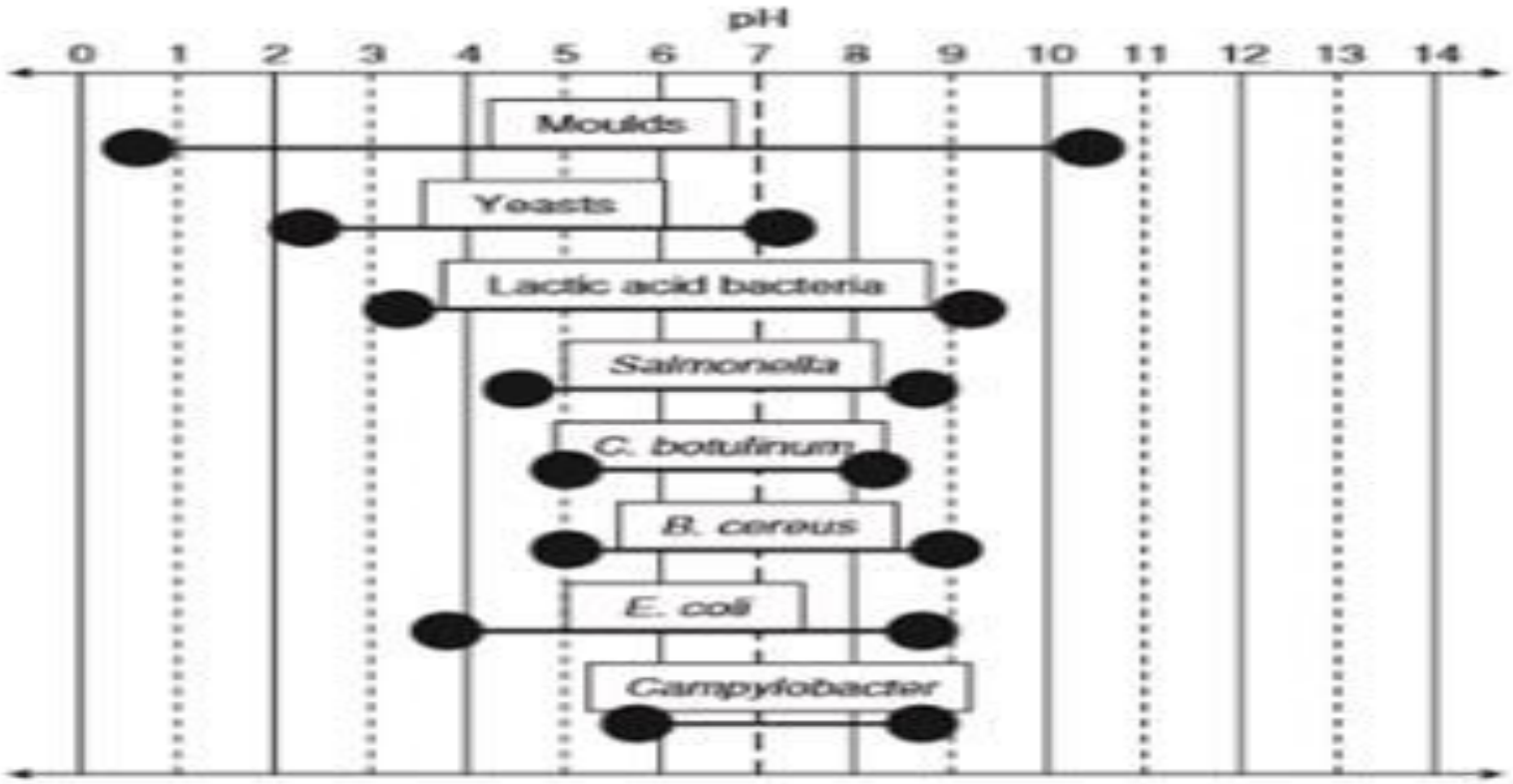
→Micro-organisms best grows at neutral pH of 7

→Below pH 4 only few micro-organism can grow

→ pH < 4.6 (no spore forming bacteria cannot grow)

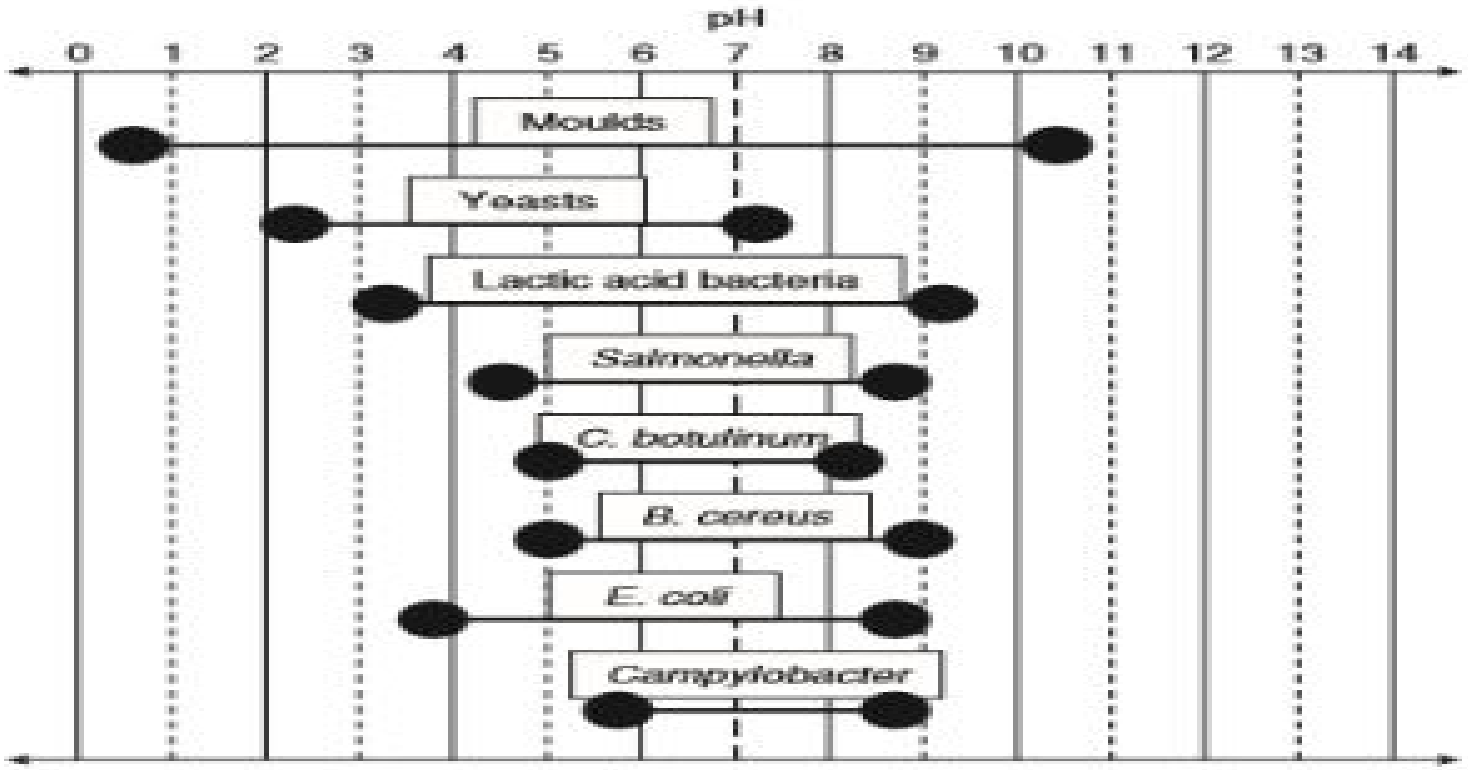
→ Bacteria are more sensitive to pH as compared to Mould and yeast

EFFECT OF pH ON MICROBIAL GROWTH



EFFECT OF pH ON MICROBIAL GROWTH

→pH growth range of selected micro-organisms



ig. 1.9 pH growth ranges for selected micro-organisms.

EFFECT OF pH ON MICROBIAL GROWTH

→ Microbial growth varies as pH varies :-

Figure 1.9 tell us that

→ Moulds can grow at (1-11 pH range)

→Yeast can grow at (2-7 pH range)

→ Bacteria can grow at (3-9 pH range)

→ So if for mould control pH above 11 , for yeast pH below 2 and above 7 , and for bacteria control pH below 3 and above 9 is set

EFFECT OF pH ON MICROBIAL GROWTH

→ Acetic Acid Fermentation is a preservative process mostly used for pickling

→ This process lowers the pH of the mixture so inhibits microbial growth

Food Biotechnology

Module 52



Food Biotechnology

Effect of Moisture for Microbial Growth

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→Micro-organisms growth greatly influence by the availability of moisture

→ If there is high moisture in a food then there are great chances of microbial growth so proper knowledge of relationship between moisture is needed in order to control microbial growth

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

Water Activity (a_w) :-

Water activity is a measure of the water that is available to micro-organisms.

→High moisture content higher will be the microbial growth hence spoilage

→ If we have to prevent microbial growth and preserve food then we have to lowers the water activity (a_w)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Classification of food on the basis of water content :-

1) Perishable Foods :-

water content (80-95 %)

e.g. Milk, Meat, Tomato

2) Semi Perishable Foods :-

water content (60-80 %)

e.g. Onion , Potato, Apple

3) Stable Foods :-

water content (10-15%)

e.g. Cereals

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Perishable foods spoil immediately (few hours) because aw is high and microbial growth is easy

→Semi perishable foods also spoil easily(few days) because of high moisture content

→ Stable foods are less prone to spoilage due to low water content and low microbial growth (months – year)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

→ Water activity and microbial growth :-

Lower the aw lower will be growth of micro-organisms

→ Pure water aw (1.0)

→ Fresh fruits aw (0.97)

→ Bacteria cannot grow at aw (<0.91)

→ Yeast and moulds need less aw to grow

→ Lowest mould growth reported at aw at (0.61)

EFFECT OF MOISTURE FOR MICROBIAL GROWTH

Methods to lowers moisture level in foods:-

→Dehydration

→Sunlight drying

→Water binding agents

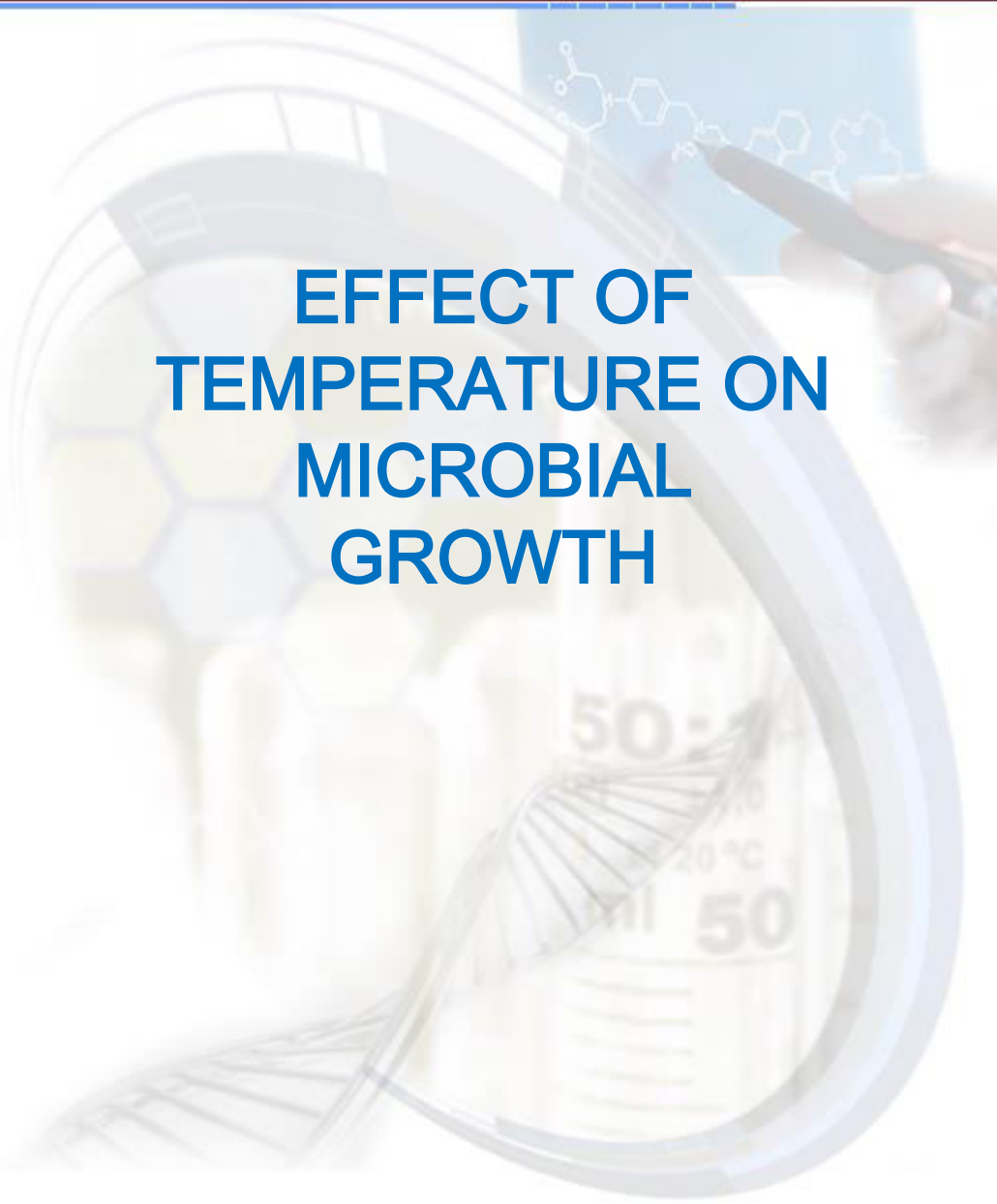
Food Biotechnology

Module 53



Food Biotechnology

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH



EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→Temperature can be considered the most important factor that affects Microbial Growth

→ Micro-organisms have been reported to grow over a wide temperature range; the lowest reported is −34° C and the highest is 90° C

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→ Types of bacteria according to temperature range

1) Thermophiles (40-55 C)

Eubacteria

2) Mesophiles (20-30 C)

Listeria

3) Psychrophiles (7-10 C)

Psychrobacter

So in order to control bacterial growth lowering and increasing temperature technique in order to stop growth of every kind of micro organism

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→Moulds are able to grow over a wide range of pH values and moisture conditions

→They can also tolerate a wider temperature range than bacteria.

→ Many moulds can grow in the refrigerator.

→ Yeasts are not usually found growing in the thermophilic temperature range, but prefer psychrotrophic and mesophilic temperature

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

→For growth control temperature should be higher than their optimum growth range temperature or extremely lowers than their optimum growth range temperature

→ But it is not suitable to store all foods in the freezer (−18° C) or even the refrigerator (<8° C), as they loose texture and other desirable features

EFFECT OF TEMPERATURE ON MICROBIAL GROWTH

Heat control methods to control Microbial growth :-

→High Temperature :-

1) Pasteurization

2) Sterilization

3) UHT

etc

→ Low Temperature :-

1) Blast freezing

2) Cryogenic freezing

etc.

Food Biotechnology

Module # 49



Food Biotechnology

Fungal Food Spoilage

Fungal Food Spoilage

Fungi are a large group of small chemoheterotrophic organisms.

They do not contain chlorophyll and therefore cannot make their own food by using sunlight.

They are, however extremely adaptable and can utilize almost any organic material.

They are yeast and molds

Fungal Food Spoilage

The conditions affecting the growth of fungi are similar to those affecting bacterial Both yeasts and molds are able to grow in an acidic environment (pH less than 7). The pH range for yeast growth is 3.5 to 4.5 and for molds is 3.5 to 8.0. The low pH of fruits is unfavourable for the growth of bacteria, but yeasts and molds can grow and cause spoilage in fruits.

Fungal Food Spoilage

Fungal growth is characterized by unicellular or multicellular filamentous hyphae, which can often be the cause of physical biodeterioration.

Fungal Food Spoilage

Moulds are responsible for significant spoilage and economic losses in the food chain.

In addition, some species belonging to the genera *Aspergillus*, *Fusarium* and *Penicillium* are mycotoxigenic

Fungal Food Spoilage

Moulds multiply by means of ascospores, zygospores or conidia. The ascospores of some moulds are particularly significant in food spoilage as they are heat resistant. Botrytis or grey molds

Rhizopus stolonifer (soft rots).

Aspergillus molds.

Fungal Food Spoilage

The presence of this diverse group of fungal secondary metabolites on feed or food presents wide fluctuation from year to year mainly because of the large number of factors affecting fungal invasion and growth.

Fungal Food Spoilage

Examples of common mycotoxins include Aflatoxin,

Ocratoxin A,

Patulin,

Fusarin,

Fumonisin, deoxynivalenol (DON) and

Zearalenone

END

Fungal Food Spoilage

The presence of this diverse group of fungal secondary metabolites on feed or food presents wide fluctuation from year to year mainly because of the large number of factors affecting fungal invasion and growth.

Fungal Food Spoilage

These contaminants may occur in a wide range of agricultural products such as cereals, fresh and dried fruits, coffee beans, cocoa, coffee and beverages such as beer and wine, while through the consumption of contaminated feed, mycotoxins can be found in meat and dairy products.

Fungal Food Spoilage

Ochratoxin A (OTA) is a secondary metabolite of several species of filamentous fungi (*Penicillium* and *Aspergillus* spp.) with nephrotoxic, immunosuppressive, teratogenic and carcinogenic effects to animals and humans.

Aflatoxin are serious safety concern.

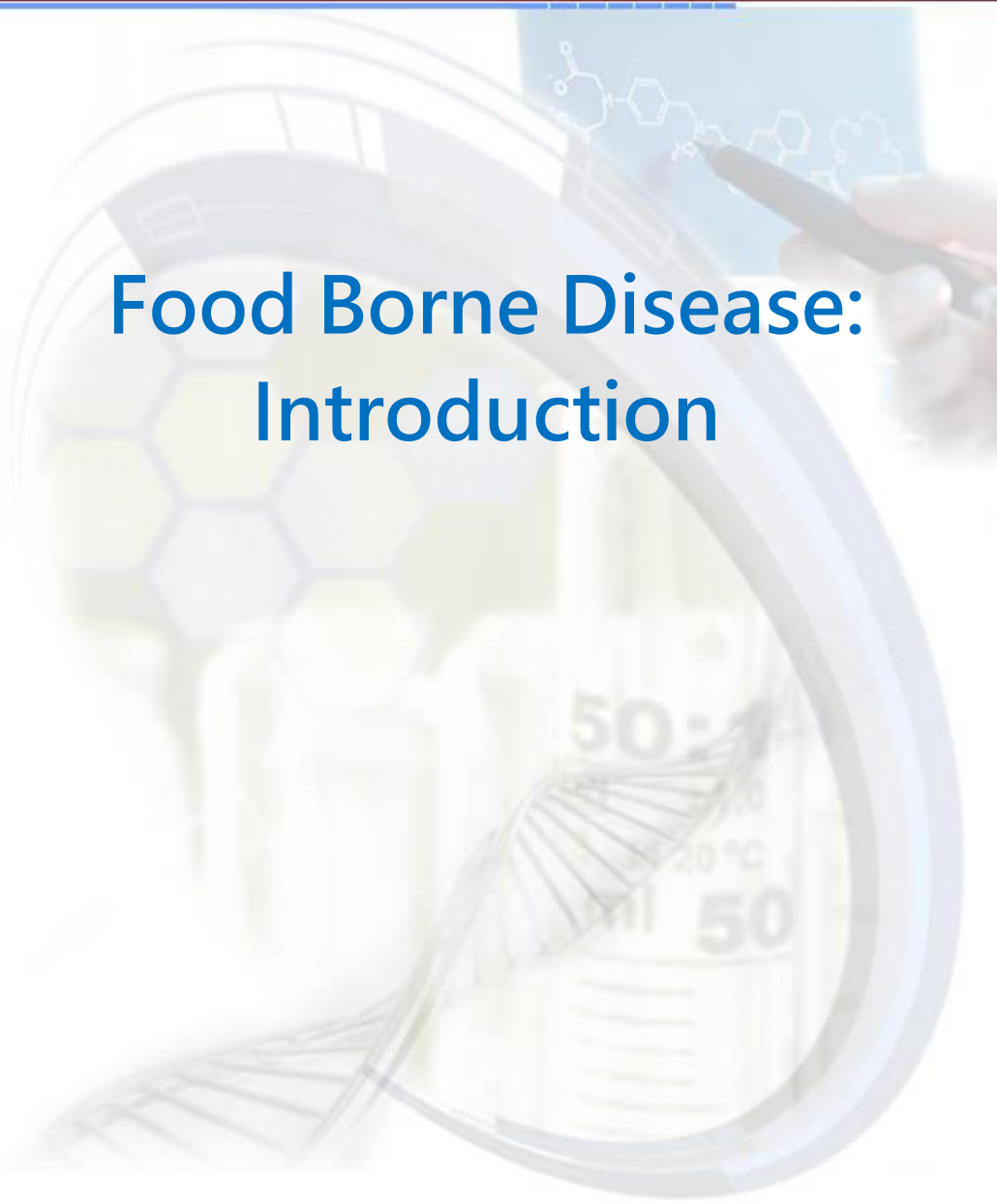
Food Biotechnology

Module # 54



Food Biotechnology

Food Borne Disease: Introduction



Food Borne Disease: Introduction

Food Borne Diseases are on the rise globally.

76 millions case, 325,000 hospitalization and 5000 deaths per annum.

different patterns of food consumption

changing consumer demand

changes in food manufacturing, retail, distribution and storage

Food Borne Disease: Introduction

emergence of new food pathogens

increasing numbers of more susceptible individuals.

Three causes

1. Biological
2. Chemical
3. Physical

Food Borne Disease: Introduction

(1). Pathogenic Bacteria:

More than forty different pathogenic bacteria are known; but Major ones incude

- *Salmonella* spp.,
- *Eschericha coli 0157:H7,*
- *L. monocytogenes,*
- *C. perfringens,*
- *C.botulinum,*
- *S. aureus,* and
- *Campylobacter jejeuni*

Food Borne Disease: Introduction

Symptoms that include headache, muscle pain, nausea, fatigue, chills or fever, stomach or abdominal pain, vomiting, and diarrhoea.

Higher risk population includes pregnant women, young children, older adults, and people with weakened immune systems.

Food Borne Disease: Introduction

The foods that are commonly involved in these food poisoning incidents include meat and poultry and their products, seafood and seafood products, egg and egg products, milk and dairy products, fruits and vegetables.

Food Borne Disease: Introduction

(2). ***Viruses***

Foods can be the medium for transmission of certain viruses. Examples of viruses that are known to be food safety hazards are the hepatitis A and E viruses, the Norwalk group of viruses, and rotavirus.

Food Borne Disease: Introduction

(3). ***Parasites***

Several human parasites can be transmitted by foods. The most common human parasites include parasitic protozoan species (e.g., *Entamoeba histolytica, Giardia lambia, Cryptosporidium parvum*), and parasitic worms (*Ascaris lumbricoides, Taenia solium, Trichinella spiralis*).

Food Borne Disease: Introduction

- Other Considerations:

1. Natuarlly occuring substances
2. Unavoidable contaminants
3. Agri residues
4. Industrial Chemicals
5. Heavy metals
6. Food Allergens.

END

Food Biotechnology

Module #55



Food Biotechnology

Plant Toxins



Plant Toxins

- A poisonous substance produced within plants
- Occurs naturally
- Secondary metabolites

Plant Toxins

- Solanine in potatoes
- Inhibits acetyl-choline esterase (a key component in the central nervous system).

•Potatoes contain about 2mg – 15mg / 100g fresh weight

•Amount higher in potatoes exposed to light (as much as 100mg/100g)

Plant Toxins

- Heat stable
- Insoluble in water
- Can cause abdominal pain and diarrhea if consumed in high amounts
- May develop signs on neurological damage
- General awareness of green potatoes is high.

Plant Toxins

- Caffeine
- Theobromine

- Pure alkaloids
- Tea, coffee, and cocoa
- Regarded as stimulants.
- Addictive in nature
- Physiological effects depend on amount consumed

Plant Toxins

Phenylethylamine :

- Found in chocolate
- Vasopressor amine
- Constrict blood vessels
- Can cause migraine in susceptible person

Plant Toxins

Hydrogen Cyanide (HCN):

- Present in tapioca
- Released from glycosides in food such as lima beans.
- Formed from amygdalin

- Varieties rich in cyanogenic glucosides need to be fermented.
- Neurological degeneration and a form of blindness

Plant Toxins

- Legumes contain inhibitors of trypsin and chymotrypsin
- Mostly inactivated by cooking
- Soy beans need to be autoclaved

Plant Toxins

- Some mushrooms can be severely toxic
- Can be fatal
- Snakeroot poisoning

Plant Toxins

- Lectins
- Haemoglutinins

- Present in kidney beans
- Bind to surface of RBC
- Causes clumping

- Toxic: 0.5mg/kg

Plant Toxins

Myristicin :

- Significant in nutmeg
- Lesser amounts in black pepper, carrot and celery
- 10 mg nutmeg can cause euphoria, hallucinations and narcosis
- Nutmeg used in very small amount for flavoring: induces sleep

Plant Toxins

- Most plant toxins are inhibited during cooking
- Some exceptions remain
- Consuming raw food increases chances of toxicity
- Must ensure raw food safety before consuming

END

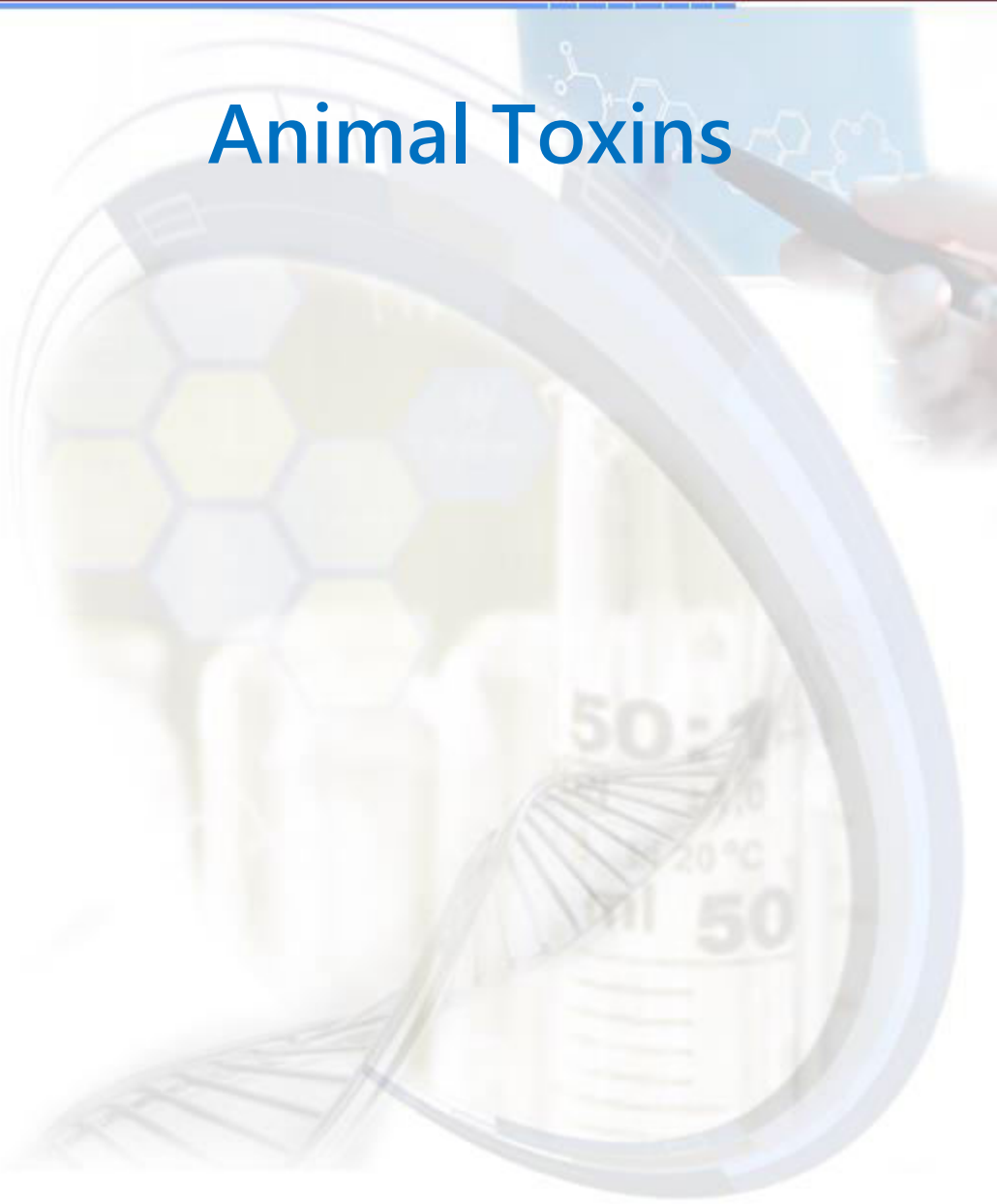
Food Biotechnology

Module #56



Food Biotechnology

Animal Toxins



Animal Toxins

- A poisonous substance produced within animals
- Produced during metabolism and growth
- Occur naturally

Animal Toxins

Tetrodotoxin :

- Minimum lethal dose: 1.5 mg - 4.0 mg
- Occurs in puffer fish(in liver and ovaries)
- Muscle and testes of puffer fish delicacy in Japan
- Highly skilled food handlers

Animal Toxins

- Block movement of Na+ across membranes of nerve fiber
- Disrupts transmission of electrical impulses
- Can lead to total paralysis and respiratory failure
- Death within 6-24 hours
- No effective treatment

Animal Toxins

- Other animal toxins

- Present in shell fish
- Such as clams, scallops
- Toxin produced due to some planktons

Animal Toxins

- Mussels and Clams
- Poisonous alkaloid
- Can cause food poisoning symptoms

Animal Toxins

- **Scrombridae**
- Tuna, Sardines, Mackerel
- Held above 10° C for some hours
- Formation of high levels of histamine

Animal Toxins

- Tryptophan converted to histamine in muscles
- Catalyzed by decarboxylase produced by bacteria in fish
- High levels before putrefaction

Animal Toxins

- Headaches, palpitations, gastrointestinal upset, skin flushes and erethrema
- Anti-histamine drugs used to treat

Animal Toxin

- Awareness of symptoms
- Residents near areas with such poisonous plant/animal usually guard against it

END

Food Biotechnology

Module #57



Food Biotechnology

Agricultural Residues

Agricultural Residues

- Residues of pesticides
- May be toxic
- Can make their way into food

Agricultural Residues

- Dichlorodiphenyltrichloro ethane (DDT)
- Pesticide
- Used for killing mosquitoes
- Found in food in very high proportions

Agricultural Residues

- Other pesticide residue found in food:

- Dieldrin
- Lindane (BHC)
- Malathion

- Used as spray on fruits, vegetables

Agricultural Residues

- Many of these pesticides have been banned
- Used for many years
- Still present in soil
- Finds its way into human food through crops

Agricultural Residues

Sterols :

- Source of residue
- Used for increased growth in animals
- Mostly for meat operations

Agricultural Residues

- Some sterols have been shown to be carcinogenic
- Residue of sterols may be present in meat

Agricultural Residues

Antibiotics :

- Can be a source of residue
- Used as growth promoters
- Used to combat diseased in animals

Agricultural Residues

- Mastitis in cows treated with antibiotics
- Milk from these cows should be discarded
- Not always the case
- Penicillin in milk can cause allergic reactions to susceptible persons

END

Food Biotechnology

Module #58



Food Biotechnology

Poisoning by Chemicals

Poisoning by Chemicals

- Toxic metals are a major source of chemical poisoning
- Use of contaminated water
- Soil on vegetables poorly washed
- Water pipes (lead)
- Equipment/Containers

Poisoning by Chemicals

Lead :

- Pipes and storage tank
- Soft water more susceptible
- Lead based solder in canning

Poisoning by Chemicals

- Most food subject to regulations of lead content
- Lead poisoning uncommon
- Absorption of lead in children is higher than of adults

Poisoning by Chemicals

Chronic Lead Poisoning

- Symptoms include:

- Anemia
- Poor I.Q performance
- Poor attention span

Poisoning by Chemicals

- Food and water intake contribute to daily lead intake
- Exhaust fumes can be an additional source
- Children from congested cities more susceptible

Poisoning by Chemicals

Mercury :

- Cereal grains treated with antifungal mercury compounds
- Industrial pollution of coast water
- Seafood contaminated with alkyl mercury compounds

Poisoning by Chemicals

- Symptoms of mercury poisoning are variable
- All point to damage to CNS
- Children more sensitive to exposure

Poisoning by Chemicals

Arsenic :

- Widely distributed in nature
- May contaminate chemicals used in food processing
- Important to check for arsenic contamination in food additives

Poisoning by Chemicals

Sodium phosphate :

- Used as stabilizer in milk powder
- Contaminated with arsenic trioxide
- Poisoning of 12,000 infants in Japan

Poisoning by Chemicals

Cadmium :

- Cadmium poisoning come from three sources

- Contaminated water
- Cadmium plated components in food processing machinery
- Zinc plating of containers used to store acidic food

Poisoning by Chemicals

- Lead, Mercury, Arsenic, Cadmium
- Bivalent in nature

- If diet is low in bivalent nutrient metals (Ca, Fe), these contaminants might enter vulnerable systems such as
- Nervous system, kidneys, bone marrow

END

Food Biotechnology

Module #59



Food Biotechnology

Poisoning by microorganisms

Poisoning by microorganisms

- Poisoning is caused by eating contaminated food by the the microorganisms.
- There are two chief kinds of food poisoning caused by bacteria:

1. Botulism
2. Staphylococcus poisoning

Poisoning by microorganisms

Botulism

Poisoning by microorganisms

- It is caused by the presence of toxin in the food
- Foods such as corn, beans, peas, meat, fish are likely to be contaminated.
- Some strains of this organism produce a toxin in the food, which is a potent poison.

Poisoning by microorganisms

- It can cause paralysis of involuntary muscles and may ultimately lead to death due to respiratory failure.
- Toxin can be destroyed by heating.
- Likely to occur if neutral foods are inadequately processed.

Poisoning by microorganisms

Staphylococcus poisoning

Poisoning by microorganisms

- It is caused by staphylococcus aureus.
- Staphylococcus contamination of the food may either be from human or animal sources.
- Nasal passages of many humans, boils and infected wounds are potential sources.

Poisoning by microorganisms

- Strains of *Staphylococcus aureus* produce an enterotoxin.
- It can cause:

1. Salivation
2. Nausea
3. Vomiting
4. Abdominal cramps
5. Diarrhoea.

Poisoning by microorganisms

- The time between the consumption of the food and the appearance of symptoms can range from 1–6 hours.
- Recovery takes about a day or two and mortality is very low.

Poisoning by microorganisms

END

- Employees suffering from *staphylococcal* infections such as colds, boils should not be allowed to handle foods.
- Prevention by refrigeration of foods immediately after making.
- Pasteurization kills all the *staphylococci* that may be present in foods.

Food Biotechnology

Module #60



Food Biotechnology

Food Infections



Food Infections

- Foodborne infection occurs by the consumption of living organisms.
- Symptoms do not appear for at least one day after ingestion.
- There is a vast number of different bacteria involved.

Food Infections

Identification of different pathogens related to food

Food Infections

- *Clostridium botulinum* is a Gram-positive, obligate anaerobe, sporeforming bacterium.
- The botulinum toxin is one of the most potent neurotoxins known which can even cause death.
- Food contaminated with only a few nano grams (10^{-9} ng) can be toxic.

Food Infections

There are five types of botulism:

1. Food-borne botulism
2. Wound botulism
3. Infant botulism
4. Adult infectious botulism
5. Inadvertent botulism

Food Infections

- **Salmonella** is a Gram-negative, rod shaped, motile bacterium, that does not form spores.
- Its is found in fresh and salt water, soil and animal faeces.
- It causes salmonellosis infection and even extremely low doses can cause death.

Food Infections

- *Listeria monocytogens:* are Gram-positive, motile bacteria. They do not form spores, but are quite resistant to desiccation and heat.
- Found in the soil and associated with birds and animals
- Cause disease called listeriosis, which can result in septicaemia, meningitis encephalitis and spontaneous abortions in pregnant females.

Food Infections

- *S. aureus* is a Gram-positive, small round bacterium (coccus), that can produce a heat-stable toxin.
- *S. aureus* is found all over most commonly via humans.
- It is found in the nasal passages, throat, on the skin and in the hair of more than half of healthy people.

Food Infections

- *Escherichia coli* are Gram-negative bacteria found in the intestines of all mammals.
- cause four different classes of illnesses.
- Outbreaks are associated with fecal contamination of water by food handlers, or are due to unprocessed or under-processed foods.

Food Infections

END

- *Campylobacter jejuni* are Gram-negative, microaerophilic, motile, rod- shaped bacteria.
- They are found on raw chickens, in raw milk, and are often carried in the intestines of healthy cattle.
- They can contaminate food via flies on farms and are also found in non chlorinated water.

Food Biotechnology

Module # 61



Food Biotechnology

Sequence of Events in Food Borne Diseases

Sequence of Events in Food Borne Diseases

A sequence of events happens in all food borne diseases.

An understanding of these sequences is helpful in investigating the cause (the source and means of transmission) of a food borne disease. It also helps in recognizing how the sequence can be broken.

Sequence of Events in Food Borne Diseases

1. Initially, there has to be a source of a pathogen.

2. the pathogen has to contaminate a food.

3. Consumption of the food contaminated with a pathogenic virus or parasites

4.It lead to viral or parasitic infection.

Sequence of Events in Food Borne Diseases

For bacterial pathogens (and toxicogenic molds) the contaminated food has to support growth and be exposed for a certain period of time at a suitable temperature to enable the pathogens to grow to produce toxins.

Sequence of Events in Food Borne Diseases

However for some potent pathogens (such as *Esc. coli* 0157:H7) growth may not be necessary to cause a food borne infection.

For intoxication the growth should reach a sufficient level to produce enough toxins so that when the food is consumed, the individual develops the symptoms

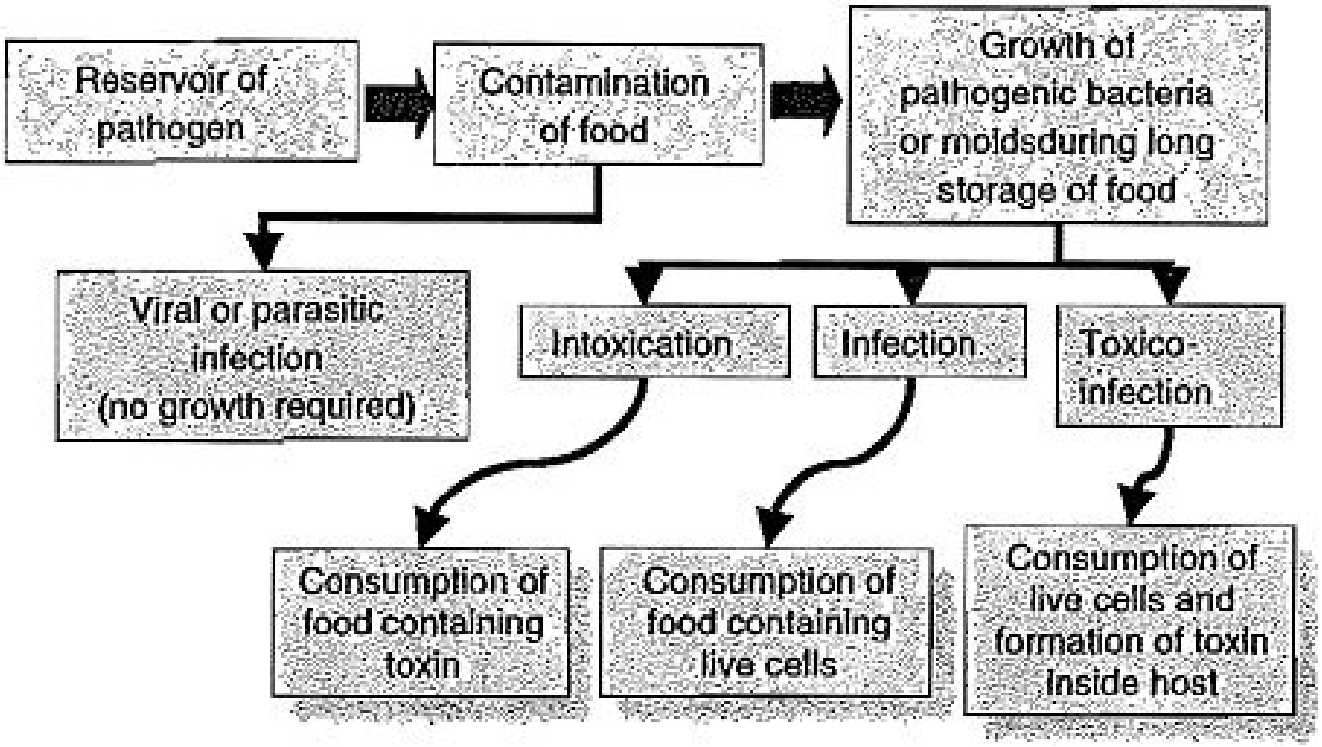
Sequence of Events in Food Borne Diseases

For bacterial infection, viable cells of a pathogen need to be consumed in sufficient numbers, which vary greatly with pathogens, to survive stomach acidity, establish in the digestive tract, and cause illness.

Sequence of Events in Food Borne Diseases

In case of toxicoinfection, viable cells should be consumed either in very high numbers (for those that cannot multiply in the digestive tract, such as *Clo. peifringens)* or in reasonable numbers (for those that can multiply in the digestive tract, such as *Vibrio* cholerae) so that toxins released by them in the digestive tract can produce the symptoms.

Sequence of Events in Food Borne Diseases



Food Biotechnology

Module # 62



Food Biotechnology

Detection of *E.Coli* in Drinking Water

Detection of *E.Coli* in Drinking Water

The Most Probable Number (MPN) method is applicable to the enumeration of coliforms, faecal coliforms and aerogenic *Escherichia coli* in foods and water, including contact water from food manufacturing plants.

Detection of *E.Coli* in Drinking Water

MPN is not intended to be used to isolate and enumerate *E. coli* serotypes associated with human illness, particularly the enterohemorrhagic serotype O157:H7. Many of the pathogenic serotypes do not give a positive faecal coliform reaction.

Detection of *E.Coli* in Drinking Water

Description The MPN procedure involves a multiple tube fermentation technique where three or more decimal dilutions of the sample are inoculated into tubes of broth medium and incubated at a specific temperature and for a specific time.

Detection of *E.Coli* in Drinking Water

The method is progressive; i.e., first determining the presence of coliforms in the tubes, then determining if these tubes also contain faecal coliforms, and then confirming whether *E. coli* is present. Based on the number of tubes indicating the presence / absence of the three groups of organisms, MPN is established

Detection of *E.Coli* in Drinking Water

3. Principle

The terms “coliform” and “faecal coliform” have no taxonomic validity and, therefore, are only meaningful when expressed in terms of the analytical test parameters of medium, time and temperature of incubation. Coliforms, faecal coliforms, and *E. coli* are considered “indicator organisms.”

Detection of *E.Coli* in Drinking Water

Which indicate one of the following possibilities:

1. inadequate processing and/or post-processing contamination; and/or
2. microbial growth. The presence of faecal coliforms and *E. coli* may indicate faecal contamination.

Detection of *E.Coli* in Drinking Water

Briefly, this method involves serially diluting out the target organisms in the sample, in 5-replicate aliquots, to extinction. The probable level of the target organisms is then statistically estimated from an MPN table.

Detection of *E.Coli* in Drinking Water

Gas production is used as an indication of ability to ferment lactose from Lauryl Sulfate Tryptose (LST) broth (presumptive coliform test); gas production from Brilliant Green Lactose Bile (BGLB) broth is considered confirmation of coliform presence.

Detection of *E.Coli* in Drinking Water

gas production at 44.5 or 45° C from EC broth is used as confirmation of faecal coliform presence; and appearance of typical nucleated, dark-centred colonies with or without metallic sheen when positive EC broths are streaked onto Levine's Eosin Methylene Blue (L-EMB) agar are indicative of *E. coli.* It must be confirmed by further biochemical tests

Food Biotechnology

Module # 63



Food Biotechnology

Food Preservation Methods

Food Preservation Methods

Inactivation, inhibition, and methods of avoiding recontamination

Food Preservation Methods

Early Methods of Preservation may include: Using chemicals, e.g. acid and salt:

- pickling in vinegar;
- salting meat;
- sugar to make jam;
- storing in alcohol.

Temperature control:

- sun drying fruit;
- using ice blocks.

Food Preservation Methods

Food technology advanced in the early 19th century with the invention of the canning process.

Since then the range of technologies used to process and preserve foods has expanded and uses a range of physical and chemical techniques

Food Preservation Methods

There have also been many technological changes in the home.

- refrigerators became widely used from the mid-1950s;
- domestic freezers in the 1970s;
- microwave ovens in the 1980s.
- Frozen and ready meals followed these.

Food Preservation Methods

Advances in Technology:

New technologies have given us a greater range of methods to package and store foods. This enables the preservation of nutrients and the extension of food product shelf-life.
Demand has increased for minimally processed foods which retain their freshness.

Food Preservation Methods

Effort By: AARISH ALI

MAP Technologies:

Modified atmosphere packaging is a way of extending the shelf life of fresh food products. The technology substitutes the air inside a package with a protective gas mix.

New Packaging technologies are reducing food wastes.

Food Preservation Methods

Nano particles of silver have anti-bacterial properties and can be used on:

- kitchen tools and tableware;
- preparation surfaces;
- plastic food storage containers;
- plastic wrap.

Food Preservation Methods

Encapsulation Technologies:

Encapsulation is "*a process where a continuous thin coating is formed around solid particles, liquid droplets or gas cells that are fully contained within the capsule wall* .

Food Preservation Methods

High Pressure Processing is a cold pasteurisation technique. Food, previously sealed in flexible and water-resistant packaging, is subjected to a high level of hydrostatic pressure (pressure transmitted by water) for a few seconds to a few minutes

Food Biotechnology

Module # 64



Food Biotechnology

Need for Food Preservation

Need for Food Preservation

The main reasons for food preservation are :

To overcome inappropriate planning in agriculture.

To overcome spoilage factors.

Need for Food Preservation

produce value-added products,

and provide variation in diet.

Food Security and food wastage

Need for Food Preservation

In food preservation, the important points that need to be considered are

- The desired level of quality
- The preservation length
- The group for whom the products are preserved

Need for Food Preservation

Detection of

E.Coli

Food Biotechnology

Module # 65



Food Biotechnology

Blanching

Blanching

Blanching is steam treatment of vegetables and some fruits to fix colour and stop enzyme activities.
Its only pre treatment.
It can be followed by heat sterilisation, dehydration and freezing.
also combined with peeling and/or cleaning of food (Chapter 3), to achieve savings in energy consumption, space and equipment costs.

Blanching

Onions and green chillies are exceptions but others suffer quality greatly if blanching is skipped.
The factors which influence blanching time are:

- type of fruit or vegetable
- size of the pieces of food
- blanching temperature
- method of heating.

Blanching

Freezing and dehydration temperatures may not inactivate enzymes properly.

In case of canning sterilization temperature may also not be sufficient to warrant blanching.

Blanching

Under blanching is equally bad because of accelerated enzyme activity.
lipoxygenase, polyphenoloxidase, polygalacturonase and chlorophyllase reduce eating and nutritional quality.

Blanching

catalase and peroxidase are found in most vegetable and heat resistant. These are used as marker enzymes particularly peroxidase.

Blanching reduces contaminating microbes helping further processing operations such as sterilization and freezing.

Blanching

Hot water bath and saturated steam chambers are routinely used.

Microwaves are also a possibility.

Need to consider effluent impact by reducing soluble components from food.

Cooling is critical for yield of final product and is efficient for steaming.

END

Food Biotechnology

Module #66



Pasteurization

- Pasteurisation is a relatively mild heat treatment, in which food is heated to below 100°C.
- In low acid foods, for example milk) it is used to minimise pathogenic micro-organisms and to extend the shelf for days.
- In acidic foods is used to minimise pathogens, and to extend the shelf life for several months.

Pasteurization

- Two types of processes are used
- 1. LTLT: 63ºCfor 30 m
- 2. HTST: 71.8ºC for 15 seconds or

88ºC for 1 s, 94ºC for 0.1 s or 100 ºC for 0.01 s for milk.

Other foods are Beer, Eggs, Fruit juices, milk, ice cream etc.

Pasteurization

Pasteurization of packaged foods:

- Some liquids like beer and fruit juices are pasteurized after filling into containers.
- Hot water is normally used if the food is packaged in glass, to reduce risk of thermal shock to the container.

Pasteurization

Metal or plastic containers are processed using steam–air mixtures or hot water as there is little risk of thermal shock.

Pasteurization

Pasteurization of unpackaged foods:

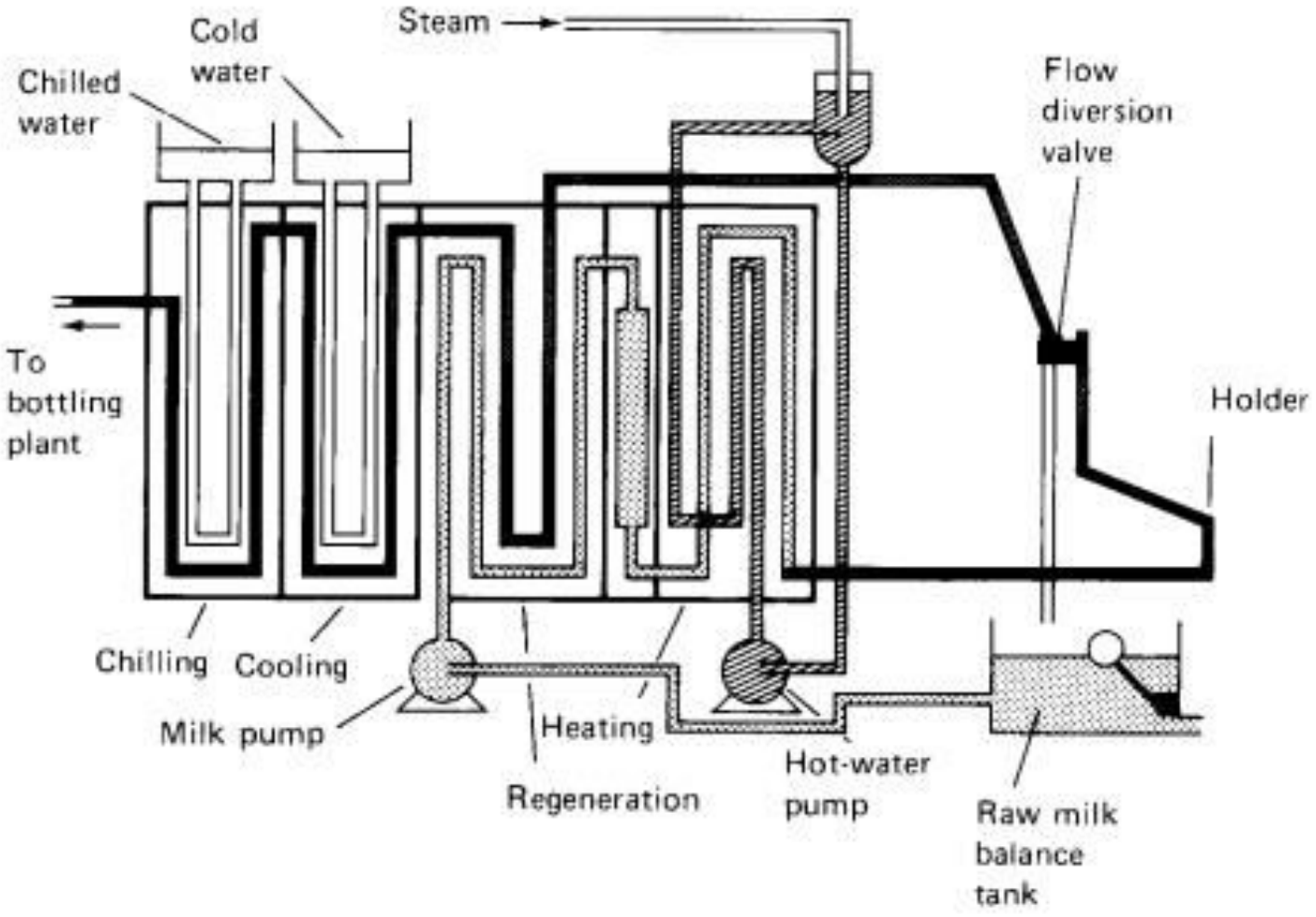
- Swept surface heat exchangers or open boiling pans for small scale batch pasteurization.
- Low viscosity liquids (for example milk, milk products, fruit juices, liquid egg, beers and wines) usually employs plate heat exchangers.

Pasteurization

Plate heat exchanger:

- Consists of a series of thin vertical stainless steel plates, held tightly together in a metal frame.
- The plates form parallel channels.
- Liquid food and heating medium are pumped through alternate channels usually in a counter-current flow pattern.

Pasteurization



Pasteurization

Other heat exchanger:

- Concentric tube heat exchanger is suitable for more viscous foods and is used with dairy products, mayonnaise, tomato ketchup and baby foods.
- Consists of a number of concentric stainless steel coils, each made from double/triple-walled tube.
- Liquid food is passed from one coil to the next for heating and cooling.

Pasteurization

- Pasteurised food is immediately filled into cartons or bottles.
- Significant levels of spoilage and risks from pathogens can arise from post-pasteurisation contamination.
- Particularly when foods (for example milk) are not re-heated before consumption.

Pasteurization: Effect on foods

- Minor changes to the nutritional and sensory characteristics of most foods.
- The shelf life of pasteurised foods is usually only extended by a few days or weeks compared with many months with the more severe heat sterilisation.

Pasteurization: Effect on foods

Colour, flavour and aroma:

- In fruit juices enzymic browning by polyphenoloxidase.
- Other pigments in plant and animal products are also mostly unaffected.
- A small loss of volatile aroma compounds of juices causes a reduction in quality.

Pasteurization: Effect on foods

Vitamin Loss:

- In fruit juices losses of vitamin C and carotene are minimised by deaeration.
- Changes to milk are confined to a 5% loss of serum proteins and small changes to the vitamin content

Pasteurization: Effect on foods

Colour, flavour and aroma:

- In fruit juices enzymic browning by polyphenoloxidase.
- Other pigments in plant and animal products are also mostly unaffected by pasteurisation.
- A small loss of volatile aroma compounds during pasteurisation of juices causes a reduction in quality unmask other 'cooked'

Food Biotechnology

Module #67



Heat Sterilization

Heat sterilisation :

foods are heated at a sufficiently high temperature and for a sufficiently long time to destroy microbial and enzyme activity.

Sterilised foods have a shelf life in excess of six months at ambient temperatures.

Heat Sterilization

The length of time required to sterilise a food is influenced by:

- the heat resistance of micro-organisms or enzymes likely to be present in the food
- the heating conditions
- the pH of the food
- the size of the container
- the physical state of the food.

Heat Sterilization

Heat resistance of micro-organisms

- Most heat resistant spores have a *z* value of around 10°C. This temperature rise will result in a ten-fold reduction in processing time needed to achieve the same lethality.

Heat Sterilization

- In low-acid foods (pH 4.5), the heat resistant, spore forming micro-organism, *Clostridium botulinum* is the most dangerous pathogen likely to be present. It can cause lethality up to 65% cases by toxin botulin in anaerobic situations.

Heat Sterilization

- Thermal destruction of micro-organisms takes place logarithmically and a sterile product cannot therefore be produced with certainty no matter how long the process time.
- Commercial sterility means in practice that the heat processing inactivates substantially all micro-organisms and spores with potential to grow in storage conditions

Heat Sterilization

A 12*D* process is used when *C. botulinum* is likely to be present in low acid foods, but in foods that contain more heat-resistant spoilage microorganisms it would result in over processing so 2D to 8D process are used for an economical operation and quality

Prior care to avoid contamination is helpful.

Heat Sterilization

Heating Methods:

- Plate heat exchangers.
- Concentric tube Consists of a number of concentric stainless steel coils, each made from double/triple-walled tube.

.

Heat Sterilization

Rate of heat penetration:

- Type of product
- Size of container
- Agitation of the container
- Temperature of the retort
- Shape of the container
- Type of container

END

Food Biotechnology

Module #68



UHT Process

- Also termed as *aseptic processing.*
- *Ultra-high temperature processing (UHT), ultra-heat treatment*, or *ultra-pasteurization* is a food processing technology that sterilizes liquid food, chiefly milk, by heating it above 135 °C (275 °F) – the temperature required to kill spores in milk – for 1 to 2 seconds.

UHT Process

- It is used to sterilise a wide range of liquid foods, including milk, fruit juices and concentrates, cream, yoghurt, wine, salad dressing, egg and ice cream mix
- Possible if the product is sterilised before it is filled into pre-sterilised containers in a sterile atmosphere.

UHT Process

Can also be used to process foods which contain small discrete particles, for example

- cottage cheese
- baby foods,
- tomato products,
- fruit and vegetables,
- soups and rice desserts.

UHT has an important additional advantage of a shelf life of at least six months at ambient temp.

UHT Process

- UHT processing is independent of container size
- Conventional retorting of A2 cans of vegetable soup requires 70 min at 121°C to achieve an F_0 value of 7min followed by 50 min cooling,
- aseptic processing in a scraped-surface heat exchanger at 140°C for 5 s gives an F_0 value of 9 min.

UHT Process

Increasing the can size to A10 increases the processing time to 218 min, whereas with aseptic processing the sterilisation time is the same. So very big containers can be used for sterilization unlike canning operations.

UHT Process

A theoretically ideal UHT process would heat the product instantly to the required temperature, hold it at that temperature to achieve sterility and cool it instantly to filling temperature.

In practice the degree to which this is achieved depends;

- part on the method used to heat the food
- on the properties of the food

UHT Process

Equipment used for UHT processing has the following characteristics:

- operation above 132°C
- exposure of a relatively small volume of product
- maintenance of turbulence
- use of pumps
- exchanger
- constant cleaning

UHT Process

Equipments are according to the method of heating into:

- direct systems (steam injection and steam infusion)
- indirect systems (plate heat exchangers, tubular heat exchangers (concentric tube or shell-and-tube) and scraped surface heat exchangers)
- other systems (microwave, dielectric, ohmic and induction heating

UHT Process

The main limitations of UHT processing are the

- Cost
- complexity of the plant,

which arises from the necessity to

- sterilise packaging materials,
- pipework/tanks,
- maintenance of sterile air and surfaces in filling machines,
- Higher skill levels required by operators and staff

End

Food Biotechnology

Module #69



Effects of heating on food

The purpose of heat sterilisation is

- to extend the shelf life of foods for storage at ambient temperatures,
- while minimising the changes in nutritional value and eating quality.

Effects of heating on food

Differences between the *D* and *z* values of micro-organisms, enzymes and sensory or nutritional components of foods are exploited to optimise processes for the retention of nutritional and sensory qualities

Effects of heating on food

COLOUR

- In meats the red oxymyoglobin pigment is converted to brown metmyoglobin, and purplish myoglobin is converted to red–brown myohaemichromogen
- Maillard browning and caramelisation also contribute to the colour of sterilised meats.

Effects of heating on food

COLOUR

- Chlorophyll is converted to pheophytin and anthocyanins are degraded to brown pigments.
- Iron or tin react with anthocyanins to form a purple pigment, or when colourless leucoanthocyanins form pink anthocyanin in pears and quinces.

Effects of heating on food

COLOUR

- In sterilised milk slight colour changes are due to caramelisation, Maillard browning and changes in the reflectivity of casein micelles.

Effects of heating on food

Flavor and Aroma

- In canned meats there are complex changes, like pyrolysis, deamination and decarboxylation of amino acids, degradation, Maillard reactions and caramelisation of carbohydrates to furfural and hydroxymethylfurfural, and oxidation and decarboxylation of lipids.

Effects of heating on food

Flavor and Aroma

- In fruits and vegetables, changes are due to complex reactions which involve the degradation, recombination and volatilisation of aldehydes, ketones, sugars, lactones, amino acids and organic acids
- Interactions between these components produce more than 600 flavour compounds in ten chemical classes.

Effects of heating on food

Flavour and Aroma

- In milk the development of a cooked flavour is due to denaturation of whey proteins to form hydrogen sulphide and the formation of lactones and methyl ketones from lipids.
- In aseptically sterilised foods the changes are better retained.

Effects of heating on food

Texture or Viscosity

- In canned meats, changes in texture are caused by coagulation and a loss of water holding capacity of proteins, which produces shrinkage and stiffening of muscle tissues.
- Softening is caused by hydrolysis of collagen, solubilisation of the resulting gelatin, and melting and dispersion of fats through the product.

Effects of heating on food

Texture or Viscosity

- Polyphosphates are added to increase tenderness.

Effects of heating on food

Texture or Viscosity

- In fruits and vegetables, softening is caused by hydrolysis of pectic materials, gelatinisation of starches and partial solubilisation of hemicelluloses.
- Calcium salts added increase the firmness of the canned products.
- Different salts are needed for different type of fruits.

Effects of heating on food

Nutritional Value

- Canning causes the hydrolysis of carbohydrates and lipid but the nutritional value is not affected.
- Proteins are coagulated and, in canned meats, losses of amino acids are 10–20%.
- Negligible vitamin losses in aseptically processed milk and lipids, carbohydrates and minerals are virtually unaffected..

Effects of heating on food

NUTRITIONAL VALUE

- Nutrient losses also occur during periods of prolonged storage.

Food Biotechnology

Module #70



Evaporation

- Evaporation, or concentration by boiling, is the partial removal of water from liquid foods by boiling off water vapour.
- It increases the solids content of a food and hence preserves it by a reduction in water activity.
- Evaporation is used to pre-concentrate foods (for example fruit juice, milk and coffee) to reduce their weight.

Evaporation

- During evaporation, sensible heat is transferred from steam to the food, to raise the temperature to its boiling point.
- Latent heat of vaporisation is then supplied by the steam to form bubbles of vapour, which leave the surface of the boiling liquid.

Evaporation

- Heat and mass balances are used to calculate the degree of concentration, energy use and processing times in an evaporator.

Evaporation

Factors influencing the rate of heat transfer:

- Temperature difference between the steam and boiling liquid
- Deposits on heat transfer surfaces.
- Boundary films

Evaporation

Factors influencing the economics of evaporation:

- Vapour recompression
- Preheating
- Multiple effect evaporation

Evaporation

Evaporators consist of:

- a heat exchanger (termed a *calandria*) which transfers heat from steam to the food
- a means of separating the vapours produced
- a mechanical or steam ejector vacuum pump. Mechanical pumps have lower operating costs but higher capital costs than steam ejector pumps.

Evaporation

Natural circulation evaporators

- *Open- or closed-pan evaporators* are hemispherical pans, heated directly by gas or electrical resistance.
- For vacuum operation, they are fitted with a lid. A stirrer or paddle is used to increase the rate of heat transfer and to prevent food from burning onto the pan.

Evaporation

Natural circulation evaporators

- *Long-tube evaporators* consist of a vertical bundle of tubes, each up to 5 cm in diameter, contained within a steam shell 3–15 m high.
- Liquor is heated almost to boiling point before entering the evaporator.
- It is then further heated inside the tubes and boiling commences.

Evaporation

Natural circulation evaporators

- The *short-tube evaporator* is an example of a tube-and-shell heat exchanger, also used in pasteurisation and heat sterilisation.
- It consists of a vessel (or shell) which contains a vertical, or less commonly horizontal bundle of tubes.

Evaporation

Forced circulation evaporators:

In forced-circulation evaporators a pump or scraper assembly moves the liquor, usually in thin layers, and thus maintains high heat transfer rates and short residence times.

- **Plate evaporators.**
- **The expanding flow evaporators.**

Evaporation

Effects on Foods

- Aroma compounds that are more volatile than water are thus lost during evaporation.
- This reduces the sensory characteristics of most concentrates;
- In fruit juices this results in a loss of flavour, although in some foods the loss of unpleasant volatiles improves the product quality.

Evaporation

Effects on Foods

- Some volatiles can be recovered and retained in the product:
- by volatile recovery by vapour condensation and fractional distillation
- by stripping volatiles from the feed liquor with inert gas and adding them back after evaporation

Food Biotechnology

Module #71



Distillation

- Distillation is separating the components or substances from a liquid mixture by using selective boiling and condensation.
- May be complete or partial separation that increases the concentration of selected components in the mixture.

Distillation

- Routine unit operation in chemical and food industry.
- Alcoholic spirits and
- Separation of volatile flavour and aroma compounds (for example, production of essential oils by steam distillation).

Distillation

- Some food contains components having different degrees of volatility.
- When heated, that have a higher vapour pressure (more volatile components) are separated first.
- These are termed the ‘distillate’ and components that have a lower volatility are termed ‘bottoms’ or residues.

Distillation

- Some industries perform batch distillation (in pots) for whiskey and other spirit distilleries.
- Most industrial distillation operations use more economical continuous distillation columns

Distillation

To enhance the separation of some components:

- a proportion of the distillate is added back to the top of the column called (reflux)
- a portion of the bottoms is vapourised in a reboiler and added to the bottom of the column

Distillation

Use of spinning cone column:

- To remove the volatile components from liquids
- It is used to recover flavours from beer, coffee, tomato products and fruit juices.
- To produce low-alcohol wines and beers
- To remove off-flavours.

Distillation

Equipment:

- A column containing a series of rotating inverted cones, which are intermeshed with stationary cones attached to the column wall.
- Steam or nitrogen is supplied to the base of the column and the feed liquor enters at the top.

Distillation

Equipment:

- Thin turbulent films are produced over the large surface area of the cones and rapid separation takes place.
- The gas passes out of the top of the column and volatile aroma compound are condensed and collected.

Distillation

Equipment:

- Separation is achieved by mechanical energy, rather than heat, there is less damage to flavours and lower energy consumption.

end

Food Biotechnology

Module #72



Dehydration

- ‘The application of heat under controlled conditions to remove the majority of the water by evaporation’ (or in the case of freeze drying by sublimation).
- Purpose of dehydration is to extend the shelf life of foods.

Dehydration

Large number of factors control the rate at which foods dry, which can be grouped into the following categories:

- those related to the processing conditions
- those related to the nature of the food
- those related to the drier design.

Dehydration

Drying using heated air:

Heat from drying air is absorbed by food and provides the latent heat needed to evaporate water from the surface.

There are three inter-related factors controling capacity of air to remove moisture:

- water vapour already carried by the air
- the air temperature
- the amount of air that passes over the food.

Dehydration

Drying using heated surfaces:

- Heat is conducted from the hot surface, through the food, and moisture is evaporated from the exposed surface.
- Slurries of food are deposited on a heated steel drum.

Dehydration

Equipment: Hot air driers:

- Bin driers
- Cabinet driers(tray driers)
- Tunnel driers
- Conveyor driers(belt driers)
- Fluidised-bed driers
- Klin driers
- Pneumatic driers
- Rotary driers
- Spray driers
- Sun and solar drying

Dehydration

Equipment: Heated surface driers:

- Drum driers(roller driers)
- Vacuum band and vacuum shelf driers

Dehydration

END

Effects on food:

aim of improved drying technologies is to minimise undesirable changes while maximising process efficiency.

- Texture
- Loss of flavour and aroma
- Changes in colour
- Nutritional value

Food Biotechnology

Module #73



Use of Low Temperature

Temperature of a food is reduced to between 1°C and 8°C.

- to reduce the rate of biochemical and microbiological changes, to extend the shelf life foods.
- It causes minimal changes to sensory characteristics and nutritional properties of foods

Use of Low Temperature

FRESH FOODS

Chilling reduces the rate of enzymic and microbiological change and retards respiration of fresh foods. The factors that control the shelf life of fresh crops in chill storage include:

- the type of food and variety or cultivar
- the part of the crop selected (the fastest growing parts have the highest metabolic rates

Use of Low Temperature

FRESH FOODS

- the condition of the food at harvest.
- the temperature of harvest, storage, distribution and retail display
- the relative humidity of the storage atmosphere, which influences dehydration losses.

Use of Low Temperature

PROCESSED FOODS

There are four broad categories of micro-organism, based on the temperature range for growth

1. thermophilic (minimum: 30–40 °C, optimum: 55–65 °C)
2. mesophilic (minimum: 5–10 °C, optimum: 30–40 °C)
3. psychrotrophic (minimum: <0–5°C, optimum: 20–30 °C)
4. psychrophilic (minimum:

Use of Low Temperature

COOK-CHILL SYSTEMS

Individual foods (for example sliced roast meats) or complete meals are produced by cook–chill or cook–pasteurise–chill processes.

The range of chilled foods can be characterised by the class of microbial risk that they pose to consumers as follows:

Use of Low Temperature

COOK-CHILL SYSTEMS

CLASS 1

Foods containing raw or uncooked ingredients, such as salad or cheese as ready-to-eat (RTE) foods (also includes chill-stable raw foods, such as meat, fish, etc.)

CLASS 2

Products made from a mixture of cooked and low risk raw ingredients

Use of Low Temperature

COOK-CHILL SYSTEMS

CLASS 3

Cooked products that are then packaged

CLASS 4

Products that are cooked after packaging, including ready-to-eat-products- for-extended-durability (REPFEDs) having a shelf life of 40+ days.

Use of Low Temperature

MECHANICAL REFRIGERATORS:

Mechanical refrigerators have four basic elements:

- an evaporator,
- a compressor,
- a condenser and
- an expansion valve

constructed from copper as the low thermal conductivity allows high rates of heat transfer and high thermal efficiencies.

Use of Low Temperature

Cryogenic Refrigerators

- A cryogen is a refrigerant that changes phase by absorbing latent heat to cool the food
- Cryogenic chillers use solid carbon dioxide, liquid carbon dioxide or liquid nitrogen.
- Solid carbon dioxide removes latent heat of sublimation and liquid cryogens remove latent heat of vaporization.

Use of Low Temperature

Control of Storage Conditions

- In all stores it is important to maintain an adequate circulation of air using fans, to control the temperature, relative humidity or atmospheric composition.
- Foods are therefore stacked in ways that enable air to circulate freely around all sides.

Use of Low Temperature

Effect on Foods

- Chilling foods to their correct storage temperature causes little or no reduction in the eating quality or nutritional properties of food.
- The most significant effect of chilling on the sensory characteristics of processed foods is hardening due to solidification of fats and oils.

Food Biotechnology

Module #74



Use of food preservatives

- compounds that help slow or prevent bacterial growth in a wide range of products, including foods, medicines, and personal care products.
- These compounds can be natural or synthetic.
- Preservatives play important roles in many products people use every.

Use of food preservatives

ANTIOXIDANTS :

Used to prevent, delay, or minimize the oxidation of the food to which they are added.

NUTRIENTS:

Losses of fractional amounts of nutrients during processing lead to the practice of adding a number of nutrients to the foods.

Flavorings :

Flavourings can be natural or synthetic, that are added to foods to produce flavors.

Flavor Enhancers :

intensify existing flavors. Monosodium glutamate (MSG) is one of the best known and most widely used flavor enhancers.

Use of food preservatives

Acidulants :

lower the pH of any food. Acidulants also enhance desirable flavors such as in pickled products.

Alkaline Compounds:

Compounds that raise the pH. Such as sodium hydroxide or potassium hydroxide.

Use of food preservatives

Sweeteners:

1. Fructose
2. Molasses
3. Honey
4. Maple sugar
5. Lactose
6. Maltose
7. Xylitol
8. Sorbitol
9. Mannitol
10. Aspartame
11. Saccharin
12. Starches

Use of food preservatives

ENZYMES:

1. Invertase
2. Pectinase
3. Cellulase
4. Proteases
5. Lipases
6. Glucose Oxidase
7. Calatase

Use of food preservatives

ENZYMES:

1. Invertase
2. Pectinase
3. Cellulase
4. Proteases
5. Lipases
6. Glucose Oxidase
7. Calatase

Use of food preservatives

Sequestrants:

Sequestrants combine with metals, forming complexes with them and making them unavailable for other reactions.

Polyhydric Alcohols

Used to improve texture and moisture retention because of their affinity for water. Many polyols are present in foods naturally, glycerine (glycerol) being the predominant one.

Use of food preservatives

Surface Active Agents:

affect the physical force at the interface of surfaces. Commonly called surfactants, they are present in all natural foods.

Leaving Agents :

Leavening agents are used to enhance the rising of dough in the manufacture of baked products.

Use of food preservatives

Ionizing Radiation :

Irradiation does not leave a residue in food and it does not make it radioactive. The levels of irradiation allowed in food processing do not induce measurable radioactivity.

Chemical Preservatives :

Added to create conditions unfavourable for food spoilage.

Use of food preservatives

EDN

COLORANTS :

Many colorants (compounds that add colors to foods) are natural, and these include the yellow from the annatto seed; green from chlorophyll; orange from carotene; brown from burnt sugar; and red from beets, tomatoes, and the cochineal insect.

Food Biotechnology

Module #75



Use of high pressure and ultrasound

ULTRASOUND:

It is used in food industry for many purposes including analysis methods and food processing such as:

- freezing,
- cutting,
- drying,
- Tempering,
- homogenization, degassing,
- antifoaming, filtration and extraction.

Use of high pressure and ultrasound

ULTRASOUND:

Ultrasound can be used as a promoter or alternative to food processing.

Use of high pressure and ultrasound

ULTRASOUND :

Ultrasound in food preservation causes:

a) Microbial inactivation
b) Enzyme inactivation

Use of high pressure and ultrasound

EFFECT ON FOOD

- Ultrasound is reported to have minimal effect on the quality of fruit juices.
- Such as oranges, guava, blackberry, and strawberry juices.
- Fruit juice colour is a primary factor in keeping the quality .

Use of high pressure and ultrasound

HIGH PRESSURE:

High pressure processing (HPP) has emerged as a commercially viable food manufacturing tool that satisfies consumers' demand for mildly processed, convenient, fresh-tasting foods with minimal to no preservatives.

Use of high pressure and ultrasound

HIGH PRESSURE:

Pressure treatment, with or without heat, inactivates pathogenic and spoilage bacteria, yeast, mould, viruses, and also spores and extends shelf life.

Use of high pressure and ultrasound

HIGH PRESSURE:

- applied at short periods of time (20minutes).
- Inactivation of Parasites,
- Plant cells.
- Vegetative micro-organisms. Some fungal spores.
- Many food borne viruses. Enzymes are selectively inactivated.
- Macro molecules can change conformation.

Use of high pressure and ultrasound

END

HIGH PRESSURE:

- Small molecules (eg: flavors) are generally unaffected
- High pressure is instantaneously and uniformly applied to the sample.
- Compression is fully reversible.

Food Biotechnology

Module #76



Electricity, radiation and microwaves

- Dielectric (microwave and radio frequency) energy and infrared (or radiant) energy are two forms of electromagnetic energy
- They are both transmitted as waves, which penetrate food and are then absorbed and converted to heat.

Electricity, radiation and microwaves

Direct methods:

- dielectric and
- ohmic heating

In which heat is generated within the product

Indirect method:

- Infrared

heat that is generated externally.

Electricity, radiation and microwaves

Microwave

consists of

- a microwave generator (*magnetron)*,
- aluminium tubes or w*ave guides*, and
- a metal chamber for batch operation, or a tunnel fitted with a conveyor belt for continuous operation.

Electricity, radiation and microwaves

Radio Frequency Heating:

similar principle to microwave heating, but at lower frequencies.

Food is passed between electrodes and a radio frequency voltage is applied across the electrodes.

Electricity, radiation and microwaves

Applications:

high rates of heating absence of surface changes.

The most important applications are

- Thawing and tempering
- Dehydration and baking.
- Other applications

(Pasteurisations and sterilization)

Electricity, radiation and microwaves

Effect On Foods

- no direct effect on micro-organisms, in contrast with ionising radiation.
- In pasteurisation and blanching applications result in reduced losses of heat-sensitive nutrients.

Electricity, radiation and microwaves

OHMIC HEATING:

'resistance heating' or 'electroheating', involves

- an alternating electric current is passed through a food, and
- the electrical resistance of the food causes the power to be translated directly into heat.

Electricity, radiation and microwaves

OHMIC HEATING:

can be used for UHT sterilisation of foods, and especially those that contain large particles (up to 2.5 cm) that are difficult to sterilise by other means.

Electricity, radiation and microwaves

OHMIC HEATING:

Commercially available:

- aseptic processing of high added-value ready meals, stored at ambient temperature
- pasteurisation of particulate foods for hot filling
- pre-heating for canning
- Value added prepared meals, distributed at chill temperatures

Electricity, radiation and microwaves

END

INFRARED HEATING:

The rate of heat transfer depends on:

- the surface temperatures of the heating and receiving materials
- the surface properties of the two materials
- the shapes of the emitting and receiving bodies.

Food Biotechnology

Module #77



Detection of yeast and moulds in dairy and bakery products

- Procedure is applicable to the enumeration of viable yeasts and moulds in foods and food ingredients
- It may also be used to material scraped from food plant equipment and the manufacturing environment.

Detection of yeast and moulds in dairy and bakery products

PRINCIPLE

A medium, containing

a. adequate nutrients for yeasts and moulds
b. Antibiotics is inoculated with a given quantity of the product or with scrapings from equipment or the manufacturing environment.

Detection of yeast and moulds in dairy and bakery products

PRINCIPLE:

incubated at 22-25°C for 3-5 days.

Colonies appearing on the medium

counted and/or examined.

Detection of yeast and moulds in dairy and bakery products

METHOD:

1. Preparation of medium
2. Preparation of dilutions
3. Plating
4. Incubation
5. Counting colonies and examining growth
6. Differentiation of colonies from interfering particles
7. Recording results

Detection of yeast and moulds in dairy and bakery products

Materials and Equipment:

Enumeration of xerophilic yeasts and moulds in grains, flours, nuts, and spices

- Dichloran-glycerol DG 18 agar (DG-18)

Enumeration of xerophilic yeasts and moulds in jams, jellies, and dried fruits

- 20% sucrose (diluent additive for osmophiles,
- Malt extract agar with50% (w/w) sucrose

Detection of yeast and moulds in dairy and bakery products

PLATING:

- surface spread-plate technique rather than with pour plates for maximal exposure of the cells to atmospheric oxygen and to avoids heat stress from molten agar.
- Agar spread plates should be dried overnight before being inoculated.

Detection of yeast and moulds in dairy and bakery products

Counting Colonies and Examining Growth:

- Colonies are counted according to their colonial morphology
- Microscopic examination with crystal violet stained smears may be used to distinguish yeast colonies from bacteria.

Detection of yeast and moulds in dairy and bakery products

END

Differentiation of Colonies from Interfering Particles:

- Food particles interfere with the enumeration of colonies.
- one extra plate of each dilution containing interfering particles, and holding it under refrigeration as a control for comparison during counting.

Food Biotechnology

Module #78



Detection of yeast and moulds in dairy and bakery products

Enumeration of Yeasts & Moulds in Foods:

These agars are suitable for foods where the a_w is above 0.95, such as fresh foods (fruit, vegetables, meat and dairy).

Detection of yeast and moulds in dairy and bakery products

1) Dichloran rose bengal chloramphenicol agar (DRBC)
2) Plate count agar with chloramphenicol (PCA-C)
3) Potato dextrose agar with chloramphenicol (PDA-C)
4) Potato dextrose salt agar with chloramphenicol (PDSA-C) (for analysis of 'spreader' moulds)

Detection of yeast and moulds in dairy and bakery products

OTHER:

- Peptone water (0.1%) (PW)
- 2% sodium citrate tempered to 45°C (diluent for high fat foods, such as cheese) (optional)
- 1N HCl and 1N NaOH
- Gram stain solutions
- Stomacher, blender or equivalent

Detection of yeast and moulds in dairy and bakery products

OTHER:

- pH meter
- Light microscope
- Colony counting device (optional)
- Incubator (darkened) capable of maintaining 22 to 25°C, 55°C
- water bath

Detection of yeast and moulds in dairy and bakery products

INCUBATION:

- Incubate plates in an upright position at 22 to 25°C for 3-5 days.
- Incubate plates in the dark.
- Count colonies on plates after 5 days.
- Or third day and if mould colonies are numerous,
- count them and then count again on the fifth day, if possible.

Detection of yeast and moulds in dairy and bakery products

INCUBATION:

- Handle the plates as little as possible when counting on day 3 so spores will not be dislodged, which may result in secondary growth.

Detection of yeast and moulds in dairy and bakery products

END

Recording Results:

- average count (arithmetic mean) is calculated
- To compute the yeast and mould count, use the formula:

N = A x D,

where N is the no. of colonies /g (mL) of product,

A is the average count/ plate, and

D is the dilution factor

Food Biotechnology

Module #79



Fermented foods

General Method of Production :

- exposing the raw or starting food materials to given conditions.
- Microorganisms grow and produce end-products with desirable acceptance qualities, many of which are attributed to the metabolic end products.

Fermented foods

Microorganisms Used:

- Variety of species and strains of bacteria, yeasts, and moulds are associated with fermentation of foods.
- Single or mixed strains or species

Fermented foods

Maximum growth of a desirable microorganism and optimum fermentation rate are dependent on environmental parameters such as:

- nutrients,
- temperature of incubation,
- oxidation-reduction potential, and
- pH.

Fermented foods

PROCESS:

Foods can be fermented in three different ways, based on the sources of the desirable microorganisms:

- natural fermentation,
- back slopping, and
- controlled fermentation.

Fermented foods

Fermented Dairy Products:

Fermented dairy products can be broadly divided into two groups:

- fermented milk products and
- cheeses.

Fermented foods

- In fermented milk products, all the constituents of the milk are retained in the final products, with the exception of those partially metabolized by the bacteria.
- In cheeses, a large portion of milk constituents is removed in whey to obtain the final products.

Fermented foods

Fermented Meat Products:

Fermented meat products are produced by first mixing

- meat,
- fat,
- salt,
- sugar,
- curing agents, and
- spices;

Filling the mixture in a casing; and fermenting it naturally or selected starter-culture bacteria.

Fermented foods

Fermented Vegetable Products:

- Almost all vegetables can be fermented through natural processes LA bacteria
- originated in the early years of human civilization and even now is widely used by many cultures.

Fermented foods

END

Fermented Vegetable Products:

sauerkraut (from cabbage), olives, cucumbers, carrots, celery, beans, peas, corn, okra, tomatoes, cauliflower, peppers, onions, citron, beets, turnips, radishes, chard, Brussels sprouts, and their blends.

Food Biotechnology

Module #80



Microbiology of fermented products

Cultured Buttermilk Fermentation:

- partially skim milk through controlled fermentation with starter cultures.
- It should have a pleasant acid taste and aroma (from diacetyl), with slight effervescence (from $C0_2$).
- It should have white colour with smooth, thick body and should pour easily.

Microbiology of fermented products

Microbiology

- *Lac. lactis* ssp. *lactis* or *cremoris* is used for acid and *Leu. mesenteroides* ssp. *cremoris* for diacetyl and $C0_2$.

Microbiology of fermented products

YOGHURT

- coagulation of milk (skim, or full fat) by starter-culture bacteria.
- sharp acid taste with a flavor similar to walnuts and a smooth mouth feel.
- The flavor is due to the combined effects of acetaldehyde, lactate, diacetyl and acetate, but 90% of the flavor is due to acetaldehyde.

Microbiology of fermented products

Effort By: AARISH ALI

Cottage Cheese:

- from low-fat or skim milk and has a soft texture with Ca. 80% moisture.
- It is unripened and has a buttery aroma due to diacetyl (along with lactic acid and little acetaldehyde).
- Mixed strains of *Lac. lactis* ssp. *cremoris* and *lactis* are predominantly used for acid.

Microbiology of fermented products

- made from whole milk. contains less than 39% moisture, 48% fat
- Starters are selected mixed strains of *Lac. lactis* ssp. *cremoris* or *lactis. Leuconostoc,* which may be added for flavor.
- Starters can be used as frozen concentrates.

Microbiology of fermented products

Swiss Cheese:

- from partially skimmed milk (cow's milk) and coagulated with acid and rennin contains Ca. 41% moisture and 43% fat
- The cheese should have uniformly distributed medium-sized eyes (openings).It has a sweet taste, due to proline, and a nutty flavor.

Microbiology of fermented products

Microbiology of Swiss Cheese:

- Starters are *Str. thermophilus* and *Lab. helveticus* as primary for acid and *Propionibacterium* sp. as secondary for eye formation, taste, and flavor.

Microbiology of fermented products

Microbiology Of Semidry Sausages:

- The average composition is Ca. 30% fat, 20% protein. 3% minerals (salts), and 47% water.
- For high temperature and low pH, *Pediococcus acidilactici* strains are preferred;
- For low temperature and high pH, *Lab. plantarum* strains are preferred.

Microbiology of fermented products

- flavor imparted by the combined effect of lactate, acetate, and diacetyl, and some breakdown components from proteolysis and lipolysis.
- The use of spices also contributes to the flavor.
- Those containing nitrite have a pinkish color in contrast to the grayish color in products without it.

Microbiology of fermented products

Microbiology Of Sauerkraut:

- fermenting shredded cabbage.
- has a sour taste with a clean acid flavor.
- The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1%).

Microbiology of fermented products

- Among the lactic acid bacteria, most are *Lactococcus* spp. and *Leuconostoc* spp and
- a small fraction is *Lactobacillus* spp. and *Pediococcus* spp.

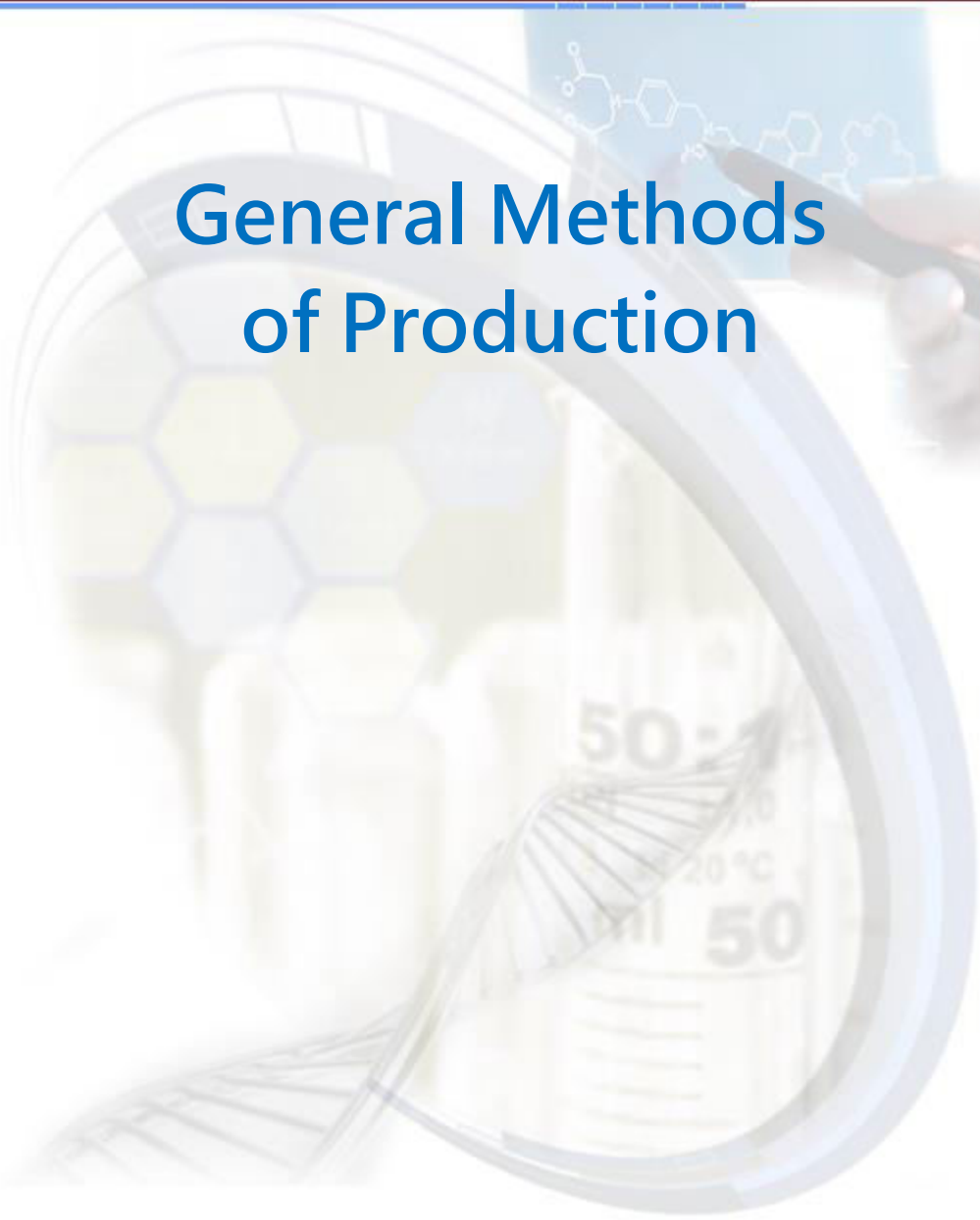
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Food Biotechnology

Module :- 81

Food Biotechnology

General Methods of Production



General Methods of Production

Fermentation :-
raw or starting food materials exposed to conditions that favor growth and metabolism of desirable microorganisms

→ Products that are made via fermentation are called fermented products
e.g. Pickles , Vinegar etc

General Methods of Production

→ Process of fermentation based on two conditions

1) Metabolic activities of micro-organisms

2) Storage condition

→ Type of micro-organisms selected for fermentation is crucial for fermentation process

General Methods of Production

Fermentation Process

Foods can be fermented in three different ways, based on the sources of the desirable microorganisms:

1) Natural Fermentation
2) Back Slopping
3) Controlled Fermentation

General Methods of Production

Natural Fermentation :-
both desired and non desired organisms are involved .

→ Rapid growth of desired one

→ Suppress the growth of undesired one

Benefits :-

Special Aroma

Drawback :-

Non-consistency in the organoleptic properties.

General Methods of Production

2) Back Slopping :-

conditions are set to facilitate the growth of the microorganisms coming from the previous product. This is still practiced in the production of many ethnic products in small volumes e.g. Yoghurt from Milk

General Methods of Production

END

3) Controlled Fermentation

starting materials (may be heat treated) are inoculated with a high population (10^6 cells/ml or more) of a pure culture of single or mixed strains or species of starter culture.

→ Larger volume can be fermented

→ Consistency

Food Biotechnology

Module :- 82

Food Biotechnology

Fermentation Process



Fermentation Process

As the desirable microorganisms grow in specific conditions on selected foods, they utilize some nutrients and produce some end products.

Fermentation Process

1) Alcoholic Fermentation :-
(Alcohol end product)

2) Acetic Acid Fermentation
(Acetic acid end product)

3) Lactic Acid Fermentation
(Lactic Acid end product)

Fermentation Process

General process

Substrate (Vegetables, Meat , Milk etc.) + Starter Culture (Bacteria, Yeast , mold) + optimum conditions (Time, Temperature etc.) -→ End products (yoghurt , Vinegar , Alcohol , acetic acid etc.)

Fermentation Process

Alcoholic Fermentation:

In this fermentation Alcohol is produced as a end product by the activity of micro-organisms (yeast)

→ Mostly used by Bakery industry for special aroma

Fermentation Process

Acetic Acid Fermentation :-

In this fermentation Acetic acid is produced as end product by the activity of Acetobacter bacteria on sugary / alcoholic food stuff

→ Used for the production of vinegar

Fermentation Process

END

Lactic Acid Fermentation

In this fermentation glucose and other six Carbon sugars are converted into cellular energy and metabolite lactate via Lactic acid producing bacteria as starter culture

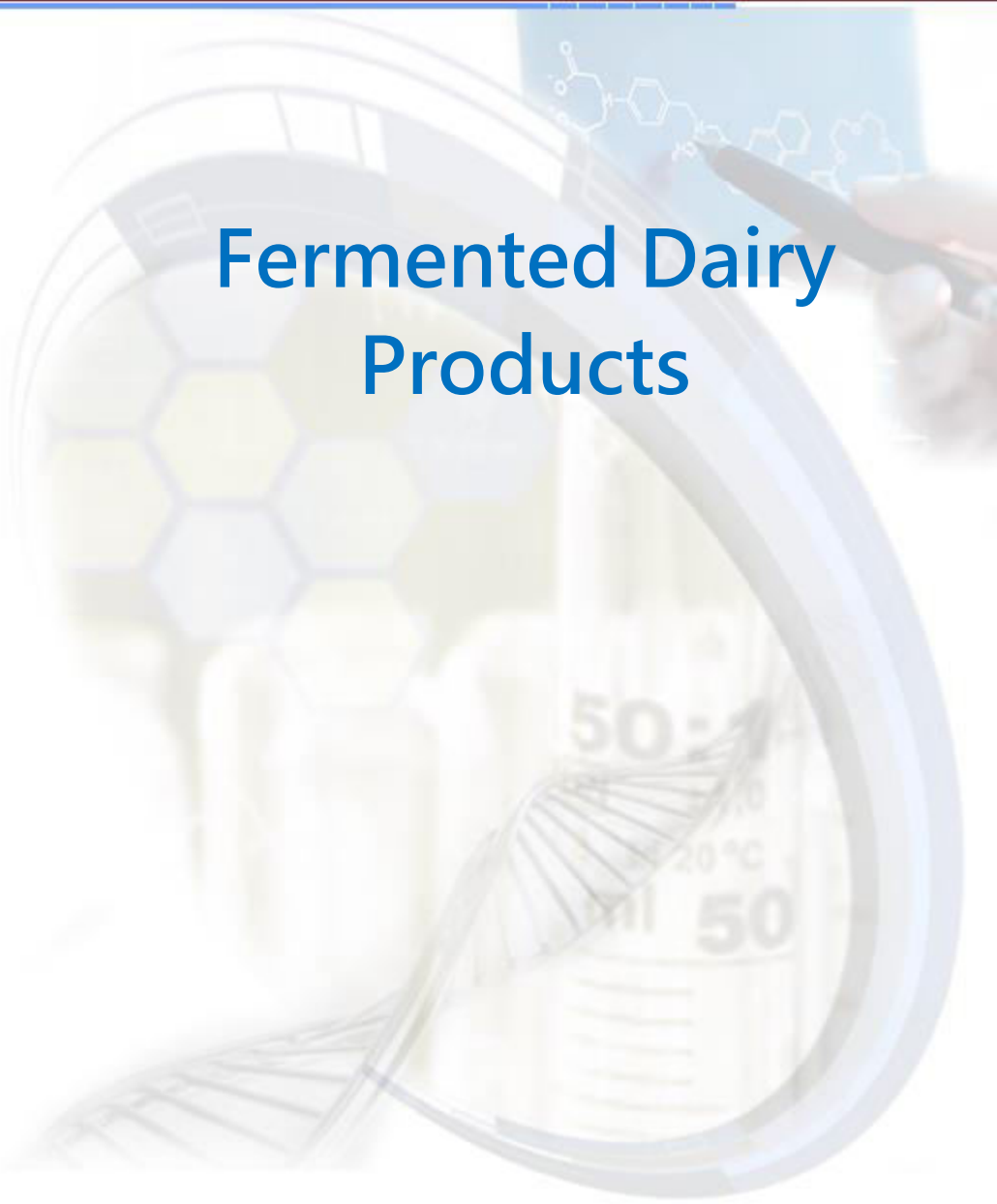
→ Mostly used for the production of yoghurt from milk

Food Biotechnology

Module :- 83

Food Biotechnology

Fermented Dairy Products



Fermented Dairy Products

Huge variety of products produced globally

A few are produced by →Controlled fermentation, and the microbial types and their respective contributions are known.

Fermented Dairy Products

→ In many others, fermented either naturally or by back slopping, the microbial profiles and their contribution are not exactly known.

→ Many types of lactic acid bacteria and some yeasts are found to predominate microbial flora in these products.

Fermented Dairy Products

1) Butter milk.:-

Made with Lactococcus species without or with Leuconostoc cremoris; . Some can have a ropy variant of Lactococcus species (landfill in Norway) or mold (Geotrichum candidum in villi in Finland)

Fermented Dairy Products

Yogurt :

Streptococcus thermophiles and Lactobacillus delbrueckii subsp. bulgaricus;
some with added Lab. acidophilus, casein, rhamnosus, and Bifidobacterium spp.;
Or
Lactococcus species and Lab. plantarum and lactose-fermentating yeasts (dahi)

Fermented Dairy Products

3)Acidophilus Milk.

Made with Lab. acidophilus.

4. Bifidus milk

Made with Bifidobacterium spp. **5.**

Yakult.

Made with Lab. casei; may contain Bifidobacterium spp.

Fermented Dairy Products

END

6. Kefir.

Made from Lab. kefir (several species of yeasts along with Leuconostoc, Lactobacillus, and Lactococcus spp.).

7. Kumiss.

Made from Lab. delbrueckii subsp. buigaricus and yeasts

Food Biotechnology

Module :- 84

Food Biotechnology

Microbiology of Yogurt

Microbiology of Yogurt

Yogurt Fermentation Steps:-

1) Processed Milk
2) Starter Culture
3) Growth of Micro-organisms
4) Metabolites production
5) Flavor Production
6) Formate Production
7) Slime Formation (Glycan)
8) Proteolysis

Microbiology of Yogurt

1) Milk Treatment :-

Firstly milk is treated

→ Heat Treated (Pasteurize , Sterilize)

→ Homogenized

→ Standardized

Microbiology of Yogurt

2) Starters (Controlled Fermentation)

Frozen concentrates or direct vat set starters can be used.

Normally, *Lab delbrueckii ssp. bulgaricus and Str. thermophilus* are used.

Microbiology of Yogurt

other species may be combined, such as *Lab. acidophilus* and *Bifidobacterium spp., Lab. rhamnosus, or Lab. casei*.. For a good product, the two starter species should be added at a *Streptococcus Lactobacillus* cell ratio of 1:1; in the final product, the ratio should not exceed 3:2.

Microbiology of Yogurt

3)Growth of Micro-organisms :-

fermentation is conducted at ca. 110°F (43.3°C). both acid and flavor compounds are produced at the desired level If the temperature is raised above 110°F, the Lactobacillus sp. predominates, causing more acid and less flavor production

Microbiology of Yogurt

4)Metabolites Production :-

Lactic Acid is produced by the action of micro-organisms on lactose

Microbiology of Yogurt

5)Flavor Development :-

The major flavor compound in yogurt is acetaldehyde (25 ppm), with some diacetyl (0.5 ppm) and acetate. Acetaldehyde is produced by Lactobacillus sp.

Microbiology of Yogurt

6) Formate Production :-
Formate (necessary for Lactobacillus growth) is produced by Str. thermophilus from pyruvate by the action of formate lyase.

Microbiology of Yogurt

7) Slime Formation (Glycan)

B- galactosidase in some strains of *Str. thermopilus* polymerizes glucose to produce oligosaccharides and glycan, which may give a viscous texture to yogurt.

Microbiology of Yogurt

8) Proteolysis :-

Proteinase (LAB) reacts with milk protein and make curdles which leads to yoghurt

Microbiology of Yogurt

Microbial Problems :-

In plain yogurt. flavor problems can be associated with the concentrations of acetaldehyde. A low concentration gives a chalky and sour flavor, and too much acetaldehyde can give a green flavor.

Microbiology of Yogurt

END

Similarly, too much diacetyl gives a buttery aroma.

Too much acid production during storage causes a sour taste.

Proteolysis and accumulation of bitter peptides during storage are associated with bitter flavor

Food Biotechnology

Module 85 :-

Food Biotechnology

Fermented Product (Cheese)

Fermented Product (Cheese)

Introduction :-

Cheese are made by coagulating the casein in milk with lactic acid produced by lactic acid bacteria, without or with the enzyme rennin, followed by collecting the casein for further processing, which may include ripening.

Fermented Product (Cheese)

Types of cheese:-

Cheese varieties have been grouped in different ways. Examples of several varieties based on starter cultures used and some important secondary flora are listed here

1) Unripened cheese

→ Soft

2) Ripened cheese

→ Soft, Semi-hard,Hard

Fermented Product (Cheese)

1) Unripened Cheese :-

Soft

- Cottage cheese with starters *Lac. lactis ssp. lactis* and cremoris and *Leuconostoc mesenteroides ssp. cremoris.*
- Mozzarella cheese with starters *Stl: thermophilus* and *Lab. delbrueckii ssp. bulgaricus*.

Fermented Product (Cheese)

2) Ripened Cheese :-

→Soft

• Brie with *starter Lac. lactis ssp.; Penicillium sp*. and yeasts are secondary flora.

→Semi - hard

• Gouda with starters *Lac. lactis ssp. and Leuconostoc spp*.; dairy Propionibacterium may be secondary flora. • Blue cheese with starter *Lac. lactis ssp., Leuconostoc spp.; Penicillium roquefortii*, yeasts, micrococci are secondary flora

Fermented Product (Cheese)

Hard cheese :-

- Cheddar cheese with starters *Lac. lactis ssp.*; some *lactobacilli* and *pediococci* (and probably *enterococci*) are secondary (or associative) flora.

Fermented Product (Cheese)

Cottage cheese :-

Cottage cheese is made from low-fat or skim milk and has a soft texture with ca. 80% moisture. It is unripened and has a buttery aroma due to diacetyl (along with lactic acid and little

Fermented Product (Cheese)

END

Production of cottage cheese :-

1. Pasteurized, cooled to 70°F (22.2°C), starter added, and incubated for 12 h at pH 4.7.
2. Firm curd set, cut in cubes, and cooked at 125°F (51.7°C) for 50 min or more.
3. Whey drained off, stirred to remove more to get dry curd.
4. Salted, creamed, and preservative added.
5. Packaged and refrigerated

Food Biotechnology

Module 86



Food Biotechnology

Fermented Meat Products

Fermented Meat Products

Meat fermentation probably originated in the Mediterranean countries and later spread to European countries and North America. In the United States, semidry sausages are most popular, although some dry sausages are also produced.

Fermented Meat Products

Fermentation of Meat :-

1)Fermented meat products are produced by first mixing meat, fat, salt, sugar, curing agents, and spices.

2) Filling the mixture in a casing; and fermenting it either naturally or by adding (during mixing) selected starter-culture bacteria

Fermented Meat Products

3) Acids produced by the starters during fermentation and the curing agents used help control the growth of pathogenic and spoilage bacteria that might be present in the meat.

Fermented Meat Products

4) Depending on the type, the fermented products may be dried to reduce Aw or smoked or heated to ensure the safety and shelf life of the products

Fermented Meat Products

Types of Fermented Meat Products :-

1) Sausages

→ Semi-dry (Heated before consumption)

→Dry (Cooked before consumption to reduce Aw)

Fermented Meat Products

Types of semi-dry and dry sausages :-

→Semidry and dry sausages include pepperoni, Genoa salami, hard salami, summer sausage, beef sticks, beef logs, Italian salami etc.

→Most made with beef and pork, but recently meat from chicken and turkey is also used.

Fermented Meat Products

END

Microbiology of salami :-

→The average composition is ca. 30% fat, 20% protein. 3% minerals (salts), and 47% water.

→ They have a tangy taste with a desirable flavor imparted by the combined effect of lactate, acetate, and diacetyl, and some breakdown of protein by proteolysis enzyme

Food Biotechnology

Module 87

Food Biotechnology

Microbiology of Semi-dry Sausages

Microbiology of Semi-dry Sausage

Sausages :-

It’s a cylindrical meat product usually made from ground meat ,often pork , beef along with salt, spices and other flavorings and breadcrumbs encased in a skin traditionally made from intestine

Microbiology of Semi-dry Sausage

Starters selection based on pH and Temperature for semi-dry sausages :-

→ For high temperature and low pH, *Pediococcus acidilactici* strains are used

→ For low temperature and high pH, *Lab. plantarum* strains are preferred

Microbiology of Semi-dry Sausage

raw meat used may contain pathogens and spoilage bacteria, it is extremely important that starter culture grows rapidly and produces acid in large amounts to reduce pH from the initial 5.7 to ca. 5.3 very quickly to retard their growth.

Microbiology of Semi-dry Sausage

Microbial Problems :-

→Slow acid production can be a serious problem if the starters used are not metabolically active or have lower numbers of viable cells or due to other factors such as low glucose and high salts in the mix.

Microbiology of Semi-dry Sausage

END

→Sour or no flavor can occur if the starter, especially *Ped. acidilactici*, grows very rapidly and reduces the pH to below 4.5.

→Gas formation can occur because of growth of *Leuconostoc spp*. during fermentation and during storage in vacuum packages

Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

Fermented Vegetable Products

→Vegetable fermentation since early human civilization and even now is widely used now a days.

→ Almost all vegetables can be fermented through natural processes, because they harbor many types of lactic acid bacteria

e.g. sauerkraut , reddish, turnips , beets , beans, peas , corn etc.

Fermented Vegetable Products

Microbiology:

→ Fermented vegetable products have Sour taste with clean acid flavor

2) Processing :-

→ Wash, Clean and shred the vegetables

→ Put in a jar and air tight it and layered with salt (Anaerobic conditions for micro-organisms)

→ Ferment it for 2-3 months (18°C)

Fermented Vegetable Products

3) Starter (Natural) and growth :-

The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1 %). Among the lactic acid bacteria, most are *Lactococcus spp.* and *Leuconostoc spp*.

FERMENTED VEGETABLE PRODUCTS

END

Microbial Problems :-

Off-flavor, soft texture, and discoloration of vegetables can occur by growth of molds and yeasts when air is not completely excluded.

Food Biotechnology

Module 86



Food Biotechnology

Fermented Meat Products

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Food Biotechnology

Module 87

Food Biotechnology

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Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

Fermented Vegetable Products

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e.g. sauerkraut , reddish, turnips , beets , beans, peas , corn etc.

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Fermented Vegetable Products

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FERMENTED VEGETABLE PRODUCTS

END

Microbial Problems :-

Off-flavor, soft texture, and discoloration of vegetables can occur by growth of molds and yeasts when air is not completely excluded.

Food Biotechnology

Module 88 :-

Food Biotechnology

Fermented Vegetable Products

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FERMENTED VEGETABLE PRODUCTS

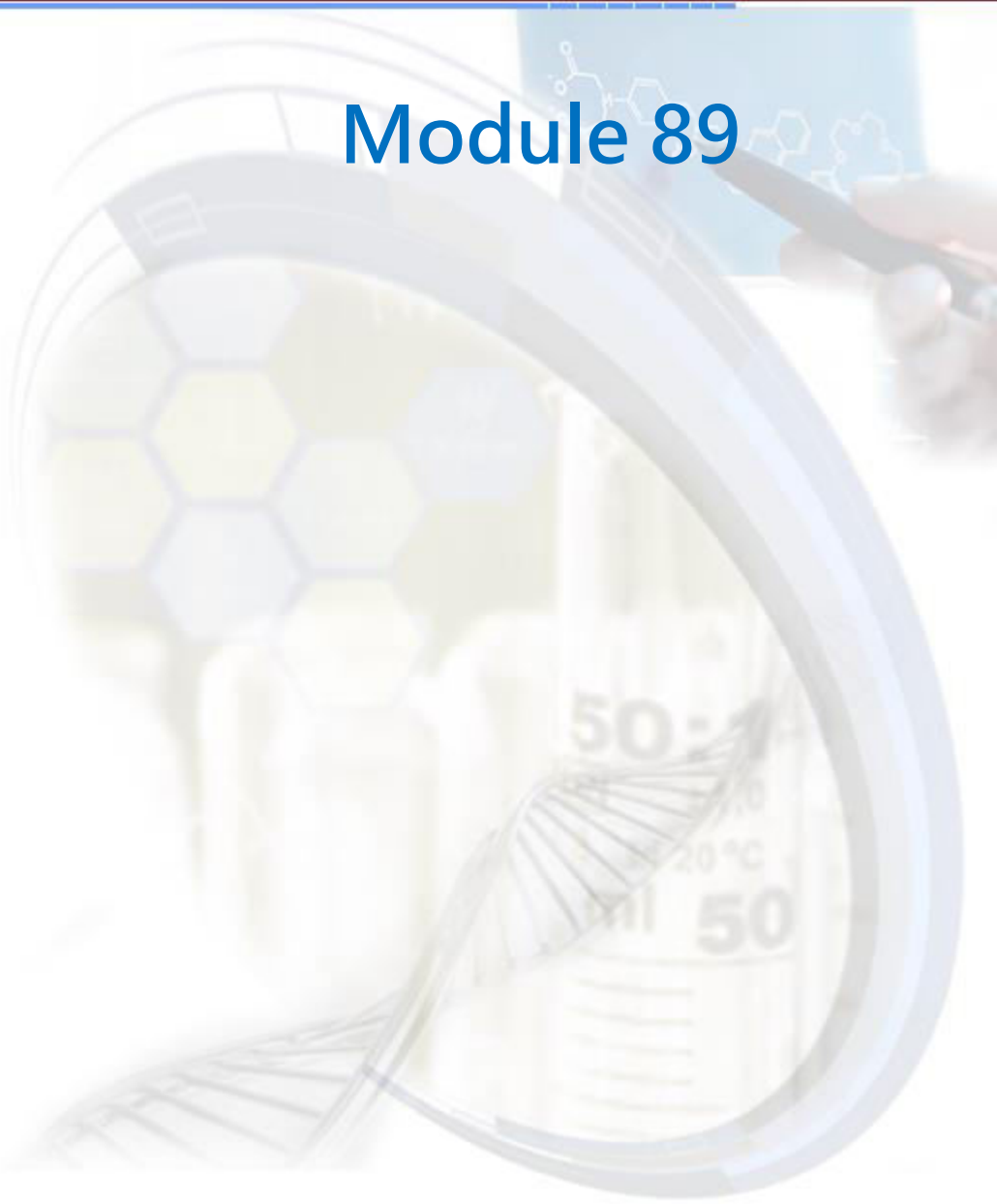
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Food Biotechnology

Module 89



Food Biotechnology

Sauerkraut

SAUERKRAUT

Sauerkraut is finely cut raw cabbage that has been fermented via various lactic acid bacteria which gives it a special sour flavor and aroma

SAUERKRAUT

Processing :-

1) Cabbage cleaned, trimmed, and shredded fine and uniform.

2) Packaged tight to exclude air in vat, and layered with salt (2.25%).

3) Top covered to exclude air, and fermented at 18°C (65°F) for 2 months.

SAUERKRAUT

3) Starter (Natural) and growth :-

The raw material has a large number of undesirable organisms and a small population of lactic acid bacteria «1 %). Among the lactic acid bacteria, most are *Lactococcus spp.* and *Leuconostoc spp.*

SAUERKRAUT

Biochemistry :-

Leuconostoc spp. metabolize sucrose, hexoses, and some pentose's in the raw material to lactate, acetate, ethanol, C02, and diacetyl

SAUERKRAUT

END

Microbial Problems :-

1)Off-flavor, soft texture, and discoloration of sauerkraut by growth of molds and yeasts when aerobic conditions..

2)A slimy texture occur due to overgrowth of *Leuconostoc spp*. in the presence of sucrose; they metabolize fructose but synthesize dextran's from glucose

Food Biotechnology

Module 90



Food Biotechnology

Vegetable Pickles

VEGETABLE PICKLES

Blends of different vegetables (Reddish, Carrot, Peas, Lemon etc.) which are fermented (Lactic Acid Fermentation) to have special aroma , flavor and increased shelf life are called vegetable pickles

Effort By: AARISH ALI

VEGETABLE PICKLES

Types of Pickling :-

Pickling can be either done by

→ Direct immersion of vegetables in the salt and vinegar solution

→Anaerobic fermentation

VEGETABLE PICKLES

Process of Vegetable Pickling :-

1) Selection of Qualitatively good vegetables

2) Preparation of vegetables (Cutting, Slicing , Dicing , Shredding etc. and addition of salt)

3) Fermentation :- Anaerobic Fermentation mostly by LAB

VEGETABLE PICKLES

4) Filling :-

Fill in a air tight container and add oil .

→ Layer of oil over pickle act as a barrier for micro-organisms moreover oil has anti fungal property so preserve it efficiently

VEGETABLE PICKLES

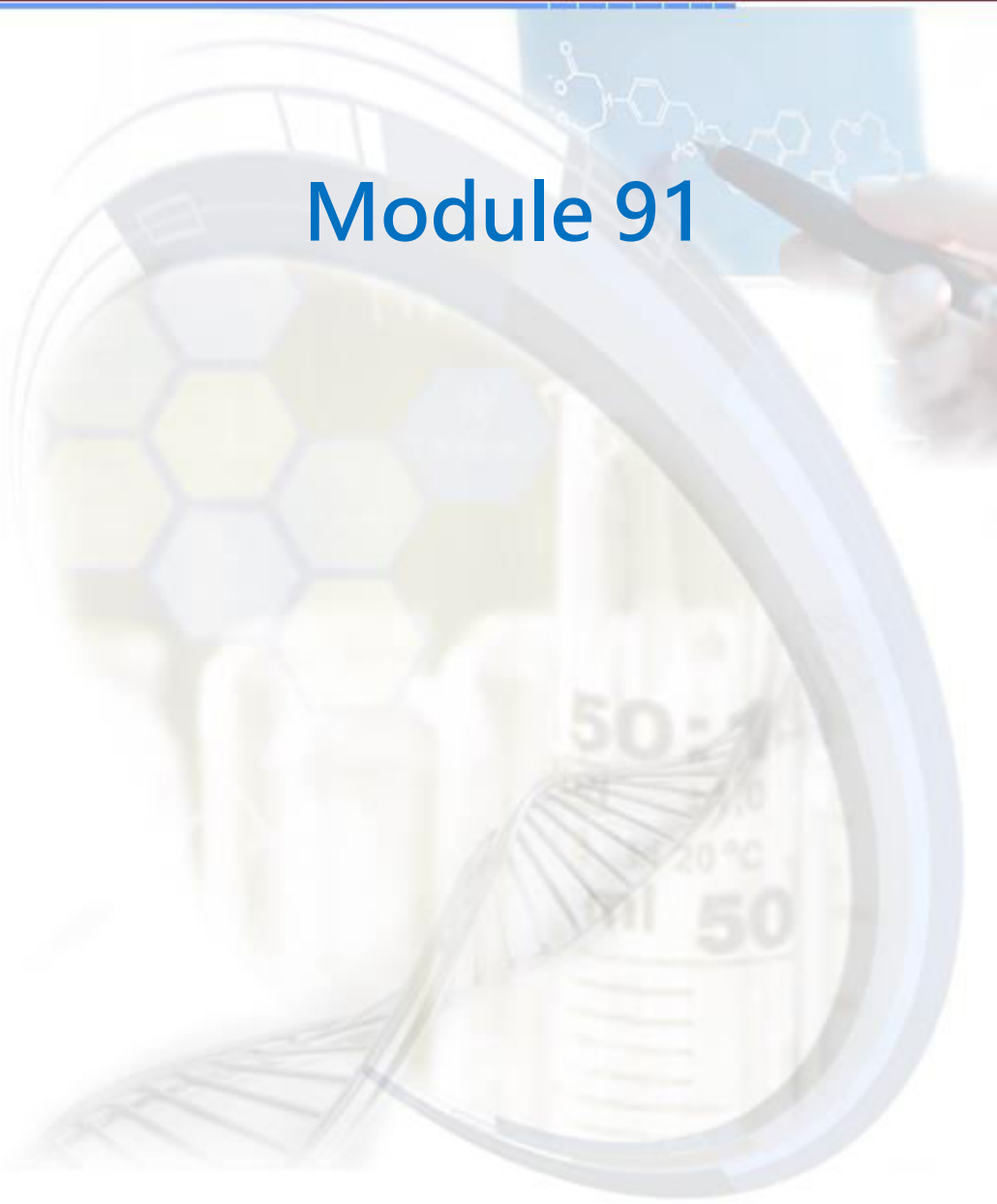
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Non-Fermented Pickles:

1) Selection of fruits
2) Preparation of Fruits (add salt /sugar)
3) Cooking
4) Containerization of fruits

Food Biotechnology

Module 91



Food Biotechnology

Production of Cheddar and Related Hard Cheese

Production of Cheddar and Related Hard Cheese

→Cheddar cheese is classified as a hard cheese, ranging in color from nearly white (particularly if made from goat or sheep milk) to yellow to orange

→ Standards of identity for Cheddar include 39% moisture and 50% fat on a dry basis.

Production of Cheddar and Related Hard Cheese

Ingredients used for cheese making :-

) Milk

2) Calcium Chloride

3) Starter Culture

4) Adjunct Culture

5)Color

6) Enzyme

7) Salt

8) Optional ingredients

Production of Cheddar and Related Hard Cheese

Preparation for cheese making :-

1) Culture Selection / Propagation:-

→ Most important

1)Liquid (for propagation of mother culture; rarely used today);

2) Deep-frozen concentrated cultures (for propagation of bulk starter);

Production of Cheddar and Related Hard Cheese

3.)Freeze-dried concentrated cultures in powder form (for propagation of bulk starter

4. Deep-frozen or freeze-dried, super-concentrated cultures

Production of Cheddar and Related Hard Cheese

2) Cheese Milk Pretreatments

→ Centrifugation prior to standardization

→ Homogenization

→ Pasteurization

Production of Cheddar and Related Hard Cheese

Steps for cheese making :-

1) Addition of Culture , Calcium Chloride and color in pre-treated milk

2) Ripening for 15-30 minutes

3) Add coagulant and check after regular interval

4) After coagulation cut into proper sizes

Production of Cheddar and Related Hard Cheese

5) Cook it for 40-50 minutes at 102 Fahrenheit

6) Drain whey via proper filtration aid

7) Trenching

8) Cheddaring

9) Milling

10) Salting (TASTE)

Production of Cheddar and Related Hard Cheese

11) Hooping

12) Pressing (HARD CHEESE)

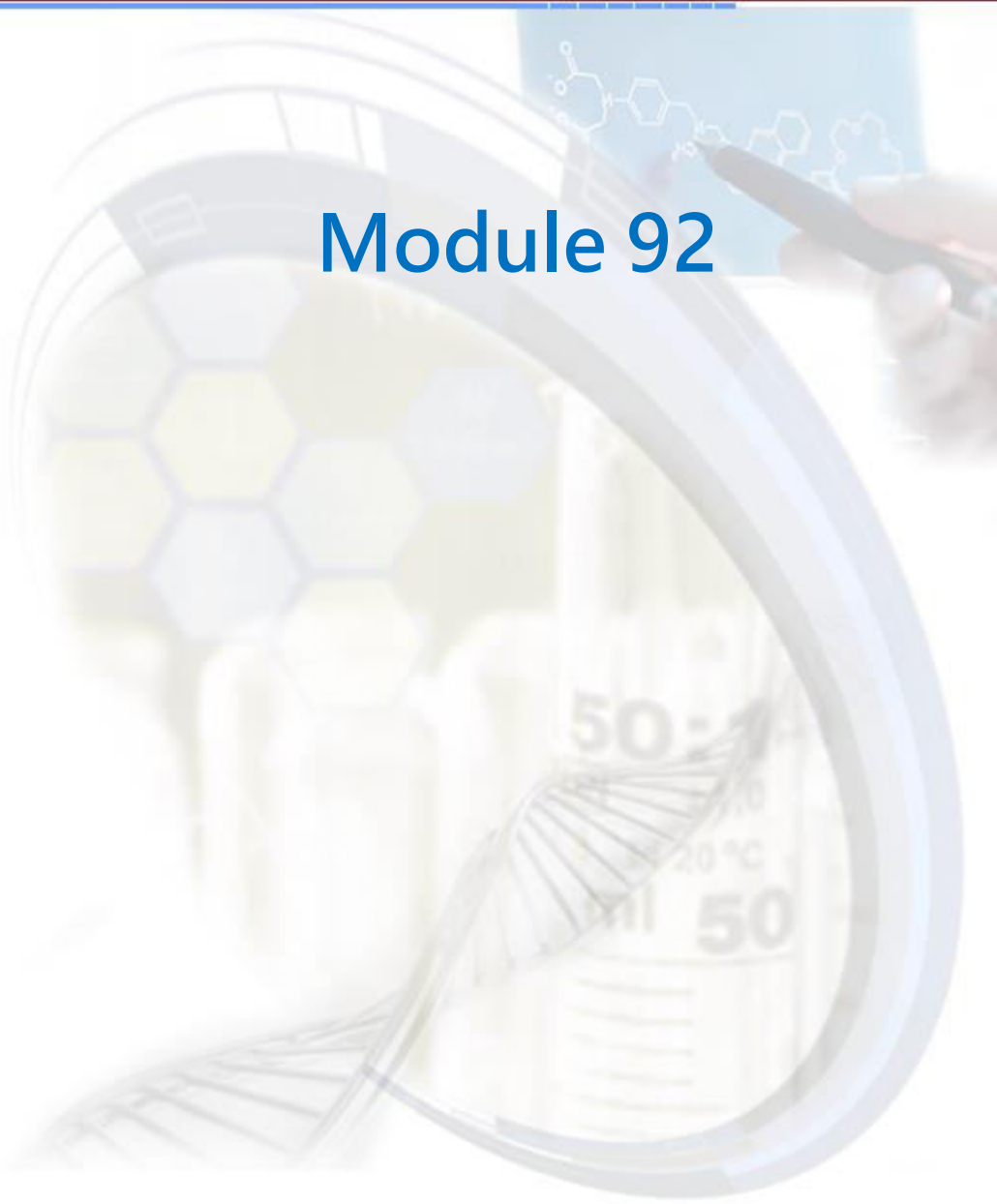
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13) Packing

14) Ageing

Food Biotechnology

Module 92



Food Biotechnology

Ingredients used for Cheese Making

Ingredients used for Cheese Making

→Good cheese requires high-quality milk and carefully selected starter cultures.

→Additional ingredients are often utilized to enhance visual appeal (annatto), coagulation properties (calcium chloride or enzymes), and flavor development (adjunct cultures or enzymes) to make great cheeses.

Ingredients used for Cheese Making

Ingredients for cheese production :-

1) Milk
2) Calcium Chloride
3) Starter Culture
4) Adjunct Culture
5)Color
6) Enzyme
7) Salt
8) Optional ingredients

Ingredients used for Cheese Making

1) Milk :-

→must start with high-quality milk.

→Cheddar and most Cheddar-like cheeses can be made from raw, heat-treated, heat-shocked, or pasteurized milk, nonfat milk or cream, alone or in combination

2) Calcium Chloride :-

→ Use as coagulant (0.02 %)

Ingredients used for Cheese Making

3) Starter Culture :-

→ Isolates that produce sufficient acid to reduce the pH of milk to 5.3 in 6 h at 30–37.8°C, and aid in curd digestion and flavor development

→ For cheddar cheese → *Lactococcus lactis Sub spp + lactococcus cremoris* is used.

Ingredients used for Cheese Making

4) Adjunct Culture :-

→An adjunct culture is a one that is added, along with starter culture, for the desirable characteristics it may impart upon the cheese other than acid.

→Adjunct cultures are select nonstarter lactic acid bacteria (NSLAB)

Ingredients used for Cheese Making

5) Color :-

→The recognizable yellow to orange color of Cheddar and Colby cheeses is derived from annatto, an extract from seeds of the “Lipstick tree,

6) Enzyme :-

1)Rennet

2) Chymosin

Other clotting enzymes

Ingredients used for Cheese Making

7) Salt :-

→Cheddar cheese typically contains 1.6–1.8% salt in the form of NaCl.

→Salt enhances flavor, encourages syneresis, and slows or stops growth of salt-sensitive bacteria

Ingredients used for Cheese Making

END

8) Optional ingredients :-

→ sodium benzoate ,
→sodium propionate,
→sorbic acid →Hydrogen peroxide is allowed, if followed by a quantity of catalase preparation sufficient to eliminate the hydrogen peroxide

Midterm Term Syllabus
Lecture 01 to 92

Efforts By

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Note:

Students must follow the course announcement for midterm syllabus & any up-gradation in the course.

REMEMBER IN PRAYERS

KEEP SMILING

SAY NO TO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

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